

PS1. Poster Session 1

September 29 (Tue), Room 301 (Grand Ballroom), BEXCO

Board No. 보드번호	Title	Presenter	Abstract No. 초록번호
S1. Polymer Synthesis (PSK50 Student Presentation Award Applicants)			
PS1_001	Polymer Chain Modification via HAT Chemistry and Its Application in Graft Copolymer Synthesis	Manish KUMAR, The University of New South Wales, Australia	POS1-0015
PS1_002	Reactive Nanofibers - Design of Stimuli-Responsive Supramolecular Bottlebrushes	Anne SKOTNICKI, University of Bayreuth, Germany	POS1-0077
PS1_003	Mechanophore-Bearing PE-like Polymers for Force-Triggered Degradation	Curie YIM, Yonsei University, Korea	POS1-0084
PS1_004	Synthesis and Amphiphilic Property of Substituted Acrylic Acid Polymer with Octadecyl Group	Syamim Syufiana Binti SAZALI, Mie University, Japan	POS1-0147
PS1_005	Polymerization-Induced Self-Assembly in Liquid Crystal Solvents: Toward Aligned Polymer Nanoobjects	Seungmin LEE, KAIST, Korea	POS1-0152
PS1_006	N-Alkylamide Functionalization of Vinyl-Addition Polynorbornene Derivatives via Post-Polymerization Modification to Introduce Interchain Hydrogen-Bonding Interactions	Jun Woo PARK, Advanced Materials Division, Korea Research Institute of Chemical Technology (KRICT), Korea	POS1-0161
PS1_007	Cocatalyst Design for Controlled Olefin and Diene Polymerization with Single-Site Catalysts: Toward Simplified Catalytic Processes	Jia YU, Hiroshima University, Japan	POS1-0180
PS1_008	High-Rate Responsive Polyborosiloxane with Exceptional Impact Stiffening and Energy Absorption for Impact Protection	Yue HUANG, Sichuan University, China	POS1-0211
PS1_009	Structure-Property Relationships in Liquid-Crystalline Epoxies Cured by Dianhydride	Hoang LE MINH, Kyungpook National University, Korea	POS1-0225
PS1_010	Development of High-Tg Reprocessable Vitrimers Through Molecular Rigidity Control of Bismaleimide Monomers	Woohyeon KWON, Kyungpook National University, Korea	POS1-0250
PS1_011	Design and Characterization of Cyanate Ester Thermosets Exhibiting Both Liquid Crystalline Ordering and Vitrimer Behavior	Eunseo NAM, Kyungpook National University, Korea	POS1-0251
PS1_012	Development of Controlled Radical Polymerization with Thioether Initiators and Nickel Complexes	Mina YONEGUCHI, Nagoya University, Japan	POS1-0256
PS1_013	Harnessing Nickel's Dual Reactivity for the One-Pot Synthesis of Polar Polyolefin Block Copolymers	Muhammad ARSLAN, University of Houston, United States	POS1-0262
PS1_014	Degradable Polymers Synthesized via Thermal Cationic Copolymerization and Their Application for Photoimageable Materials	Mingyeong KIM, Inha University, Korea	POS1-0341
PS1_015	Degradable 3-Arm Polymers Using Acid-Cleavable Chain Transfer Agent	Hyein LEE, Inha University, Korea	POS1-0342
PS1_016	Photodegradable Poly(caprolactone-co-urethane)s for Dual-Tone Photoimaging Materials	Yedam KIM, Inha University, Korea	POS1-0347
PS1_017	Design of Branched Polymers Based on Silyl Radicals in Radical Polymerization of Si-H Containing Monomers	Yuki ISHIKAWA, Nagoya University, Japan	POS1-0375

PS1_018	Organobse-Catalyzed Metal-Free Proton Transfer Anionic Polymerization Using C-H Bonds as Dormant Species	Mei OKANO, Nagoya University, Japan	POS1-0376
PS1_019	Synthesis of Optically Active β -Peptides by Stereospecific Step-Growth Anionic Polymerization of Various Acrylamide Derivatives	Satoki YAMAMOTO, Nagoya University, Japan	POS1-0379
PS1_020	Design and Synthesis of Rigid Bio-Derived Isosorbide-Based Hyperbranched Polymers	Hyunhee LEE, Yonsei University, Korea	POS1-0380
PS1_021	Development of Photocontrolled Nitroxide-Mediated Polymerization by Photothermal Conversion	Seima KONDO, Nagoya University, Japan	POS1-0381
PS1_022	Development of Controlled Ring-Opening Polymerization of Lactams Using Lewis Acid	Yinan GAO, Nagoya University, Japan	POS1-0382
PS1_023	Effect of Intramolecular Interactions on the Morphologies of Bifunctional Polymers	Yeonji LEE, POSTECH, Korea	POS1-0385
PS1_024	Precision Synthesis and Marine Biodegradability Evaluation of Monodisperse Vinyl Oligomers Using Living Polymerization and Chromatographic Separation	Okamoto SHOKEI, Nagoya University, Japan	POS1-0390
PS1_025	Synthesis of Sequence-Controlled Degradable Polymers Based on Specific Radical Copolymerization of Thioamide Derivatives	Rikuta NOGUCHI, Nagoya University, Japan	POS1-0391
PS1_026	Styrenic 1,2,4-Triazolium-based Polymers: From Molecular Design to Functional Materials	Marius HINGEL, Stockholm University, Sweden	POS1-0393
PS1_027	Sequence Distribution Design for Tunable Miscibility and Self-Assembly in Amphiphilic Random Copolymer Mixtures	Jimin YOO, Korea Advanced Institute of Science and Technology (KAIST), Korea	POS1-0414
PS1_028	Anionic Ring-Opening Polymerization of D-Glucal-Derived Bicyclic Lactone Toward Sustainable Degradable Polyesters	Hyun NAMKOONG, Yonsei University, Korea	POS1-0428
PS1_029	Anionic Homo- and Copolymerization of Thioamide Derivatives	Momo ISHIDA, Nagoya University, Japan	POS1-0444
PS1_030	Sequence Distribution Control in Polyacrylate Upcycling via (Thio)urea-Catalyzed Transesterification	Eunseo CHO, Korea Advanced Institute of Science and Technology (KAIST), Korea	POS1-0456
PS1_031	Chain-Length-Accelerated Polyacrylate Deoxygenation for Functional Polypropylene	Youngho KIM, KAIST, Korea	POS1-0479
PS1_032	Separation and Quantification of Living Chains in Photoiniferter RAFT Block Copolymers via Solvent-Gradient HPLC	Sungmin CHOI, Pusan National University, Korea	POS1-0487
PS1_033	Synthesis of Light-Responsive Miktoarm Copolymers via Aqueous Polymerization-Induced Self-Assembly	Yujin JANG, Gwangju Institute of Science and Technology, Korea	POS1-0499
PS1_034	Enhancing the Thermal Stability of Directed Self-Assembly Through Dehalogenation of Bottlebrush Block Copolymers	Jaewon JEONG, Korea University, Korea	POS1-0512
PS1_035	Real-Time Observation of Photo-Induced PISA in Architecturally Inverted Miktoarm Copolymers	Chaeyeon PARK, Gwangju Institute of Science and Technology, Korea	POS1-0514
PS1_036	Sulfur-Containing Cardo Structures for High Refractive Index, Low Birefringence Polymers	Minju RYU, Kyung Hee University, Korea	POS1-0547
PS1_037	Crosslinker-Free Vitrimer Networks for Tunable Mechanics and Bond-Exchange-Assisted Blending	Youseop LEE, Korea University, Korea	POS1-0575

PS1_038	Molecular Twisting of Poly(butylene terephthalate) Through Spirocyclic Comonomer Design	Jin Young LEE, Chungnam National University, Korea	POS1-0692
PS1_039	Scalable Synthesis of [4.4.4]Propellane-based Polyphenylenes with Suppressed Aggregation	Yesl KIM, Ewha Womans University, Korea	POS1-0712
PS1_040	Synthesis of High Density Nanospheres via Early-Stage Curvature Stabilization from Polymerization-Induced Microphase Separation of Bottlebrush Block Copolymer	Seungjin HA, KAIST, Korea	POS1-0874
PS1_041	Ball-Milling-Assisted Silicone Ring-Opening Polymerization for Structure-Controlled Synthesis and Mechanistic Understanding	NaGyeong LEE, Kumoh National Institute of Technology, Korea	POS1-0899
PS1_042	Development of Sequence-Controlled Densely Hydroxylated Polymers via Ring-Opening Metathesis Polymerization of Cyclic Olefins and Post-Functionalization	Dabon LEE, KAIST, Korea	POS1-0932
PS1_043	Topology-Enabled Star Polyelectrolyte Coacervates for Adaptive and Robust Enzyme Immobilization	Jie ZHU, Fudan University, China	POS1-1081
PS1_044	Library Synthesis of Methacrylate-Styrene Alternating Copolymers Using a Highly Side-Chain Transformable Monomer and Their Sequence-Specific Properties	Haruto NITANDA, Kyoto University, Japan	POS1-1185
PS1_045	Precision Synthesis of Graft Alternating Copolymers: Sequence Impact on Self-Assembly Behaviors	Aoto ISHIKAWA, Kyoto University, Japan	POS1-1231
PS1_046	Sustainable Upcycling of Waste Tires via Sunlight-Driven Carbonization for High-Performance Fuel Cells	Hyeonseong KIM, Chung Ang University, Korea	POS1-1311
PS1_047	Chemical Recycling and Upcycling of Functionalized Cyclic Olefin Copolymers	Jimin CHOI, Dongguk University, Korea	POS1-1313
PS1_048	Topology-Controlled Polyglycerol-g-Polylactide Copolymers: Synthesis and Stereocomplex Formation	Junui WI, Dongguk University, Korea	POS1-1314
PS1_049	Synthesis and Characterization of Novel High Thermal Polymer Resin	Minseo PARK, Dongguk University, Korea	POS1-1400
PS1_050	Topology-Dependent Self-Assembly of BTA-Based Supramolecular Templates for Au Nanostructure	ChaeRim KIM, Dongguk University, Korea	POS1-1401
PS1_051	Cross-Linked Polyphosphazene Binders for Silicon Anodes	Myeongji LEE, Seoul National University, Korea	POS1-1436
PS1_052	Compositional Design of an PEO-PPO-PEO Copolymer-Based Support Matrix for Embedded 3D Printing of Liquid Metal	Nayeon KIM, KyungHee University, Korea	POS1-1582
PS1_053	Hexadecimal Digital Information Storage via Sequence-Defined Polymers Synthesized by Passerini Iterative Exponential Growth (P-IEG)	Jeongyun CHOI, Seoul National University, Korea	POS1-1588
PS1_054	Fluorine-Free Polymer Coatings Based on Plant-Derived Cardanol for Durable Antifouling and Antibacterial Adhesion Resistance	Sungyeon CHO, Donga University, Korea	POS1-1595
PS1_055	Kinetic Interception of Crosslinker-First Ring-Opening Metathesis Polymerization Enables Architecture-Tunable Star Polymers	Hanjin CHOI, Kyunghee University, Korea	POS1-1668
S2. High-End Characterization/Polymer Physics/Properties (PSK50 Student Presentation Award Applicants)			
PS1_056	The Role of Water in Controlling Molecular Recognition of Glycopolymers	Yoko KITSUKI, Kyushu University, Japan	POS2-0137

PS1_057	Ultra-Robust Hydrogels Enabled by Transient Entanglements	Zhaoyang YUAN, Sichuan University, China	POS2-0210
PS1_058	Local Orientation of Polyetherimide at Carbon Interfaces Probed by Sum-Frequency Generation Spectroscopy	Kazuma AOKI, Kyushu University, Japan	POS2-0229
PS1_059	Quantitative Defect Engineering Governs Crystallization Pathways in Polymer Crystals	Gyusung HWANG, Inha University, Korea	POS2-0247
PS1_060	Strippable Catechol-Terminated Polyurethane Coating for Large-Area Radioactive Decontamination	Donghyun KIM, Ulsan National Institute of Science and Technology, Korea	POS2-0272
PS1_061	Nanomechanical Characterization of Fibrillar Structures in Liquid Crystalline Polymers Under Uniaxial Stretching by in Situ AFM Nanomechanics	Yuto MIZUNO, Institute of Science Tokyo, Japan	POS2-0308
PS1_062	Synthesis and Self-Assembly of Glucose-Derived High- χ Diblock Copolymers with Tuneable Nanoscale Morphologies	Jiwon KIM, KAIST, Korea	POS2-0315
PS1_063	Desert Beetle-Inspired Fog-Harvesting Surfaces Integrating Buckled Microchannels and Alternating Wettability	Ting Yu XU, National Chung Hsing University, Taiwan	POS2-0328
PS1_064	Tacticity Driven Intrachain Hydrogen Bond Selection Encodes Single Chain Conformational Metastability in Water	Donggeun CHOI, Sogang University, Korea	POS2-0335
PS1_065	Effect of Terminal-Group Compatibility on Chain Stretching and Self-Assembled Morphologies in Block Copolymers	Namjun KIM, Pohang University of Science and Technology, Korea	POS2-0377
PS1_066	Controlling Initial Interfacial Adsorption to Direct Interfacial Chain Exchange Dynamics in Polymer Nanocomposites	Eunjin CHOI, Seoul National University, Korea	POS2-0384
PS1_067	Nonclassical Domain Morphologies in Comb-Linear Block Copolymers and Their Crystallization Behavior	Dongyeol KUM, KAIST, Korea	POS2-0400
PS1_068	Architectural Effect on the Microphase Separation Behavior of Bottlebrush Block Copolymers	Sungjun LEE, Korea University, Korea	POS2-0430
PS1_069	Surface Free Energy Change of Poly(dimethylsiloxane) Under Elongational Deformation	Lynn BEKKI, Kyushu University, Japan	POS2-0464
PS1_070	Photoswitchable Coordination Competition for In-Situ Tunable Damping in Stimuli-Responsive Supramolecular Networks	Sieun JE, Pusan National University, Korea	POS2-0502
PS1_071	Surface-Engineered Nanoneedles via iCVD for Electroporation-Assisted Intracellular Delivery of Gene-Editing Tools	Sowon LEE, Korea Institute of Science and Technology, Korea	POS2-0558
PS1_072	Deformation Behavior of Vitrimers Under Strain and Temperatures	Ji Eun HYUN, Seoul National University, Korea	POS2-0593
PS1_073	Understanding Stress Relaxation in Covalent Adaptable Networks Through Decoupled Kinetic and Diffusion Timescales	Yeomyung YOON, Pusan National University, Korea	POS2-0628
PS1_074	Additive-Induced Segmental Dynamics in PR-like Polymer Thin Films Probed by Broadband Dielectric Spectroscopy	Seongeun KIM, Seoul National University, Korea	POS2-0639
PS1_075	Time-Aligned Operando GIWAXS on Organic Mixed Ionic Electronic Conductors for Probing Transient Structural Evolution	Taewoong YOON, Sungkyunkwan University, Korea	POS2-0645
PS1_076	Tuning the LCST Behavior of Poly(2-isopropyl-2-oxazoline) Through Molecular Weight Control	Seoeun LEE, Dankook University, Korea	POS2-0786

PS1_077	Multi-Angle DLS-Microrheology of Poly(acrylic acid): Separating Chain-Charge and Added-Salt Contributions to Viscoelasticity	Hokyeong JEONG, Hanyang University, Korea	POS2-0790
PS1_078	Bridging the Strain-Rate Gap in Polyethylene Creep: All-Atom Molecular Dynamics Simulation with Burgers and Ree-Eyring Models	Seoyeon KO, Sogang University, Korea	POS2-0792
PS1_079	Balancing Toughness and Self-Healing in Disulfide-Containing PI/PU Blends Through Dynamic Network Engineering	Somi YOON, Hanyang University, Korea Institute of Science and Technology, Korea	POS2-0801
PS1_080	The Complex Coacervation Phase Behavior of Polyampholytes	Gyumin YIM, Sogang University, Korea	POS2-0862
PS1_081	A Reaction-Regulated Ionic State Governed by Decarboxylation-Driven Ionic Remodeling	Ji Hong KIM, Hanyang University, Korea	POS2-0947
PS1_082	Machine Learning Clustering of Chain Conformations Resolves Glass Transition Temperature (T _g) Shifts in Polymer Thin Films	Gabriella Pasya IRIANTI, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Korea	POS2-0949
PS1_083	Stoichiometric Control of Shear Alignment and Photochemical Locking for Stable Actuation in Aza-Michael Liquid Crystal Elastomers	Sungwoo YANG, Seoul National University, Korea	POS2-0991
PS1_084	pH-dependent Conformational Properties of poly(alkylacrylic acid)s in good/poor Solvents	Seungbin HONG, Seoul National University, Korea	POS2-0994
PS1_085	Stress-Dissipative, Self-Healing Polyurethane Network Crosslinked by Tri-Disulfide Junctions for Transparent Optical Coatings	Seongho HONG, UNIST, Korea	POS2-1039
PS1_086	Multi-Responsive Behaviors and Configurational Mesophases of Amorphous-Liquid Crystalline Block Copolymers Containing Azobenzene Units	Suhyeon BAEK, Yonsei University, Korea	POS2-1131
PS1_087	Epoxy Vitrimers with High Strength and Rapid Stress Relaxation by Controlling Effectively Permanent Network Crosslink Density Toward Semiconductor Packaging	Hyun Woo PARK, Dankook University, Korea	POS2-1164
PS1_088	Synthesis and Characterization of Spiro-Structured Polyimides for Reduced Dielectric Constant	Juhyeon LEE, Yeungnam University, Korea	POS2-1258
PS1_089	CO ₂ -Curable Silicone Elastomers with Tunable Mechanical Properties Based on Amine-Functionalized PDMS	Kazuma OKADA, Gifu University, Japan	POS2-1310
PS1_090	One-Pot Synthesis of Ternary PMMA/PBA/Block Copolymer Blend for Transparent Toughened PMMA	JongWook AHN, Pusan National University, Korea	POS2-1368
PS1_091	Reconfigurable Network Using TBEMA-IEM Crosslinker in 1K Waterborne Polyurethane Coating	Youngjae CHEON, Pusan National University, Korea	POS2-1422
PS1_092	Development of a Smart Bilayer Polymer Film with Reversible Pore Modulation for Thermal Management	Yikyung SHIM, Dankook University, Korea	POS2-1451
PS1_093	Standardisation of Gate-Dependent Mobility Measurement Methods in Organic Electrochemical Transistors	Youhyun NAM, Seoul National University, Korea	POS2-1504
PS1_094	Dynamic Heterogeneity in the Trommsdorff Effect of Bulk Polymerization Probed by Diffracted X-ray Blinking	Eisuke SASAKI, The University of Tokyo, Japan	POS2-1522
PS1_095	Development of Mealworm-Enriched Rice Starch Soft Foods for Dysphagia Care	Yeryeong LEE, Konkuk University, Korea	POS2-1525

S3. Processing / Fabrications (Emerging Horizons in Polymer Processing and Fabrication) (PSK50 Student Presentation Award Applicants)

PS1_096	Biodegradable Porous Scaffold Platform via Green Solvent Processing of P3HP/PLA Blends	Seungho BAIK, Kyung Hee University, Korea	POS3-0087
PS1_097	Polyaniline as a Conducting Physical Intercalant for Multilayered Nanohybrid Electrodes in Flexible All-Solid-State Supercapacitors	Hyeonseung YOON, Chonnam National University, Korea	POS3-0410
PS1_098	Hierarchical Porosity Engineering in Bio-Derived Carbon Electrodes via SWNT-Modified Precursors	Gihoon KIM, Chonnam National University, Korea	POS3-0416
PS1_099	Textural Engineering of Carbon Nanofibers Derived from SWNT/Polymer Precursors for High-Performance Supercapacitor Applications	Soyeon KIM, Chonnam National University, Korea	POS3-0417
PS1_100	Overcoming the High- χ Mixing Dilemma: High-Entropy-Inspired Quaternary Systems for Ordered Block Copolymer Monoliths	WonJune YEO, KAIST, Korea	POS3-0450
PS1_101	Architected Biopolymer Nanosheets as Building Blocks for Ultralight Aerogels	Byeunggon KIM, North Carolina State University, United States	POS3-0473
PS1_102	High-Performance Silicone Glue Etchant Enabled by Linear Alkane Swelling Solvents for Wafer Support System Debonding	Jaehyoung PARK, Inha University, Korea	POS3-0570
PS1_103	Reliable Low-Temperature Cu/SiO ₂ Hybrid Bonding via Selective Modification on Metal and Dielectric Surfaces	Jae-Yong LEE, Inha University, Korea	POS3-0585
PS1_104	Joule Heating-Triggered Shape Memory Polymer Composites via CNT Fiber Integration and Multi-Resin Patterning for Self-Deployable Structures	Jung Uk YUN, Pusan National University, Korea	POS3-0621
PS1_105	UV-Blocking Pressure-Sensitive Adhesives for Optoelectronic and Display Applications	Dahyun JIN, Ulsan National Institute of Science and Technology, Korea	POS3-0682
PS1_106	Shape-Programmable and Recyclable Epoxy Covalent Adaptable Network Substrates for Free-Form Thin-Film Electronics	Gyeong Hwan CHOI, Pusan National University, Korea	POS3-0731
PS1_107	Molecular Sensing on Curved Surfaces Enabled by Shape-Memory Nanogaps	Goum MIN, Seoul National University, Korea	POS3-0740
PS1_108	Design of a Photothermal Conversion Optical System and Evaluation of Photothermal Polymerization Behavior	Tomoki WATANABE, Laboratory for Chemistry and Life Science, Institute of Integrated Research, Institute of Science Tokyo,	POS3-0750
PS1_109	Simultaneous Resin Impregnation and CNT Network Formation via VANCI Process for Nanoconfined Epoxy-CFRP with Superior Thermal Stability	Yuha JANG, Pusan National University, Korea	POS3-0780
PS1_110	Pulsed DC-Assisted Resin Impregnation of CNT Films Using Ionic Liquid	Sumin HEO, Pusan National University, Korea	POS3-0810
PS1_111	Synthesis of Hydrochar Supported Fe Aminoclay Composites via Hydrothermal Carbonization for Efficient Anionic Dye Adsorption	Minseong JO, Chonnam National University, Korea	POS3-0816
PS1_112	Investigating Structure and Properties of PC/PMMA Blends Through Transesterification During Reactive Melt Blending	Sihyeong LEE, Korea Research Institute of Chemical Technology (KRICT), Yonsei University, Korea	POS3-1230
PS1_113	Rapid, High-Resolution Visible-Light DLP 3D Printing Using 4DP-IPN Photoredox Chemistry and a Liquid-Cooled Vat	Juhyeon SONG, Ulsan National Institute of Science and Technology, Korea	POS3-1277
PS1_114	Density-Controlled Radical Chemisorption for Tailoring Carbon Nanotube-Solvent Interfaces	Eunho RYU, Pohang University of Science and Technology, Korea	POS3-1362
PS1_115	Relationship Between Rheological Properties and Foamability in PLA/LDPE Blends Containing Different Fillers	Minchan KIM, Pusan National University, Korea	POS3-1410

S4. Colloids, Interfaces, and Molecular Assemblies for Functional Soft Materials (PSK50 Student Presentation Award Applicants)			
PS1_116	Azobenzene-Encoded Polymerizable Deep Eutectic Solvent Networks for Counteranion-Programmed Intelligent and Architected Ionotronic Elastomers	Chun Chi CHANG, National Yang Ming Chiao Tung University, Taiwan	POS4-0070
PS1_117	CCNC-Modulated Bacterial Cellulose Networks for High-Performance UV-Shielding Sunscreen Platforms	Nayoung KIM, Kyung Hee University, Korea	POS4-0088
PS1_118	Large-Area Single-Grain Hexagonally Perforated Lamella of Block Copolymers via Composition-Programmed Directed Self-Assembly	Sang Hyun CHOI, Seoul National University, Korea	POS4-0119
PS1_119	From Self-Assembled Block Copolymers to Nitrogen-Doped Mesoporous Carbons with Tunable Nanostructures	Youngwon KONG, Institute of Science Tokyo, Japan	POS4-0131
PS1_120	DMSO-Mediated Assembly of Lipid-PLGA Hybrid Nanoparticles	Sang Ryeol JEONG, POSTECH, Korea	POS4-0166
PS1_121	Multilevel Optical PUF in Color-Programmable Microparticle Monolayers	Jinmin LEE, POSTECH, Korea	POS4-0174
PS1_122	Synthesis of Glucose-Responsive Polymersomes Using PDMS Microfluidic Device for Biochemical Compartmentalization	Minji CHAE, POSTECH, Korea	POS4-0178
PS1_123	Dual-Responsive Hydrogel Interferometric Patches for Multi-Factor Optical Encryption	Hyein JIN, POSTECH, Korea	POS4-0179
PS1_124	Effect of End-Group Chemistry on Interfacial Tension in Short-Chain Complex Coacervates	Yeonseo KIM, UNIST, Korea	POS4-0206
PS1_125	Azo-Functionalized Polymersomes for Light-Triggered Membrane Permeabilization	Hyun Su LEE, POSTECH, Korea	POS4-0208
PS1_126	Blending-Assisted Formation of PLA-b-PS Polymer Cubosomes Through Liquid-Liquid Phase Separation Interface Stabilization	Jinyeong OH, KAIST, Korea	POS4-0253
PS1_127	Molecular-Level Investigation of Multi-Component Peptide Assembly	Nayeong JEON, Max Planck Institute for Polymer Research, Germany	POS4-0260
PS1_128	Regulating Azobenzene Photodynamics Through Cyclodextrin Cavity Size Under Confinement	Geonwoo LEE, Ulsan National Institute of Science and Technology, Korea	POS4-0271
PS1_129	Wood-Mimetic Hollow Microbeads Using Cellulose Nanofiber-Stabilized Pickering Emulsion Polymerization	Gyuhwan KIM, Kyushu University, Japan	POS4-0275
PS1_130	Molecular Design of Side-Chain Liquid Crystalline Azopolymers with Tunable Dielectric Polarization for Highly Sensitive Thermoresponsive Triboelectric Sensors	Chaeyeon WOO, Hanyang University, Korea	POS4-0317
PS1_131	Tuning of the Flory-Huggins Interaction Parameter via Monomer Functionalization for Tailored Self-Assembly	Chanhyeok MOON, Korea Advanced Institute of Science and Technology, Korea	POS4-0318
PS1_132	Directional Liquid Transport on Position-Selective Wettability-Patterned Surfaces	Seoyoung KIM, Dankook University, Korea	POS4-0322
PS1_133	Donor-Acceptor Conjugated Polymer Nanoparticle@CuO Core-Shell Nanohybrids for Solar-Driven CO ₂ Conversion	SunJu HONG, Chung-Ang University, Korea	POS4-0323
PS1_134	Sequential Interfacial Transfer Printing of Three-Dimensional Quantum Dot Structures	Jeonghun SHIN, Pusan National University, Korea	POS4-0332

PS1_135	Solution-Processed PMMA Mold Patterned Au Nanoparticles Enabling Localized Surface Plasmon Resonance in Inverted CQD Infrared Photodetectors	Minjeong KIM, Pusan National University, Korea	POS4-0333
PS1_136	Particle Collision-Mediated Assembly of Polymer-Inorganic Hybrid Supraparticles	Geonho LEE, Ulsan National Institute of Science and Technology (UNIST), Korea	POS4-0334
PS1_137	Micellar Morphological Evolution-Driven Hierarchical Network Composites for Long-Lived and Multifunctional Room-Temperature Phosphorescence	Seongeun CHO, Yonsei University, Korea	POS4-0398
PS1_138	Surface-Functionalized MOF Nanoagents for Ice-Water Interface Control in Low-Temperature Preservation	Leeseo KIM, Gwangju Institute of Science and Technology, Korea	POS4-0422
PS1_139	Ice-Plane-Selective Binding of AFP-Mimetic Peptides Across Atomic-to-Macroscopic Length Scales	Minyoung SON, GIST, Korea	POS4-0429
PS1_140	Click Chemistry-Mediated LNP Surface Modification for Targeted Extrahepatic Delivery	Seeun KIM, POSTECH, Korea	POS4-0431
PS1_141	Tunable First Network Design of Double Network Hydrogels Using Polyether-Based ABA Triblock Copolymer	Byungwoo YOO, Yonsei University, Korea	POS4-0432
PS1_142	Complexation-Mediated Diffusion-Limited Crystal Growth: A General Framework for Anisotropic Crystal Growth in Cu-Based Perovskites	Hyunmin LEE, Hanyang University, Korea	POS4-0443
PS1_143	Nanoparticle-Assembled Nanobridges for Preferential Surface Modification of Hydrogels	Hyeseoung LIM, Gachon University, Korea	POS4-0451
PS1_144	Pine Cone-Inspired Water-Responsive Soft Actuators via Interfacial Phase Separation and Bilayer Design	Yoonsung NOH, Yonsei University, Korea	POS4-0460
PS1_145	Raft-like Assemblies of Bottlebrush Polymers at Liquid-Liquid Interfaces	Dupyo JEON, UMass Amherst, United States	POS4-0467
PS1_146	3D-Printed Step-Emulsification Platform for Scalable Production of Thermally Solidified Droplets	Taesoo JANG, POSTECH, Korea	POS4-0484
PS1_147	Orientation-Encoded Structural Color for Programmable Mechanochromics	Jin Hyeok CHO, UNIST, Korea	POS4-0492
PS1_148	Selective PVA Adsorption for Tuning the Stacked Structure and Rheological Properties of MXene/GO Dispersions	Eunho CHO, Seoul National University, Korea	POS4-0498
PS1_149	Soft Actuators Based on Conductive Poly(N-isopropylacrylamide) Cryogels	Heesu PARK, Chung-Ang University, Korea	POS4-0521
PS1_150	Real-Time Observation of Order-Disorder Transitions in Block Copolymer Nanostructures via in Situ TEM	Gayeon MOON, Gwangju Institute of Science and Technology, Korea	POS4-0554
PS1_151	Highly Processable and Adhesive PFAS-Free Bottlebrush Copolymer Binder for Dry Electrode Fabrication	Soonho JANG, Korea University, Korea	POS4-0562
PS1_152	Multimodal Mechanochromic Sensing of Stress and Cracks via Polydiacetylene-Upconversion Nanoparticle Composites	Jongkyu KIM, UNIST, Korea	POS4-0588
PS1_153	Nanoconfined Polyurea Membrane Reactors with PEG-Engineered Pore Interfaces for Chemoenzymatic Cascade Catalysis	Elsayed MOSTAFA, GIST, Korea	POS4-0594
PS1_154	Triazole-to-Triazolium Conversion Switches Smectic-to-Columnar Ordering in PEO-Based Lithium-Ion Conducting Liquid Crystals	Jeongwon HWANG, Dankook University, Korea	POS4-0623

PS1_155	Water Dynamics Governed by Interfacial Aggregation States in Sequence-Controlled Fluorinated Copolymers	Kazuma KUMAMOTO, Kyushu University, Japan	POS4-0625
PS1_156	Angle-Dependent Photonic Microdome Arrays for Multimodal Color Graphics	Jun-Gu KANG, KAIST, Korea	POS4-0663
PS1_157	Smart Adhesive Incorporating Visible-Light-Driven Photoswitches for Micro-Spatially Controllable and Efficient On-Demand Debonding	Minkyong GWON, Pusan National University, Korea	POS4-0670
PS1_158	Alcohol-Triggered Interfacial Assembly of BNNTs for Simultaneous Film Formation, Alignment, and Purification	Yujin KIM, Ulsan National Institute of Science and Technology, Korea	POS4-0672
PS1_159	Crystallization-Stabilized Thermoresponsive Gelation of a PiPOx-containing Hydrogel	Dahye KIM, Dankook University, Korea	POS4-0685
PS1_160	Oligopeptide-Functionalized MOF Nanogents for Ice-Water Interface	Leeseo KIM, Gwangju Institute of Science and Technology, Korea	POS4-0688
PS1_161	Neck Angle-Dependent Formation of Nucleic Acid-Loaded Lipid Nanoparticles in Toroidal Ring Micromixers	Jiwoo CHOI, Dankook University, Korea	POS4-0690
PS1_162	Non-Spherical Reflow-Formed Microstructures for Retroreflective Structural Coloration	Jiwoo LEE, KAIST, Korea	POS4-0693
PS1_163	Captive Bubble Tensiometer for Evaluating Time-Dependent Interfacial Behavior of Polymeric Pulmonary Surfactants	Yeajin LEE, Dankook University, Korea	POS4-0701
PS1_164	Porous Photonic Papers Composed of Non-Close-Packed Colloidal Crystals Created by Polymerization-Induced Phase Separation	Hyeonbin JO, KAIST, Korea	POS4-0733
PS1_165	MOF-Mediated Ligand Coverage Regulation for Defect-Suppression of CsPbX ₃ (X= Cl, Br, I) Perovskite Quantum Dots	Jun Ho CHOI, Chung-Ang University, Korea	POS4-0743
PS1_166	Phase-Separation-Engineered Conductive Photoresist for Ultra-High-Resolution Micro-LED Transfer	Sehoon JUNG, Pusan National University, Korea	POS4-0745
PS1_167	Emulsion Derived MOF Coated Fire Retardant Microcapsules for Enhanced Thermal Safety in Polymer Based Solid State System	Nann Aye MYA MYA PHU, Chonnam National University, Korea	POS4-0748
PS1_168	Interfacial Polydiacetylene at Nanoemulsion Droplet: A PEG Hydrogel Platform from Thermochromic to Hydrogen-Responsive Colorimetric Films	Dowon KIM, Ulsan National Institute of Science and Technology (UNIST), Korea	POS4-0766
PS1_169	In Vivo Patterning of Hydrogels Using Low Intensity Focused Ultrasound	Jeongin EOM, Seoul National University, Korea	POS4-0772
PS1_170	Solvent-Selectivity-Driven core/corona Inversion in Stereoregular PLLA-b-PS Block Copolymer Micelles	Daung JEON, Korea Advanced Institute of Science and Technology, Korea	POS4-0788
PS1_171	Correlation Between the Photodimer Content Within Stimuli-Responsive Molecular Layers at Adhesion Interfaces and Peel Behavior	Kento HIGA, Laboratory for Chemistry and Life Science, Institute of Integrated Research, Institute of Science Tokyo,	POS4-0812
PS1_172	Crown Ether-Induced Room-Temperature Formation and Phase Evolution of Copper-Based Perovskite Nanocrystals via Dissolution-Recrystallization	Minji KIM, Hanyang University, Korea	POS4-0844
PS1_173	A Grafting-Through Strategy Towards Polymer Nanodiscs	Elizabeth LAI, The University of Sydney, Australia	POS4-0858
PS1_174	Structural Optimization of Cellulose Nanocrystal-Stabilized Rosin Amine Pickering Emulsions	Jeongki KIM, Kyungpook National University, Korea	POS4-0859

PS1_175	Effects of Hydrophobe Addition Strategy and Seed Composition on Surfactant-Free Nanogel Formation for PFAS-Free Aqueous Coatings	Jihun JEONG, Kyungpook National University, Korea	POS4-0863
PS1_176	Orthogonal Mechanical and Thermal Fluorescence Amplification in a Single Polydiacetylene Platform	Songseok LEE, Ulsan National Institute of Science and Technology, Korea	POS4-0909
PS1_177	Molecular Origin of Anisotropic Growth of 2D Conducting Polymer Nanosheets	Sangwon BAE, Seoul National University, Korea	POS4-0922
PS1_178	HPC/Ploxamer Binary Liquid Crystals for Thermally Tunable Structural Colors Across a Wide Temperature Window	Sanghyeok LEE, UNIST, Korea	POS4-0935
PS1_179	Polymerization-Induced Formation of Ordered Complex Coacervate Hydrogels	Kangseok KIM, KAIST, Korea	POS4-0961
PS1_180	Specifying the Origin of Chiral Sensitivity Through Conformal Nanogap Engineering in a Single Helicoid Gold Nanoparticle	Donghyun KIM, Sungkyunkwan University, Korea	POS4-1075
PS1_181	Annealing-Controlled Nanogap Distribution Modulates Plasmonic Field in 2D-Pack Chiral Nanoparticles for Ultrafast PCR	Minjae KIM, Sungkyunkwan University, Korea	POS4-1078
PS1_182	Enhanced Network Morphology Development Through Isothermal Annealing in PS-b-PI Block Copolymers	Mingeun PARK, Yonsei University, Korea	POS4-1120
PS1_183	Multifunctional Sunscreens Incorporating Polydopamine Nanoparticles for UV Protection	Se Rin KOO, Chungnam National University, Korea	POS4-1188
PS1_184	Writing and Reading Chirality at the Nanoscale: Chiral Nanopaint and Chiroptical Entropy Harvesting	Wookjin JUNG, Korea Advanced Institute of Science and Technology, Korea	POS4-1208
PS1_185	DNA-Gold Nanorod Composite Films as Anisotropic Light-Scattering Layers	Jiwon SHIN, KAIST, Korea	POS4-1214
PS1_186	Morphology-Controlled Dimensional Tuning for Blue-Emissive CsPbBr ₃ Quantum Dots	Huieun KIM, Sungkyunkwan University, Korea	POS4-1230
PS1_187	Solvent-Controlled Microstructure Engineering of Biodegradable PCLDA Microneedles for Sustainable Protein Delivery	EunBi KIM, Chonnam National University, Korea	POS4-1233
PS1_188	Dual-Functionalization Strategies for Enhancing Proton Conductivity in Sulfonated Covalent Organic Frameworks	Yong Moon HONG, Korean Institute of Science and Technology, Korea	POS4-1235
PS1_189	Bioinspired Adhesive Nanofiber Electronics with Three-Dimensional Suction Cups for Efficient Transdermal Drug Delivery	Minwoo SONG, Sungkyunkwan University, Korea	POS4-1241
PS1_190	Super-Adaptive Metastructured Adhesive Bioelectronics with Conductively Soften Suction Cups for Highly Deformable Biosurfaces	Gyun Ro KANG, Sungkyunkwan University (SKKU), Korea	POS4-1243
PS1_191	Expanding the Design Space of Covalent Peptide Assemblies: From Amino Acid Modularity to Kinetic Control	Yun-Mi HUR, Kyungpook National University, Korea	POS4-1245
PS1_192	Peptide-Functionalized Photothermal Nanomaterials for Recovery of Cryopreserved Cells	Jinwoo SHIN, Korea University, Korea	POS4-1248
PS1_193	Conformable Ultrasound Patch with Suction-Cavitation Modality for Enhanced Needle-Free Transdermal Delivery	Chang Rok PARK, Sungkyunkwan University, Korea	POS4-1250
PS1_194	Tunable Pore Architectures in PEGDA Microparticles via Photopolymerization-Induced Phase Separation for Enhanced Biocatalytic Efficiency	Seung Hui AN, Chonnam National University, Korea	POS4-1252

PS1_195	Mechanically Stable Coacervate Coatings for Drug-Eluting Sutures	Seoyun CHOI, Seoul National University, Korea	POS4-1254
PS1_196	Fabrication of Liquid Crystal Elastomer/h-BN Composites with Enhanced Thermal Conduction via Alignment Control	Bukyung KIM, Pusan National University, Korea	POS4-1262
PS1_197	Evaluation of Biodegradable Polymer-Based Nanocarriers for Lipophilic Bioactive Compounds	Heewon PARK, Gachon University, Korea	POS4-1270
PS1_198	Metal-Free Synthesis and Self-Assembly of P4VP-b-PLA Block Copolymers via Sequential O-ATRP and Ring-Opening Polymerization	Kainat, Gachon University, Korea	POS4-1272
PS1_199	Spindle-Shaped Hierarchical Polymeric Microparticles as Chiral Templates for Full-Color CPL	Geunjung LEE, KAIST, Korea	POS4-1284
PS1_200	Precursor Aromaticity-Dependent Nitrogen Speciation and Photoluminescence in N-Doped Carbon Dots	Hajin SONG, Gachon University, Korea	POS4-1285
PS1_201	Microchiral Pinwheel Arrays Based on Achiral Molecules	Jeong Yeon HAN, Korea Advanced Institute of Science and Technology (KAIST), Korea	POS4-1288
PS1_202	Capillarity-Driven Particle Absorbing Air Filter with Dynamic Bond Adhesive	Hyeri JEON, Chung-Ang University, Korea	POS4-1298
PS1_203	Light Modulated Kinetic Control of Covalent Peptide Assembly Toward Hierarchical Hollow Nanostructures	Yun-Mi HUR, Kyungpook National University, Korea	POS4-1326
PS1_204	Preservation of Frozen HSC-3 Cells Using Peptide-Colloids	Euijin CHA, Korea University, Korea	POS4-1353
PS1_205	Automated Droplet Detection and Morphological Quantification in W/O/W Double Emulsions via Segment Anything Model 2	Beom Kwon JANG, Sungkyunkwan University, Korea	POS4-1357
PS1_206	A Self-Assembled Particle Platform for Low-Cost DNA Sequencing	Jaeyoung KIM, POSTECH, Korea	POS4-1375
PS1_207	Visible Photonic Bandgap Enabled by Symmetry Control in Giant Block Copolymer Gyroids	Daeyoung KIM, Yonsei University, Korea	POS4-1383
PS1_208	Stationary Liquids Encapsulated in Hydrogel Marbles for Efficient Gas Absorption	Seyeon LEE, Chung Ang University, Korea	POS4-1385
PS1_209	Dual-Catalysts-Embedded Spontaneously Propelling Asymmetric Micromotors Using Triple Emulsion Microfluidic Synthesis for Highly Efficient nano/microparticle Removal	Vinh HO HOANG GIA, Sungkyunkwan University, Korea	POS4-1433
PS1_210	Selective Growth of ZIF-8 on Block Copolymer-Templated Au Nanodot Arrays as a Hybrid Platform for Reproducible SERS Sensing	Seungmin KIM, Chungnam National University, Korea	POS4-1458
PS1_211	Thermoresponsive Alginate-Starch Composite Emulsion Gels for Mimicking the Thermal Softening Behavior of Animal Fat	Shinjaee LEE, Seoul National University, Korea	POS4-1462
PS1_212	Development of Sponge-like Cellulose Nanofiber Hydrogels via Freeze-Concentrated Structuring and Acid-Mediated Interfibrillar Assembly	Hyosan AHN, Seoul National University, Korea	POS4-1464
PS1_213	Silicon-Urea Network Nanocomposite Anodes with Enhanced Structural Connectivity via Thermal Rearrangement	Daehyeon JANG, GIST, Korea	POS4-1518
PS1_214	Temperature-Responsive Morphological Evolution and Viscoelastic Transition of Wax-Based Phase Change Microparticles	Yujin JEONG, Konkuk University, Korea	POS4-1526

PS1_215	Swelling-Controlled Polystyrene Colloids for Structurally Stabilized UVA Absorber Loading	Jimin KWAK, Ewha Womans University, Korea	POS4-1533
PS1_216	Counterion-Mediated Like-Charge Attraction of PMAA at Amorphous Silica Interfaces	Chaimae LAOUZI, University of Genoa, Italy	POS4-1539
PS1_217	Hierarchical Engineering of Polymorphic Supramolecular Chirality Through Co-Assembly of Peptides	Damhee YUN, Kyungpook National University, Korea	POS4-1556
PS1_218	Polymeric Photonic Crystal Platform with Shape-Memory and Mechanochromic Properties	Chi Yeung OH, Ajou University, Korea	POS4-1588
PS1_219	Controlling Structural Color in Colloidal Photonic Crystals Through Binary Blends of Scattering and Absorbing Building Blocks	Jun Ha PARK, Ajou University, Korea	POS4-1589
S5. Polymers for Electronics and Photonics (PSK50 Student Presentation Award Applicants)			
PS1_220	Anion-Gated Fluidic Memristors Enabled by Ion-Specific Coacervation in Cationic Polyelectrolyte Hydrogels	Jowon SHIN, Sogang University, Korea	POS5-0190
PS1_221	Polymeric Carbon Nitrides Produced from Different Gaseous Conditions and Their Photocatalytic Performance for Degrading Organic Pollutants	Yongwoo KIM, Inha University NanoMaterialChemistryLab, Korea	POS5-0241
PS1_222	Excited-State Modulation by Sparse Acceptor-Mediated Energy Transfer in Donor-Rich Linear Conjugated Polymer Nanoparticles for Photocatalytic Hydrogen Evolution	Sowon KIM, Kyungpook National University, Korea	POS5-0287
PS1_223	Engineering Active Layers for Synaptic Plasticity in Organic Electrochemical Transistors	Soohyun NOH, Korea Research Institute of Chemical Technology, Korea	POS5-0351
PS1_224	Effects of Self-Assembly and Molecular Arrangement in Organic Semiconductor Device Applications	Isaac LIN, National Taiwan University, Taiwan	POS5-0356
PS1_225	Molecular Design of Fluorine-Free Bio-Based Positive-Type Photosensitive Polyimides with Low Dielectric Properties	Yu LIU, National Taiwan University, Taiwan	POS5-0357
PS1_226	Stretchable PDMS-Based Dielectrics with Tailored Electromagnetic Properties for Flexible Antenna Technologies	Tommy Steffen CEJMER, Hamburg University of Technology, Germany	POS5-0386
PS1_227	Molecular Design of Diketopyrrolopyrrole-Based Donor-Acceptor Copolymers via Backbone and Polar Side Chain Engineering for Enhanced Thermoelectric Properties	Taewoong HAN, Sungkyunkwan University, Korea	POS5-0387
PS1_228	Multi-Valued Logic Circuits Using P-N Heterojunction Electrolyte-Gated Transistors	Jaehyun HAN, Inha University, Korea	POS5-0488
PS1_229	Ambipolar Organic Electrochemical Transistors and Logic Circuits via Vertical Architecture	Kyuyeon KIM, Inha University, Korea	POS5-0490
PS1_230	Effect of Formation Temperature on SEI and Performance of Elastomeric Lithium Metal Batteries	Hyeonseok SEONG, Korea Advanced Institute of Science and Technology, Korea	POS5-0516
PS1_231	Morphology and Crystallinity of Transparent Capacitive Ion Storage Layer for Electrochromic Displays	Seungwon LEE, Chung-Ang University, Korea	POS5-0546
PS1_232	Thermal Warpage Control of Epoxy Underfill Materials	Minsong JEON, Inha University, Korea	POS5-0557
PS1_233	Light-Programmed, Optically Multiplexed Memory Lifetimes in a Monolithic Ionogel	Yubin LEE, UNIST, Korea	POS5-0601

PS1_234	Thermally Crosslinked PEDOT:PSS Hole Transport Layers for Efficient and Stable Organic Solar Cells	Myungjin KIM, KyungPook National University, Korea	POS5-0603
PS1_235	Chalcogenophene Substitution Enables Doping-Tolerant Diketopyrrolopyrrole Mixed Ionic-Electronic Conductors for Low-Power Organic Electrochemical Transistors	Shinbee OH, Korea Advanced Institute of Science and Technology, Korea	POS5-0604
PS1_236	Helical Columnar Supramolecular Polymers Exhibiting Both Polarization Retention and Ferroelectric Switching at Room Temperature	Seongwon PARK, Dankook University, Korea	POS5-0614
PS1_237	Tannic Acid-Modified Silica Nanoparticles as Versatile Interfacial Agents for UV-Stable, High-Sensitivity NO ₂ Detection in P3HT-Based Organic Field-Effect Transistors	Divya PANIKAR, Chonnam National University, Korea	POS5-0616
PS1_238	Stretchable Organic Electrochemical Transistors Based on Non-Destructive Microlithography	Hayoung OH, Hanyang University, Korea	POS5-0617
PS1_239	Polymer-Specific Polarity Inversion in Heavily Doped Conjugated Polymers	Hoimin KIM, Organic Semiconductor Laboratory (OSL), SKKU Advanced Institute of Nano Technology, Sungkyunkwan University, Korea	POS5-0633
PS1_240	Ion-Retentive Semi-IPN Framework for Bio-Stable, Anisotropic Ion Transport	Jong Hwi KIM, Hanyang University, Korea	POS5-0641
PS1_241	Chain-End-Dependent Free Volume, Lithographic Roughness, and Etch Resistance of RAFT-Synthesized Poly(ECPMA-co-HOST) Photoresists	Yejin KIM, Seoul National University, Korea	POS5-0642
PS1_242	Photoactivated Doping Enables Structural Ordering and Efficient Charge Transport for Organic Thermoelectrics	Hyunji LEE, POSTECH, Korea	POS5-0683
PS1_243	Effect of Glass Fiber Membrane Support and PDA Interfacial Modification on NO ₂ Sensing in P3HT-Based OFETs	Gaeun JEONG, Chonnam National University, Korea	POS5-0687
PS1_244	Circularly Polarized Light-Sensing Artificial Synapses Enabled by a Single p-Type Polymer Channel in Organic Electrochemical Transistors	Yuseok SONG, Chung-Ang University, Korea	POS5-0746
PS1_245	Programming of Anisotropic Responses in Piezoelectric Composite Sensors Towards Perception-Embodied Soft Robots	Yubin KIM, Gwanju Institute of Science and Technology, Korea	POS5-0756
PS1_246	Synthesis and Properties of Low-Dielectric, Low-Loss Hydrocarbon Network	Jae Hyun KIM, Kyung Hee University, Korea	POS5-0761
PS1_247	A Relaxor Ferroelectric Thermoplastic Elastomer for High-Performance Electroadhesion on Diverse Surfaces	Dongjun LEE, Sogang University, Korea	POS5-0764
PS1_248	Conjugated Polymer Network Nanoparticles as Stable Photoactive Layers for Organic Photodetectors	Minjun CHOI, Daegu Gyeongbuk Institute of Science and Technology (DGIST), Korea	POS5-0767
PS1_249	Ultrathin Transformable Polymer Nanomembranes for Conformal Bioelectronic Interfaces	Daeyeon LEE, Ulsan National Institute of Science and Technology (UNIST), Korea	POS5-0773
PS1_250	Fast-Response 1310 Nm Photomultiplication Organic Photodetectors via Perovskite Quantum Dot Interlayer Engineering	Minyoung JEONG, POSTECH, Korea	POS5-0778
PS1_251	Geometry-Engineered Bidentate Phosphines for Effective Surface Passivation and High-Performance CsPbI ₃ Quantum-Dot Photodetectors	Byeongchan PARK, POSTECH, Korea	POS5-0782
PS1_252	Influence of Heteroatom Substitution on the Spin and Charge Transport Properties of Isatin-Based Open-Shell Quinoidal Polymers	Youngju JUNG, Gwangju Institute of Science and Technology, Korea	POS5-0789
PS1_253	Aggregation-Induced Supramolecular Chirality and NIR-II Absorption in Quinoidal Chiral π -Conjugated Molecules for CPL Photodetection	Jeongjin HONG, Gwangju Institute of Science and Technology, Korea	POS5-0798

PS1_254	Rational Design of High-Sensitivity Organic Photodetectors Using a Perylene Diimide Cathode Interlayer for Wireless Optical Communication	Min Soo KIM, Chung-Ang University, Korea	POS5-0802
PS1_255	A Versatile Molecular Buried Interlayer for Wide-Bandgap and Lead-Free Perovskite Solar Cells	Juan Anthony PRAYOGO, Pukyong National University, Korea	POS5-0804
PS1_256	Control of Chirality Induction in n-Type Conjugated Polymer by Printing Systems Using Chiral Solvents for CISS-Based Organic Spintronics	Jeongwoo BEAK, Gwangju Institute of Science and Technology, Korea	POS5-0813
PS1_257	Bridging Sustainability and High Performance in Organic Solar Cells Through Bio-Derived Solvents	Seungho LEE, KAIST, Korea	POS5-0828
PS1_258	Over 20.6% Efficiency Organic Photovoltaics Enabled by Co-Assembled Monolayer Interface Engineering	Changhun YU, KAIST, Korea	POS5-0832
PS1_259	Core-Linked Star-Shaped Trimer Design Enables Highly Stretchable and Efficient Organic Solar Cells	Kihyun BAE, KAIST, Korea	POS5-0838
PS1_260	Molecularly Controlled Thermoplastic Elastomers for Highly Efficient and Stretchable Organic Solar Cells	Jongmin OH, KAIST, Korea	POS5-0841
PS1_261	Polymerized Small-Molecule Acceptors with Electron-Withdrawing Substituents Enabling Efficient Charge Transport and Passivation for Inverted Perovskite Solar Cells	Seungjun LEE, KAIST, Korea	POS5-0856
PS1_262	Decoupling Ionic Accessibility and Electronic Energetic Disorder in Organic Mixed Ionic-Electronic Conductors	Jaehoon LEE, SKKU Advanced Institute of Nanotechnology (SAINT) and Sungkyunkwan University, Korea	POS5-0872
PS1_263	Chiral Plasmonic Supramolecular Architectures Based on Star-like Block Copolymers for Circularly Polarized Neuromorphic Devices	Seohyun JUNG, Ewha Womans University, Korea	POS5-0874
PS1_264	Low Dielectric Polymers with High Thermal Conductivity for High-Frequency Applications	Youngjae WI, Jeonbuk National University, Korea	POS5-0878
PS1_265	Photo-Switchable Organic Synaptic Transistors for Integrated Reservoir Computing	Yohyeon JANG, Seoul National University of Science and Technology, Korea	POS5-0895
PS1_266	Enhancing Synaptic Memory Through an Ion Reservoir Layer in Carbon Nanotube Synaptic Transistors	Hyeran OH, Seoul National University of Science and Technology, Korea	POS5-0896
PS1_267	Ether-Linkage-Mediated Ion Stabilization in Organic Artificial Synapses for Memory Regulation	Hyoik JANG, Seoul National University of Science and Technology, Korea	POS5-0900
PS1_268	Crystallinity-Controlled ZIF-8 Heterogeneous Nucleation for Suppressed Interfacial Recombination in Perovskite Optoelectronics	Yumin LEE, Chung-Ang University, Korea	POS5-0901
PS1_269	Mixed Ion-Electron Conductors for Dendrite-Free Aqueous Zinc Batteries	Minjae KIM, Ewha Womans University, Korea	POS5-0912
PS1_270	Systematic Cyano-Substitution Engineering of Dithienopyrrole-Based Non-Fullerene Acceptors for Near-Infrared Organic Photodetectors	Wonseok CHOI, KRICT, Korea	POS5-0915
PS1_271	Additive-Assisted Morphology Engineering for High-Performance Near-Infrared Organic Photodetectors	Jongwon CHOI, KRICT, Korea	POS5-0920
PS1_272	Highly Stretchable Bulk Heterojunction Thin Films for Stretchable Organic Photodetectors	Soyoon KIM, Ewha Womans University, Korea	POS5-0925
PS1_273	Backbone-Controlled Ion-Side Chain Accessibility in Conjugated Polymers for Organic Electrochemical Synaptic Transistors	Byeongjun JEON, Seoul National University of Science and Technology, Korea	POS5-0929

PS1_274	Aqueous Chiroptical Organic Electrochemical Transistors for Circularly Polarized Light Sensing	Chaeun HYUN, Chung-Ang University, Korea	POS5-0931
PS1_275	Realizing Ternary Logic in a Single Polymer Semiconductor via Zero-Differential Transconductance	Donghyun PARK, Inha University, Korea	POS5-0945
PS1_276	Dopant-Induced Charge-Trapping Synaptic Transistors via Nanoscale Patterning	Jae Eun KIM, Korea Maritime and Ocean University, Korea	POS5-0959
PS1_277	Self-Validating Conjugated Polymer Nanowire Gas Sensors for Reliable Environmental NO ₂ Monitoring	Bum Hwan KIM, Korea Maritime and Ocean University, Korea	POS5-0960
PS1_278	Master Mold Assisted Printing Enables Morphology Controlled Conjugated Polymer Transistors	Jin Seok YOON, Korea Maritime and Ocean University, Korea	POS5-0962
PS1_279	Leakage-Suppressed Polymer FeFETs via Nanowire Patterning for High-Precision Artificial Synapses	Useong JIN, Korea Maritime and Ocean University, Korea	POS5-0965
PS1_280	Fe ₃ O ₄ -Functionalized Ni Aerogel/Polymer Composites for Absorption-Enhanced Electromagnetic Interference Shielding Applications	Hyunju PARK, Gwangju Institute of Science and Technology, Korea	POS5-0978
PS1_281	Magnetoelastic Hemispherical Tactile Skin with GMR Spin Valve Readout for In-Plane Contact Direction Sensing	Minsun OH, Gwangju Institute of Science and Technology, Korea	POS5-0983
PS1_282	Comparative Analysis of Warpage Behavior of CTBN-Modified Epoxy-Based Underfill for Flip-Chip Packaging	Minsong JEON, Inha University, Korea	POS5-0990
PS1_283	Bromine-Functionalized Carbazole-Based Phosphonic Acids and Phosphonic Esters as Efficient Hole-Transporting Materials for Sn-Based Perovskite Solar Cells	Hsinyu CHEN, National Yang Ming Chiao Tung University, Taiwan	POS5-1004
PS1_284	Self-Healable Zwitterionic Ionogel via Ion Interaction and Dynamic Boronic Ester Networks	Yunhyung CHOI, UNIST, Korea	POS5-1041
PS1_285	Back-End-of-Line Compatible Wafer-Scale InSe Transistors	Jaeyun LEE, POSTECH, Korea	POS5-1067
PS1_286	Ligand Engineering for Precise Control of Ultrathin CsPbI ₃ Nanoplatelet Superlattices for Efficient Light-Emitting Diodes	Jongbeom KIM, Ulsan National Institute of Science and Technology, Korea	POS5-1071
PS1_287	Fused Selenophene and Side Chain Elongation Synergistically Enable Absorption Broadening and Morphology Control in Polymer Donors for Organic Solar Cells	Junmo KANG, Ulsan National Institute of Science and Technology (UNIST), Korea	POS5-1097
PS1_288	Large-Scale Indium Selenides Transistors for High-Performance Flexible Electronics	Heeseol JEON, Korea Research Institute of Chemical Technology, Korea	POS5-1100
PS1_289	Conductive Hydrogel Optical Fibers with Controlled Swelling Fabricated via Microfluidics	Arti SINGH, Pusan National University, Korea	POS5-1115
PS1_290	Anion-Trapping Ionic Hydrogel for Enhanced Thermodiffusion-Driven Energy Conversion	Taehoon KANG, University of Seoul, Korea	POS5-1117
PS1_291	Polymer Network Wrapping for Stable Pure-Blue PeLEDs	Hun CHO, Sungkyunkwan University, Korea	POS5-1129
PS1_292	Molecularly Designed PFAS-Free Flexible Transparent Films with High UV Stability and 1,000,000-Cycle Folding Reliability	Chaeun MOON, Korea Institute of Industrial Technology (KITECH), Korea	POS5-1138
PS1_293	Enhanced Stretchability of Azide-Functionalized Semiconducting CNT Thin Films via UV-Crosslinking	Doyoung KWON, Ewha Womans University, Korea	POS5-1144

PS1_294	Enhanced Thermal Stability of Self-Assembled Block Copolymer Distributed Bragg Reflectors Through Metal-Ion Doping	Shangxian DING, Yonsei University, Korea	POS5-1152
PS1_295	Decoupling Temperature Sensitivity and Humidity Interference in PEDOT:PSS Thermistors	Sunwoo CHO, Seoul National University, Korea	POS5-1159
PS1_296	Metallic Nanowire-Film Hierarchical Structures for Stretchable Bioelectronic Systems	Sunah CHEONG, POSTECH, Korea	POS5-1166
PS1_297	Effects of Gel-Gel Interface Structure on Ion Transport in Bilayer Ionic Hydrogels	Jihyun NAM, University of Seoul, Korea	POS5-1170
PS1_298	Simultaneous Enhancement of Hole Mobility and Volumetric Capacitance in PEDOT:PSS via Proton Conductor Induced Phase Separation for Biocompatible Low-Voltage OECTs	Jongmin LIM, Kookmin University, Korea	POS5-1174
PS1_299	N-Type Organic Electrochemical Transistors Gated by High-Capacitance Solid-State Ionogels	Rihyeon KIM, Inha University, Korea	POS5-1176
PS1_300	Preparation and Thin Film Properties of a Solution-Based TiO ₂ - Polyimide Nanocomposite for High-k Dielectric	Han Byul CHOI, Kyungsung Univ., Korea	POS5-1183
PS1_301	Counter Anion-Induced Conformational Modulation in Ionic Organic Nonlinear Optical Crystals	Haneul KIM, Ajou University, Korea	POS5-1190
PS1_302	Isomorphic Organic Nonlinear Optical Crystals by Space-Filling Strategy for Efficient Terahertz Wave Generation	Seungmin KIM, Ajou University, Korea	POS5-1192
PS1_303	Synthesis and Electrochromic Characterization of Polyviologens with Varied Substituent Positions and Electronic Characteristics	Kyung Yun KANG, Yeungnam University, Korea	POS5-1207
PS1_304	Robust Biodegradable Multilayer Artificial Synapse with Sub-Biological Energy and Extended Memory for Intelligent Reflexive Systems	Hyeji OH, Ulsan National institute of science and technology, Korea	POS5-1211
PS1_305	Cation Engineering of Nonlinear Optical Crystals by Introducing Fluorine Substituent for Terahertz Wave Generation	Jiwon YOO, Ajou University, Korea	POS5-1236
PS1_306	Dibenzofuran-Incorporated Hole Transport Materials Featuring High Bond Dissociation Energy for Efficient and Durable Quantum Dot Light-Emitting Diodes	Youngjun HWANG, Daegu Gyeongbuk Institute of Science and Technology, Korea	POS5-1252
PS1_307	Strategic Doping for Enhanced Supramolecular Chirality in Asymmetric PDI Derivatives	Sangwook LEE, Seoul National University, Korea	POS5-1257
PS1_308	Hydrogen Bond-Assisted Dual Passivation for Blue Perovskite Light-Emitting Diodes	Cheolsun PARK, Sungkyunkwan University, Korea	POS5-1271
PS1_309	Development of a High-Performance Solar-Driven Seawater Desalination System Using a Three-Dimensional Ionic PEC-T Hydrogel and a PEDOT/GO Photothermal Layer	Hyunji HWANG, HANNAM UNIVERSITY, Korea	POS5-1308
PS1_310	Synthesis and Characterization of Open-Shell π -conjugated Near-Infrared Active Radical Polymers	Meenakshi, Pusan National University, Korea	POS5-1316
PS1_311	Reflow-Resistant Self-Healing Microstructured Nanocomposite Conductors for Robust Bioelectronic Interfaces	Dohyun LIM, Sungkyunkwan University, Korea	POS5-1333
PS1_312	Antisolvent Engineering with Small Molecule Additives for Efficient and Stable Near-Infrared Perovskite Light-Emitting Diodes	Seungwoo SONG, Gwangju Institute of Science and Technology (GIST), Korea	POS5-1354
PS1_313	Design and Synthesis of D-A-A-Type Organic Semiconductors for Vacuum-Processable NIR Organic Photodetectors	Na Yoon KIM, Nanophotonic System Research Center, Korea Institute of Science and Technology, Korea University, Korea	POS5-1359

PS1_314	Enhancing the EHD Jet Printing Stability of Optical Clear Adhesive (OCA) via Partial Curing	Yongsu SON, Dankook University, Korea	POS5-1365
PS1_315	Retention-Engineered Synaptic Devices for Temporal Signal Processing in In-Sensor Computing	Sung Hyeon PARK, Yonsei University, Korea	POS5-1366
PS1_316	Polymer Matrix-Dependent Photoluminescence and Color Conversion Properties of Organic Dye-Based Films	JuneHyung BAE, Dankook University, Korea	POS5-1373
PS1_317	Long-Retention Wavelength-Tunable Tactile Neuromorphic Display for Vision-Assisted Robotic Manipulation	Woojoong KIM, Yonsei University, Korea	POS5-1388
PS1_318	Mechanism Transition of Polymer-Sorted Single-Walled Carbon Nanotube/Pd Nanoparticle Networks for Wafer-Scale, Reliable Hydrogen Sensing	Minjae CHOI, POSTECH, Korea	POS5-1389
PS1_319	Efficient Hole Injection and Transport in Solution-Processed OLEDs via Thermally Cross-Linkable Poly(iminoarylene)	Yu Kyeong KIM, Pusan National University, Korea	POS5-1391
PS1_320	Chalcogen Atom Substitution in Organic Semiconductors for Developing Efficient and Stable Blue Phosphorescent OLEDs	Yongjun KWON, Ulsan National Institute of Science and Technology, Korea	POS5-1395
PS1_321	Magnetically Responsive Polymer Composite Soft Robot Integrated with a Flexible Quantum Dot Light Emitting Diode	Jioh YOO, Yonsei University, Korea	POS5-1447
PS1_322	Molecularly Engineered Self-Assembled Monolayer Interfaces for Efficient and Stable Green Perovskite LEDs	Yeonji SON, Gwangju Institute of Science and Technology, Korea	POS5-1488
PS1_323	Contact Resistance in Organic Electrochemical Transistors	Tae Hoon KIM, Seoul National University, Korea	POS5-1493
PS1_324	Accurate Determination of Volumetric Carrier Density in Organic Electrochemical Transistors	Seung Hyeon BACK, Inha University, Korea	POS5-1500
PS1_325	A Photonic Smart Window for Season-Adaptive Energy Harvesting and Thermal Regulation	Seonju LEE, Korea Institute of Science and Technology, Korea	POS5-1501
PS1_326	Modular Design of Ion-Conducting Polymer Films with Tunable Nanostructures	Dunja ASCERIC, University of Graz, Austria	POS5-1508
PS1_327	Crystallinity-Driven Enhancement of Electronic and Optical Properties in Mixed Pb/Sn Perovskite	Junpyo LEE, Seoul National University, Korea	POS5-1509
PS1_328	Understanding Swelling Mechanisms for Controlling the Mechanical Properties of Channels in OECTs	Keonho KIM, Seoul National University, Korea	POS5-1528
PS1_329	Cationic Effect of P-Type OECT with Thiophene Based OMIECs	Tae Yeon LEE, Seoul National University, Korea	POS5-1529
PS1_330	Laser-Induced Graphene Ionoskins Enabled by Aromatic Copolymer Networks	Jae Yun KIM, KRICT, Korea	POS5-1563
PS1_331	Electrolyte-Regulated Ion Transport and Memory Dynamics in OMIEC-Based Organic Electrochemical Diodes	Minhu HUANG, Gwangju Institute of Science and Technology (GIST), Korea	POS5-1611
PS1_332	Tunable Synaptic Plasticity in p-n Type OMIECs for Vertical Organic Electrochemical Diodes via Dopant-Ion Size Modulation	Alexander Mauricio TIPAN QUISHPE, Gwangju Institute of Science and Technology, Korea	POS5-1613
PS1_333	Density-Tunable PEDOT:PSS Fibrous Sponges via Ice-Induced Self-Fusion for Soft and Stretchable Organic Batteries	Jongwon LEE, Gwangju Institute of Science and Technology, Korea	POS5-1614

PS1_334	Sugar-Templated 3D GnP@PDMS/CoNi@C Composites for Absorption Dominant EMI Shielding and Efficient Thermal Management	Tahreem ZAHRA, Korea National University of Transportation, Korea	POS5-1632
PS1_335	Beyond Crystalline Order: Structural-Electronic Homogeneity of Transport Pathways as an Efficient Design Framework for Polymer Semiconductor Electronics	Taehoon HWANG, Hanyang University, Korea	POS5-1633
PS1_336	Upcycling Photovoltaic Silicon Waste into Si@MgO Fillers via Magnesiothermic Reduction for Thermally Conductive and Electrically Insulating Epoxy Composites	Jaewon NA, INHA University, Korea	POS5-1659
PS1_337	Pressure-Light Dual-Sensory Ferroelectric Phototransistor Synapse Enabling Pressure-Dependent Modulation of Optical Synaptic Plasticity for Sensory-Adaptive Reservoir Computing	Seungjun PARK, Yonsei University, Korea	POS5-1669
PS1_338	Post-Functionalization Engineering of Semiconducting Polymers Bearing Multiple Hydrogen-Bonding Motifs for Robust Stretchable Transistors	Dinda BAZLIAH, National Taiwan University of Science and Technology, Taiwan	POS5-1679
S6. Emerging Polymer and Hybrid Materials for Advanced Energy Storage and Conversion (PSK50 Student Presentation Award Applicants)			
PS1_339	Dual-Function Tetrazine-Based Covalent Organic Frameworks for Stable Lithium Metal Anodes: Synergistic Regulation of Anion Trapping and Electronic Insulation	Yeong Hun JEONG, Seoul National University, Korea	POS6-0191
PS1_340	Highly Adhesive, Fluorine-Free Binders Enabled by Chemical Anchoring for Durable High-Nickel Cathodes in Lithium-Ion Batteries	Seunghyeon KIM, Seoul National University, Korea	POS6-0192
PS1_341	Chemically Anchored Organic-Inorganic Hybrid Separators for Safe and Durable Lithium-Ion Batteries	Min Seo JO, Seoul National University, Korea	POS6-0194
PS1_342	Bio-Derived Phytic Acid-Based Organic-Inorganic Hybrid Coating for Flame-Retardant and Thermally Stable Separators for Lithium-Ion Batteries	Gwangbin WON, Seoul National University, Korea	POS6-0195
PS1_343	A Multifunctional Organic-Inorganic Hybrid Flame-Retardant Separator for High-Safety Lithium-Ion Batteries	Sinyoung SEO, Seoul National University, Korea	POS6-0198
PS1_344	Quasi-Solid-State Thermogalvanic Cells Based on TEMPO-Oxidized Cellulose Nanofibril/Poly(vinyl alcohol) Double-Network Hydrogels	Sae Young LEE, Chungnam National University, Korea	POS6-0234
PS1_345	Mechanically Robust Intrinsically Stretchable Organic Solar Cells Enabled by Interfacial Mechanical Harmonization	Yunsu LIM, KAIST, Korea	POS6-0236
PS1_346	Accurate Evaluation of Hydrogen Crossover in Water Electrolysis Systems: Elucidating Ion Channel Morphology for Hydrated Membranes	Seung Hwan KIM, Kyung Hee University, Korea	POS6-0274
PS1_347	Distinct Roles of Long-Lived Charge Carriers in Organic Photovoltaics and Photocatalytic Hydrogen Evolution	Gayoung HAM, Kyungpook National University, Korea	POS6-0276
PS1_348	Tailoring Bicontinuous Elastomeric Electrolytes for High-Performance Lithium Metal Batteries	Jaejoon LEE, KAIST, Korea	POS6-0295
PS1_349	Anion Exchange Membranes Synthesized via a One-pot Monomer Reaction for Water Electrolysis Applications	Yujin NAM, Korea University, Korea	POS6-0304
PS1_350	Optimization of Indoor Light Organic Photovoltaics and Development of Smart Sensor Control System Through Aging Chamber-Based Evaluation	Bogyeong JEON, Pukyong National University, Korea	POS6-0326
PS1_351	Dual-Functional Graft Copolymer Binders for High-Nickel Cathode Materials in Lithium-Ion Batteries	Jimin PARK, Seoul National University, Korea	POS6-0345
PS1_352	Inverse Vulcanized Polymers with Controlled Sulfur Allotropes for Lithium-Sulfur Batteries	Woongsik CHOI, Postech, Korea	POS6-0378

PS1_353	Deborah Number-Governed Surface Roughness Control of Sulfur-Rich Polymer Film via Micro-Scale Surface Tearing for High-Performance Triboelectric Nanogenerator	Minji BAK, Hanyang University, Korea	POS6-0408
PS1_354	Site-Selective Fluorination of Bathocuproine Derivatives for Enhanced Performance and Stability in Perovskite Solar Cells	Dong-Geon KWUN, Pusan National Univ, Korea	POS6-0425
PS1_355	Mechanically Optimized Fluorinated Elastomer Electrolytes for Stable Operation of Lithium Metal Batteries at Low Temperature	Dongkyu LEE, Korea Advanced Institute of Science and Technology, Korea	POS6-0475
PS1_356	Seed- and Vacuum-Assisted, Anti-Solvent-Free Crystallization for Uniform Perovskite Mini-Modules	Sunju KIM, Pusan National University, Korea	POS6-0497
PS1_357	Directional Gas Evacuation Through Out-of-Plane Gas Pathways for Enhanced Mass Transport in Water Electrolyzers	Seunghyun LEE, Ulsan National Institute of Science and Technology, Korea	POS6-0500
PS1_358	Droplet-Based Electricity Generator Using Superhydrophobic Fiber-Reinforced Plastic for Self-Powered Structures	Jaejin KIM, Ulsan National Institute of Science and Technology, Korea	POS6-0515
PS1_359	Development of a Self-Healing and Reprocessable Waterborne Polyurethane Binder for Battery Electrodes	Anjum Hanief KOHLI, Kyungpook National University, Korea	POS6-0517
PS1_360	Molecular Design of Self-Assembled Electrolytes for Stable Lithium Metal Batteries	Taekyung WON, Gwangju Institute of Science and Technology, Korea	POS6-0532
PS1_361	Controlled Synthesis of Solution-Processable Oligomeric Nano-Borosiloxane Resins with Enhanced Thermal Stability and Flame Retardancy	Jieon KIM, NationPusan al University, Korea	POS6-0535
PS1_362	Conductivity and Mechanical Strength Control of Self-Healing Ionic Thermoelectric Polymers	Hyeonsu CHOI, Chungnam National University, Korea	POS6-0549
PS1_363	Synergistic Effect of Ionic Liquid and ZIF-8 to Enhance Ion Transport in PEO-Based Solid Polymer Electrolytes	Sanghee PARK, Chonnam National University, Korea	POS6-0561
PS1_364	Large-Area Carbon Nanotube Sheets with Controlled Doping for Flexible Thermoelectric Generators	Kanghyun JO, Chungnam National University, Korea	POS6-0565
PS1_365	In Situ Grown Hydrochar@ZIF-8 Hybrid Fillers for High-Performance PEO-Based Solid Polymer Electrolytes	Serim AHN, Chonnam National University, Korea	POS6-0572
PS1_366	PEO/LiTFSI-Based Composite Solid Polymer Electrolytes with Lithium Phosphate Fillers for Solid-State Lithium Batteries	Jaewon KIM, Chung-Ang University, Korea	POS6-0582
PS1_367	UiO-66-NH ₂ MOF-Integrated Cellulose Scaffold for High-Performance Solid Polymer Electrolytes	JiHyuk KIM, Chung-Ang University, Korea	POS6-0591
PS1_368	Hybrid Polymer Electrolytes with Lithium Silicate for All-Solid-State Lithium Metal Batteries Batteries	Jihwan KIM, Chonnam National University, Korea	POS6-0592
PS1_369	Fluorinated Polymer Interphases for Enhanced Cycling Stability of Lithium Metal Batteries	Seongmin KIM, KAIST, Korea	POS6-0631
PS1_370	Multifunctional Small-Molecule-Reinforced Polyacrylic Acid-Based Primer Binders for Low-Temperature Interfacial Adhesion in Dry-Processed LiFePO ₄ Electrodes	Yoo Jeong HUH, Pohang University of Science and Technology (POSTECH), Korea	POS6-0638
PS1_371	Ionic Functionalized SEBS-Based Protective Layers for High-Performance Lithium Metal Batteries	HyunSoo KWON, KAIST, Korea	POS6-0662
PS1_372	Improving Li ⁺ Transport in PTFE-Based Dry Thick Electrodes Through PEG Co-Binder Engineering	Hyeonji JUNG, Chung-Ang University, Korea	POS6-0676

PS1_373	Morphology-Driven Conductive Network Design for Low-Tortuosity Solvent-Free Thick LiFePO ₄ Electrodes	Jeehoon YU, Chung-Ang University, Korea	POS6-0697
PS1_374	Molecular Interface Engineering of Organic Photoanodes for Efficient Photoelectrochemical Water Splitting	Wonjun KIM, UNIST, Korea	POS6-0702
PS1_375	From CO ₂ to a Sustainable Aqueous Binder for Silicon/Carbon Anodes	Soyun PARK, Pusan National University, Korea	POS6-0715
PS1_376	Anisotropic PVDF/MXene Foam with Aligned Microchannels for High-Sensitivity Directional Triboelectric Nanogenerators	Eunji KIM, Pukyong National University, Korea	POS6-0779
PS1_377	Development of H ₂ TiO ₃ -incorporated PVA/SA/PSS Hydrogel Beads for Lithium Recovery	Daeyeong CHO, Chonnam National University, Korea	POS6-0815
PS1_378	Ir-Efficient Nb ₂ O ₅ @IrO ₂ Core-Shell Electrocatalysts for Acidic Oxygen Evolution Reaction	Soomin KIM, Inha University, Korea	POS6-0825
PS1_379	A Multifunctional Solid Electrolyte Design for Efficient Lithium-Ion Transport and Dendrite-Resistant Lithium-Metal Batteries	Fazal Ur REHMAN, Chonnam National University, Korea	POS6-0861
PS1_380	Imidazolium-Functionalized Polyethylenimine as an Additive for PEO-based Solid Electrolytes Toward Enhance All-Solid-State Battery Performance	Minseo KIM, Chonnam National University, Korea	POS6-0868
PS1_381	Development of Polymer Blend Systems for Synergistic Enhancement of Thermoelectric Characteristics	Seokhwan JANG, KRICT, Korea	POS6-0886
PS1_382	Cellulose-TiO ₂ Hybrid Aerogels for Interfacial Control of Au Dispersion in Photocatalytic CO ₂ Cycloaddition	Hyun A MYOUNG, Inha University, Korea	POS6-0937
PS1_383	Effect of Magnetic Filler Geometry on Soft Magnetoelastic Generators Performance	Hyeokju KWON, Gwangju Institute of Science and Technology, Korea	POS6-0980
PS1_384	Low-Temperature ALD for Process-Robust and Stable p-i-n Perovskite Solar Cells via SnO ₂ Buffer and Al ₂ O ₃ Encapsulation	Uk Jin CHOI, Pusan National University, Korea	POS6-1025
PS1_385	Mixed Ionic-Electronic Conductors Enabled by Dynamic Hydroxo Complexation for Organic Thermoelectric Devices	Jihyun KIM, Chungnam National University, Korea	POS6-1055
PS1_386	Sustainable Hydrophilic Porogen Leaching Process for Novel PPS Membrane Applied to Lithium Metal Batteries	Hoyun JEON, Korea Institute of Science and Technology, Korea	POS6-1087
PS1_387	Macromolecular Engineering Approaches Towards Advanced Polymer Mixed Ionic/Electronic Conductors for Solid-State Li-Metal Batteries	Seong Woo JEONG, Sungkyunkwan University, Korea	POS6-1088
PS1_388	A Synergistic Polymer Design Approach for Enabling Low-Temperature Performance in Energy Storage Devices	Daniel Sanghyun CHO, Korea Advanced Institute of Science and Technology (KAIST), Korea	POS6-1089
PS1_389	Development of a Sustainable PPS Membrane with Tailored Porosity Using Crumpled Graphene Oxide Containing Highly Interconnected Pore Architecture	Min-gyeong SHIN, Korea Institute of Science and Technology, Korea	POS6-1107
PS1_390	Architecturally Confined PEO-Based Composite Solid Electrolytes with Transport-Active Polymer-Ceramic Interphases for Stable Lithium Metal Batteries	Yujin KIM, Chonnam National University, Korea	POS6-1139

PS2. Poster Session 2

September 30 (Wed), Room 301 (Grand Ballroom), BEXCO

Board No. 보드번호	Title	Presenter	Abstract No. 초록번호
S6. Emerging Polymer and Hybrid Materials for Advanced Energy Storage and Conversion (PSK50 Student Presentation Award Applicants)			
PS2_001	Unlocking All-Solid-State Lithium Conductivity with Dynamic Poly(Triazolium) Electrolytes	Sirin KAMARULZAMAN, Nanyang Technological University, Singapore	POS6-1153
PS2_002	Cross-Linkable Polyester-Based Solid Polymer Electrolytes with Tailored Molecular Design for High-Performance Solid-Sate Li Metal Batteries	Jeongyoon KIM, Sungkyunkwan University (SKKU), Korea	POS6-1155
PS2_003	Coordination-Polymer-Derived Molecular Gluing for Self-Supported Hybrid Electrocatalysts Toward Sustainable Water Splitting	Xiangyu ZHANG, Sungkyunkwan University, Korea	POS6-1180
PS2_004	Highly Crystalline Donor-Acceptor Polymer Nanowires via Crystallization-Driven Self-Assembly for Efficient Photocatalytic Hydrogen Evolution	Hyojun KIM, Gwangju institute of science and technology, Korea	POS6-1233
PS2_005	Composite Membranes with Dual-Piperidinium Functionalization Across Polymer Matrix and ZrO ₂ Surface for High-Performance Anion Exchange Membrane Water Electrolysis	Joon Ho JANG, Sungkyunkwan University, Korea	POS6-1247
PS2_006	Spray-Coated Bipolar Membranes for Water Electrolysis: Effects of Catalyst Loading and a Novel Anion-Conductive Ionomer	Yumi HIDAKA, Tokyo Metropolitan University, Japan	POS6-1260
PS2_007	Efficient Water-Splitting with Electroconductive Cellulose Nanowire Structure-Based Electrodes	Sumin KANG, Sungkyunkwan University, Korea	POS6-1286
PS2_008	Fabrication, Ion Transport Properties, and Battery Applications of Anion-Conductive Network Gel Polymer Electrolytes	Yamato NUMADA, Tokyo Metropolitan University, Japan	POS6-1320
PS2_009	Gradient Janus Hydrogel Electrolyte Regulating Interfacial Ion Transport for Simultaneous Cathode Utilization and Zn Anode Stability in Flexible Batteries	Yubeen LEE, POSTECH, Korea	POS6-1323
PS2_010	Conformational Engineering of Polypeptide Binders for Thick Cathodes	Hayoung PARK, POSTECH, Korea	POS6-1324
PS2_011	Balancing Rigidity and Elasticity in Polymer Binders for Long-Life Silicon Anodes	Jisoo LEE, Seoul National University, Korea	POS6-1329
PS2_012	Phosphorylated Cellulose Nanofiber Composite Membrane Containing Acid-Functionalized Fumed Silica for Polymer Electrolyte Membranes	Hyuntaek HAN, Seoul National University, Korea	POS6-1361
PS2_013	Synergistic Coupling of Host and Electrolyte Achieving 1270 Wh L ⁻¹ in Anode-Free Lithium Metal Batteries	Sang-Won LEE, POSTECH, Korea	POS6-1371
PS2_014	Mechanically Robust and Ion-Conductive Polyampholyte Elastomers	Taebin KIM, Yonsei University, Korea	POS6-1376
PS2_015	Linker-Free Design of Naphthalenediimide Polymer Cathodes for Large Capacity and High Rate Capability	Jiyeon LEE, KIST, Korea	POS6-1381
PS2_016	Self-Supported Polypyrrole Capped CuFe LDH/NF for Corrosion Resistance Kilo-Hour Seawater Electrolysis	Bibi NAFEESA, Chung Ang University, Korea	POS6-1386
PS2_017	Carbon Fiber-Reinforced Redox-Active Polymer (CFRRAP) Composites for Structural Battery Cathodes	Joohwan EO, Korean institute of scienc and technology(KIST), Korea	POS6-1390

PS2_018	Synthesis and Characterization of Shutdown-Enabled and Flame Retardant Polyethylene Lithium-Ion Battery Separator Using Phosphorus-Rich Poly (vinyl acetate-acrylonitrile) Microparticles	Kyoung-Jin CHOI, Korea Research Institute of Chemical Technology (KRICT), Kyonggi University, Korea	POS6-1396
PS2_019	Multifunctional ROMP-Derived Polymer Binders for Stable and High-Performance Silicon Anodes	Cheol SHIN, Hanyang University, Korea	POS6-1398
PS2_020	Spectral Gap-Filling Pt(NHC-phenoxo) ₂ Sensitizer Suppresses Triplet Recombination in High-Performance D-A-A-D Ternary OSCs	Priyanka YADAV, Pusan National University, Korea	POS6-1412
PS2_021	Sulfonated Fe-MOF-Polytriazole Hybrid Membranes with Enhanced Proton Conductivity: Structure-Property Relationships and Application Potential	Riddhi KAMBLE, Indian Institute of Technology Kharagpur, India	POS6-1429
PS2_022	Thiol-Ene Functionalization of a PS-b-PI-b-PS Binder for Enhanced Interfacial Adhesion and Rate Performance in Sulfide-Based All-Solid-State Batteries	Hyeongeun RYU, Chungnam National University, Korea	POS6-1438
PS2_023	Development of a Gas Transport Characterization Methodology for Ion-Solvating Membranes and Adjacent Liquid Layers in High-Pressure Alkaline Water Electrolysis Systems	Kyuwon HWANG, Kyung Hee University, Korea	POS6-1448
PS2_024	Chemical Functionalization of Graphene Oxide for Enhanced Triboelectric Charge Density in Textile-Based Triboelectric Nanogenerators	Muhammad TOUQEER, University of Ulsan, Korea	POS6-1480
PS2_025	Characterization of Photophysical Properties in Sub-Micron Perovskite Single-Crystal Thin Films	Junmo PARK, Gwangju Institute of Science and Technology, Korea	POS6-1485
PS2_026	Activating the Lattice Oxygen Mechanism in NiCo ₂ O ₄ Through Fe Single-Atom Engineering for the Oxygen Evolution Reaction	Suyeon KIM, Sungkyunkwan University, Korea	POS6-1514
PS2_027	Acid-Medium-Controlled Loading and Nanoparticle Formation of Ni and Cu in Vertically Oriented P4VP Nanodomains	Jieun LEE, Chungnam National University, Korea	POS6-1551
PS2_028	Highly Durable Superaerophobic Hydrogel Coating for Efficient Anion Exchange Membrane Water Electrolysis	Siheon LEE, Seoul National University, Korea	POS6-1590
PS2_029	Transparent Nanostructured Electrodes for High-Performance Liquid Solar Energy Harvesters Toward Autonomous Wildfire Detection Systems	Myeong LEE, Kyunghee Univ., Korea	POS6-1602
PS2_030	Single Electrode Piezo-Enhance Triboelectricity: A Stable Hybrid Device with Vertically Aligned ZnO Nanoflakes and MXene/Ecoflex Encapsulation	Abdul SABOOR, University of Ulsan, Korea	POS6-1608
PS2_031	Unraveling the Molecular Origin of Charge Transport and Stability in Non-Halogenated Phenothiazine Self-Assembled Monolayers for High-Performance Organic Solar Cells	Mutiara SIRAIT, Pukyong National University, Korea	POS6-1633
PS2_032	Polymer-Coated Cu ₃ P/Cu ₂ S/CF NWs Heterostructure for Photo-Driven Soft Seawater Hydrogen Evolution and Dye Degradation	Bibi NAFEESA, Chung Ang University, Korea	POS6-1680
S7. Innovations in Polymeric Composites: From Design and Processing to Industrial Applications (PSK50 Student Presentation Award Applicants)			
PS2_033	Enhancing the Mechanical Performance of Bio-Based Poly(ester-amide) Nanocomposites with Aramid Nanofibers	Yibeen NA, Kyung Hee University, Korea	POS7-0083
PS2_034	Interfacial Engineering of Aramid Fibers with Tannic Acid-Epoxy Core-Shell Coatings for Flame Retardancy and Electrical Insulation	Wonhui JEONG, Korea Institute of Science and Technology, Korea	POS7-0169
PS2_035	Highly Anti-Impact Polydiboronsiloxane Nanocomposites via Interfacial Interaction Modulation	Yujia HOU, State Key Laboratory of Advanced Polymer Materials, Sichuan University, China	POS7-0212
PS2_036	Vinylogous Urethane and Disulfide Dual-Dynamic Epoxy Vitrimer with Rapid Stress Relaxation and Multiple Recycling Routes	Hu XU, Sichuan University, China	POS7-0213

PS2_037	Ordered and Reprocessable Smectic Liquid Crystal Elastomers via Bulk Thiol-Ene Polymerization	Thu Loan DANG, Kyungpook National University, Korea	POS7-0224
PS2_038	Improvement of Interfacial and Mechanical Properties of Carbon Fiber/PBAT Composites Through Biomass-Based Sizing Treatment	Hwan-Gi DO, Kumoh National Institute of Technology, Korea	POS7-0292
PS2_039	Surface-Modified Cellulose as Nucleating Agents for Enhanced Crystallization of Biodegradable Polymer Composites	Jisu KIM, Chonnam National University, Korea	POS7-0324
PS2_040	Surface-Modified Cellulose for High-Performance Cellulose-Based Adhesive	Jihwan RYU, Chonnam National University, Korea	POS7-0325
PS2_041	Effect of Copper Oxidation States on the Interfacial Chemical Composition of Epoxy Resins	Shintaro SAEKI, Kyushu University, Japan	POS7-0407
PS2_042	Segregated hBN Networks Enabling Flame-Retardant, Thermally Conductive, and Reprocessable Covalent Adaptable Network Composites	Hyunju KIM, Pusan National University, Korea	POS7-0420
PS2_043	Accelerating Interfacial Chain Dynamics in Polylactic-Acid Nanocomposites via Densely Grafted Cellulose Nanocrystals: Translating to Macroscopic Property Breakthroughs	Hyojin JUNG, Seoul National University, Korea	POS7-0503
PS2_044	Improving the Thermal Shock Durability of Fiber-Reinforced Polymer Composites via Bond Exchange Reactions	Yunghee JUNG, Ulsan National Institute of Science and Technology, Korea	POS7-0504
PS2_045	Flexible Low-Dielectric Thermotropic Liquid Crystalline Polymer Films Reinforced with Low-Loading Aramid Nanofibers for High-Frequency Substrate Applications	Min-cheol KIM, Chungnam National University, Korea	POS7-0505
PS2_046	Interfacial Engineering of Urushiol-Based Coatings Using Functionalized Silica Nanoparticles for Enhanced Mechanical Durability	Jiwon PARK, Chungnam National University, Korea	POS7-0568
PS2_047	Magnetic Field-Directed Alignment of MOF-Derived Superparamagnetic Fillers for Thermal Management in Epoxy Molding Compounds	Gayeong GIM, Inha Manufacturing Innovation School, Korea	POS7-0571
PS2_048	Dual Heat-Pathway Epoxy Beads for Constructing Continuous Thermal Transport Networks in Epoxy Molding Compounds	Seong Uk KANG, Inha University, Korea	POS7-0583
PS2_049	Multifunctional Kaolin/PDMS Composite Bandage with Hemostatic, Absorptive, and Passive Radiative Cooling Properties	Yubeen OH, Chung-Ang University, Korea	POS7-0584
PS2_050	Hybrid Reinforcement of EPDM Composites Using Carbon Black and Aramid Fibers for Enhanced Mechanical Performance	Sung-Hyuck PARK, Chungnam National University, Korea	POS7-0597
PS2_051	Hierarchical Free-Volume Engineering of UV-Curable Cycloaliphatic Epoxy Composites for Low-Dielectric Insulating Applications	Min-Young HWANG, Chungnam National University, Korea	POS7-0599
PS2_052	Hierarchical BN/Al ₂ O ₃ Filler Networks for Enhanced Thermal Transport in Epoxy-Based Non-Conductive Films	Sihyeon AHN, Chung-Ang University, Korea	POS7-0673
PS2_053	Hydroxyl-Controlled MOF-808@Silica-based Aerogel Hybrid Catalysts for Recyclable CO ₂ Cycloaddition	Chaeyeon LEE, Inha University, Korea	POS7-0681
PS2_054	Hierarchically Porous chitosan/MWCNT- Derived Bimetallic Carbon Aerogel for Electromagnetic Interference Shielding	Siwon LEE, Inha University, Korea	POS7-0718
PS2_055	Materials Design of Thermally Conductive Adhesives for Interfacial Thermal Transport in Electronics	Jinsoo NA, Seoul National University, Korea	POS7-0719
PS2_056	Multidimensional Natural Filler-Reinforced Polyurethane Composite Foams with Hierarchical Poroacoustic Pathways	Eunsu GONG, Seoul National University, Korea	POS7-0757

PS2_057	NiCo-C/CNF Aerogel-Based Composites for Absorption-Dominant Electromagnetic Interference Shielding	Jihun SEO, Pukyong National University, Korea	POS7-0770
PS2_058	Highly Aligned Dynamic Covalent Nanolaminates for Advanced Flame Retardancy	Minjeong SHIN, Pusan National University, Korea	POS7-0774
PS2_059	Electrically Detachable Multifunctional Thermal Interface Materials Based on Pickering Emulsion Architectures	Min Jun LEE, Pusan National University, Korea	POS7-0806
PS2_060	CNT Fiber-Hybrid CFRP for Lightning Strike Protection Against Joule Heating and Shock Wave Damage	Su Jeong HEO, Pusan National University, Korea	POS7-0830
PS2_061	Hybridization of Functionalized Nanofibers with Porous Coordination Frameworks for Enhanced U(VI) Adsorption Capacity	TaeYoung JEONG, Chungnam National University, Korea	POS7-0840
PS2_062	Functionalized Biomass Derived Hydrochar Microspheres for Selective Dye Removal from Wastewater	Iqra QAYYUM, Chonnam National University, Korea	POS7-0848
PS2_063	Anisotropic EMI Shielding Performance of Unidirectionally Aligned $Ti_3C_2T_x$ MXene/PDMS Composites	Jimin KIM, Pukyong National University, Korea	POS7-0850
PS2_064	TA-APTES/EDTA-Modified UHMWPE Membranes for Trace Metal-Ion Removal	Jinho KIM, Yonsei University, Korea	POS7-0918
PS2_065	Rheological Engineering of Bimodal h-BN/Al ₂ O ₃ -Filled Epoxy Composites for Directional Thermal Management	Hyeseo CHOI, Hanyang University, Korea	POS7-0924
PS2_066	Melt Crystallization for Fabrication of Conductive PDMS Films with Controlled Filler Alignment	Haemin LEE, Chung-Ang University, Korea	POS7-0928
PS2_067	Effect of Silane-Modified Silica on the Curing Behavior and Outgassing Properties of Epoxy-Based Sealants	Kyeong Jun PARK, Kyungpook National University, Korea	POS7-0939
PS2_068	Self-Reinforced Aramid Composites with Enhanced Crack Resistance and Dimensional Stability: A Platform for Multifunctional Thermal Management	Hyeonji KIM, Hanyang University, Korea	POS7-1010
PS2_069	Conductive Cellulose Composites with Low Poisson's Ratio via Thermoresponsive Shrinkage	Seoeun WI, Chung-Ang University, Korea	POS7-1093
PS2_070	RF Plasma-Induced Surface Modification of Polyethylene: Chemical Properties and Gas Barrier Enhancement	Serim YOUN, Yonsei University, Korea	POS7-1132
PS2_071	A Food-Compatible Photocurable Soy Protein Resin for DLP 3D Printing	Kisung KIM, Gachon University, Korea	POS7-1175
PS2_072	N-Doped Graphene-Based Aerogel/Epoxy Composites with Isotropic Networks for Multifunctional Thermal Management Applications	JungHyun KIM, Kyungpook National University, Korea	POS7-1217
PS2_073	Influence of Nanofiller Size and Concentration on the Photo-Induced 3D Printing Performance of Elastomer Composites	Gwanghoon CHOI, Gachon University, Korea	POS7-1231
PS2_074	Thermo-Mechanically Activated Liquid Metal Bridging in Epoxy Composites for Next-Generation Electronic Packaging	Ye Eun LEE, Korea Institute of Science and Technology (KIST), Korea	POS7-1238
PS2_075	Mechanistic Quantification of Property Enhancement in Heat-Resistant CFRTs by Carbon Fiber Sizing Agents	Satoru HAMAJIMA, Gifu University, Japan	POS7-1239
PS2_076	Multiscale MgO Filler Design for Thermally Conductive Epoxy Molding Composites	Chaemin YEO, Kyungpook National University, Korea	POS7-1251

PS2_077	A Multifunctional Heterocycle-Based Additives for Matrix Reinforcement and Interfacial Mediation in Biodegradable Materials	Subin PARK, Korea Institute of Industrial Technology, Korea	POS7-1254
PS2_078	Ni-Zn-Co Ferrite/SBR Multifunctional Nanocomposites with Data-Driven Correlation of Vulcanization Kinetics and Optimal Filler Loading	Sungmo CHOI, Gachon University, Korea	POS7-1264
PS2_079	Hydrophilic Polyethersulfone Nanofiber Membrane Produced Through Coaxial Electrospinning with Polyvinyl Alcohol	Tae Woo LEE, Kyungpook National University, Korea	POS7-1274
PS2_080	Tone-Changable Sn-Based Hybrid Photoresists for EUV Lithography	Seunghyun LEE, Ajou University, Korea	POS7-1307
PS2_081	Lightweight, Surface-Engineered MXene/TPU Composites for Paintable Microwave Absorbers with Environmental Durability	Hyun Seong YANG, Korea Institute of Science and Technology, Korea	POS7-1369
PS2_082	Incorporation of Acrylated Chitosan into a Photocurable Resin for Enhanced Mechanical Performance in Digital Light Processing (DLP) 3D Printing	Pawarit CHUMPON, Seoul National University, Korea	POS7-1370
PS2_083	Polymer-Passivated MXene Heaters for Ultrafast High-Temperature Heating and Autonomous De-Icing	Yeonji KIM, Yonsei Univ., Korea	POS7-1471
PS2_084	Dual Metal-Ligand Coordination Reinforcement in Thermoplastic Polyurethane Nanocomposites via Zn ²⁺ Bridging and Functional Nanofillers	Won Wook LEE, Chungnam National University, Korea	POS7-1582
PS2_085	Delamination-Free Multifunctional Composites Enabled by Sugar-Templated Mechanical Interlocking as Thermal Interface Materials with EMI Shielding and Electrical Insulation	Muhammad YASIR, Korea National University of Transportation, Korea	POS7-1631
PS2_086	Thermally Conductive Adhesive Acrylate Composites with Strain-Induced Alignment of Silver-Coated Carbon Fibers	Seongbin KIM, Hanyang University, Korea	POS7-1641
S8. Frontiers of Functional Polymers in Biology and Medicine (PSK50 Student Presentation Award Applicants)			
PS2_087	Synthesis of Discrete Glyco-Oligomers Mimicking GM1 Glycans	Ai CHUGANJI, Kyushu University, Japan	POS8-0145
PS2_088	Development of a MIP-Functionalized PVA/PVP Biphasic Electrospun Nanofibrous Platform for Non-Invasive Glucose Monitoring in Sweat	Shivani SHARMA, Korea University of Technology and Education, Korea	POS8-0164
PS2_089	Thermoresponsive Poly(glycidyl ether)-Modified Lipid Nanoparticles as a PEG Alternative for Temperature-Dependent Stealth Modulation and mRNA Delivery	Mingyu LEE, Inha University, Korea	POS8-0246
PS2_090	Onium-Based Cationic Polymer-Lipid Nanocomplexes for Enhanced Gene Delivery and Therapeutic Applications	Dongki LEE, The Catholic University of Korea, Korea	POS8-0270
PS2_091	Modular Design of Crosslinked Network Polymers with Tunable Mechanical and Adhesive Properties	Tang WEIJIA, KAIST, Korea	POS8-0283
PS2_092	Engineering Cell Sheets from Anchorage-Dependent Cells for Burn Therapy	Jumi KANG, Kyungpook National University, Korea	POS8-0294
PS2_093	Amine-Functionalized Polymer-Coated Paper for Capture and Colorimetric Detection of Volatile Aldehyde Biomarkers	Minju KIM, Kyungpook National University, Korea	POS8-0296
PS2_094	Polymer-Enhanced SELEX for the Identification of Aptamers with Higher Affinity	Insoo BAIK, Interdisciplinary Program in Bioengineering, Seoul National University; Bionanotechnology Research center, Korea Research Institute of Bioscience and Biotechnology (KRIBS)	POS8-0339
PS2_095	Fabrication of Copper Nanoparticle-Loaded Multifunctional Hydrogels with Tunable Adhesion and Antibacterial Activity via Repeated Swelling-Deswelling Cycles	Seoyoon KWON, Sookmyung Women's University, Korea	POS8-0353

PS2_096	Device-Free Injectable Electroceutical Hydrogel Restores Neurovascular Function via Targeted Electrostimulation in Diabetic Erectile Dysfunction	Sung-eun HEO, Yonsei University, Korea	POS8-0442
PS2_097	Ion-Immobilizing Low-Conductivity Hydrogel Reservoirs for Electrically Gated Anti-Inflammatory Drug Delivery in Diabetic Wounds	Yongho LEE, Yonsei univ., Korea	POS8-0452
PS2_098	Library-Immobilized Selection on a Three Dimensional Microporous Cryogel for Breast Cancer Exosome-Targeting Aptamer Discovery	Yeongseo KIM, Ewha Womans University, Korea	POS8-0455
PS2_099	Development of a Fluorinated Low-Molecular-Weight PEI for Delivery of Txndc5-Targeting CRISPR-Cas9 Plasmids Toward Atherosclerosis Gene Therapy	Wei Jen HUANG, National Taiwan University, Taiwan	POS8-0472
PS2_100	A Soft Biomaterial That Stiffens near Body Temperature for Tendon Reinforcement	Seojin CHOI, Seoul National University, Korea	POS8-0510
PS2_101	Stimuli-Responsive Mn-Doped Molecularly Imprinted Polymers for Targeted Chemodynamic Therapy	Min Seok KANG, Hanyang University, Korea	POS8-0581
PS2_102	Size and End-Group Engineering of Hyperbranched Polyglycerol for Enhanced Cell Cryopreservation	Dayoung KIM, Inha Univ., Korea	POS8-0644
PS2_103	N-Oxide-Based Zwitterionic Polymer Coatings for Antifouling Surface Modification	Minjae KIM, Chungnam National University, Korea	POS8-0664
PS2_104	A Polymer Hydrogel Design Framework for Advanced Tissue Processing and Phenotyping	Sangjae KIM, Seoul National University, Korea	POS8-0714
PS2_105	A Highly Aligned, Strong and Conductive Directional Nanocomposite Hydrogel for Artificial Tendons	Sanghyun PARK, Seoul National University, Korea	POS8-0728
PS2_106	Injectable Soft Nanogel for Sustained Drug Release	Youngjun LEE, Seoul National University, Korea	POS8-0833
PS2_107	Multifunctional Platform Based on Porous Coordination Frameworks for Ratiometric Fluorescent Quantification of a Gut Microbial Metabolite	MinGyeong KANG, Chungnam National University, Korea	POS8-0835
PS2_108	Polymeric Multilayer Films for Stabilization of Copper Nanostructures in Non-Enzymatic Glucose Sensing	Han kyu PARK, Chungnam National University, Korea	POS8-0836
PS2_109	Neutral Buoyancy-Based Simulated Microgravity Enhances Angiogenic Potential of MSC/HUVEC Spheroids	Yujung CHOI, Dankook University, Korea	POS8-0846
PS2_110	Development of an LSS/HepaRG Hybrid Spheroid System for Liver Toxicity Evaluation	Yoonhyuck LEE, Dankook University, Korea	POS8-0852
PS2_111	Enzyme-Free Oxygen-Supplying Containers for Wound Healing with Immunomodulation	Gwang Ryeol HEO, Incheon National University, Korea	POS8-0875
PS2_112	Polyphenol-Based Biointerfaces for Selective Lipoprotein Interference Control in Complex Biofluids	Nayoung SON, DGIST, Korea	POS8-0887
PS2_113	Oxygen-Controllable Gelatin-Based Hydrogel Platforms for Engineering Hypoxic Atopic Dermatitis Microenvironments	Jae Sung JUNG, Incheon National University, Korea	POS8-0889
PS2_114	Hyaluronic Acid and Zinc Ion-Based Cryoprotective Complex for Enhancing Post-Thaw Cell Viability	Gwang Ryeol HEO, Incheon National University, Korea	POS8-0891
PS2_115	Hydrophobic Modification of mRNA for Enhancing Lipid Nanoparticles (LNPs)-mediated mRNA Delivery Efficiency	Choeun PARK, Korea University, Korea	POS8-0906

PS2_116	MOF-incorporated Hydrogel for On-Demand Wound Healing	Jae Sung JUNG, Incheon National University, Korea	POS8-0916
PS2_117	Engineering Multivalent GalNAC-Conjugated Lipid Nanoparticles for High-Efficiency Liver Targeting	Ji Yoon LEE, Korea Research Institute of Bioscience and Biotechnology, Korea	POS8-1037
PS2_118	Polymer-Based Microstructural Biomaterials for the Local Delivery of Cytotoxic Agents	Kamil RUSZTYN, University of Warsaw, Poland	POS8-1051
PS2_119	Biomacromolecular Gold Nanocomplexes Induce Membrane Rupture and Immunogenic Cell Death	Jaeeun LEE, Chemistry, Korea	POS8-1130
PS2_120	Robust Seawater-Resistant O/W Sunscreen Formulation via Alginate-Cation Crosslinking	Jongsun BAEK, Cosmetic Chemistry Laboratory, Jeju National University, Korea	POS8-1138
PS2_121	Stable Integration of Photosensitive Therapeutics into Photocrosslinked Polymeric Microneedles	Yeong Min KANG, Chungnam National University, Korea	POS8-1148
PS2_122	Tendon-Bone Interface Regeneration Facilitated by 3D Architectures Guiding Spatial Cell Alignment and Multiphase Tissue Integration	Jinwoo NA, Hanyang University, Korea	POS8-1165
PS2_123	Exercise-Mimetic Multi-Organ Device Recapitulates Myokine-Mediated Inter-Organ Metabolic Crosstalk in Obesity-Induced T2DM	Yerim JUNG, Incheon National University, Korea	POS8-1179
PS2_124	NIR-Triggered MXene/Doxorubicin Supramolecular Hydrogel for Localized Synergistic Cancer Therapy	Seung Min YANG, POSTECH (Pohang University of Science and Technology), Korea	POS8-1184
PS2_125	Influence of Branched Molecular Architecture on Polyion Complex Stability Under Reduced Net Charge Conditions	Masayuki NAKASHOYA, Kyoto Prefectural University of Medicine, Japan	POS8-1196
PS2_126	Ultrasound-Activated MXene@MnO ₂ -Embedded Adhesive Hydrogel for Postoperative Skin Cancer Therapy	Junhee KIM, Pohang University of Science and Technology, Korea	POS8-1220
PS2_127	An Informative and Therapeutic Smart Contact Lens Platform for Correcting Ocular Motor Anomalies via Wireless Frame Resonance	Dongchul CHO, POSTECH, Korea	POS8-1221
PS2_128	In Situ-Forming Injectable Hyaluronic Acid Hydrogels via Michael Addition with Tunable Rheological Properties	Sua CHOI, Chungnam National University, Korea	POS8-1222
PS2_129	Surface Acoustic Wave Platform for Rapid and Contactless Cell Patterning in Hydrogel	Myeonghun JANG, Yonsei University, Korea	POS8-1236
PS2_130	Development of a Lyophilized Self-Healing Adhesive Hydrogel Patch for the Treatment of Oral Mucositis	Youngbin KIM, Yonsei University, Korea	POS8-1237
PS2_131	Drug Releasing and Antifouling Coacervate Interface for Medical Implants	Young KIM, Seoul National University, Korea	POS8-1246
PS2_132	Responsive Moiré Hydrogel-IOL Platform for Intraocular Biomarker Monitoring of Neurodegenerative Diseases	Won Tack OH, Yonsei University, Korea	POS8-1253
PS2_133	Targeting ER Unfolded Proteins via Enzyme-Activated Disulfide Oligomerization for Selective Cancer Therapy	Min-Seok SEU, Ulsan National Institute of Science & Technology (UNIST), Korea	POS8-1256
PS2_134	Ir(III)-Based Photosensitizer Inducing Immunogenic Cell Death via Photodynamic Therapy	Jihyeon YUN, Ulsan National Institute of Science and Technology, Korea	POS8-1263
PS2_135	Cancer Prevention via Oral Delivery of Glycocholic Acid- and Mannose-Coated Nanoparticles	Aram KIM, The Catholic University of Korea, Korea	POS8-1265

PS2_136	Glycocholic Acid-Chondroitin Sulfate Conjugates-Coated Solid Lipid Nanoparticles for Oral Metronomic Chemotherapy	Sujeong HAN, The Catholic University of Korea, Korea	POS8-1271
PS2_137	Tailored Enzyme-Mimicking Diblock Polymer Brushes for Antifouling and NO-Mediated Antibacterial Interfaces	Seung Won YANG, Center for Biomaterials, Biomedical Research Institute, Korea Institute of Science and Technology (KIST), Korea	POS8-1299
PS2_138	Development and Characterization of Non-Cell Adhesive Acyl Glycol Chitosans for 3D Cell Culture	Hyein KANG, Chungnam National University, Korea	POS8-1321
PS2_139	Functional Polymer Coatings for Enhanced Electromediated Eradication of Staphylococcus Aureus Biofilms	Hanna ZHUKOUSKAYA, Institute of Macromolecular Chemistry, Czech Academy of Sciences (IMC), Czech Republic	POS8-1340
PS2_140	Injectable Glycol Chitosan Thermogels with Programmable Hydrolytic Degradation for Biomedical Applications	Daeun KIM, Chungnam National University, Korea	POS8-1367
PS2_141	Bioactive Composite Hydrogel Based on Porcine Skin-Derived Extracellular Matrix and Pectin Methacrylate for Wound Healing	Hosoo SHIN, Jeonbuk National University, Korea	POS8-1440
PS2_142	Visible Light-Photocrosslinked GelMA Hydrogel Enables Sustained Polydeoxyribonucleotide Delivery for Enhanced Cartilage Regeneration	Nakyung KIM, Jeonbuk National University, Korea	POS8-1450
PS2_143	Viscoelastic GelMA-based Nanocomposite Hydrogels for Regulating Cell Behavior and Drug Responses	Minha PARK, Kyungpook National University, Korea	POS8-1474
PS2_144	Influence of Photocrosslinking Mechanisms on the Gelation Kinetics of Methacrylated Silk Fibroin	Chaewon LEE, Kyungpook National University, Korea	POS8-1475
PS2_145	Construction of Hydrogels Incorporating Polymerized DNA Nanostructures	Heeyeon KIM, Yonsei University, Korea	POS8-1478
PS2_146	Enhanced Chemical Stability and Oral Bioavailability of Retinol Encapsulated in O/W Nanoemulsion Stabilized by Cycloamylose-Catechin Complexes	In-Geun JUNG, Seoul National University, Korea	POS8-1505
PS2_147	A Polymer Matrix-Supported 3D Co-Culture Model Recapitulating Amyloid- β Accumulation for Alzheimer's Drug Screening	Jiyeon LEE, Seoul National University, Korea	POS8-1516
PS2_148	Rational Head-Group Engineering of Ionizable Lipids for High-Performance mRNA Delivery	Viet Cuong HO, Infectious Disease Research Center, Korea Research Institute of Bioscience and Biotechnology, Korea	POS8-1524
PS2_149	Multifunctional Amphiphilic Hydrogel Composites Reinforced with Natural Illite for Skin-Adhesive Biomedical Patches	Hwain LEE, Konkuk University, Korea	POS8-1527
PS2_150	Direction-Dependent Stretch Drives Stage-Specific Remodeling in a COPD-Relevant Muscle Model	Taehoon LEE, POSTECH, Korea	POS8-1558
PS2_151	Swelling Behaviour and Phenolic Release from Buccal Films Based on Biodegradable Polymers and Acerola By-Products Extract	Leticia CARVALHO, University of São Paulo, Brazil	POS8-1585
PS2_152	Real-Time, Label-Free Monitoring of Tissue Stiffness in a Lung-on-a-Chip Under Breathing-Mimetic Cyclic Strain	Giheon HA, POSTECH, Korea	POS8-1586
PS2_153	Engineering of a Microneedle-Based Drug Delivery System via a Reaction-Diffusion Process	Eun Ji SON, Ajou University, Korea	POS8-1587
PS2_154	NO-Shielded T Cells for Enhanced Antitumor Immunotherapy	Chaerin SHIN, POSTECH, Korea	POS8-1630
PS2_155	Bioengineering a Needle-Free Intranasal Nanovaccine with TLR7/8 Agonist for Potent Antimetastatic Immunity	Byung Deok KIM, Sungkyunkwan Univ., Korea	POS8-1652

S9. Polymer Technology for Sustainability (PSK50 Student Presentation Award Applicants)			
PS2_156	Programmable Degradation and LCST Transitions in Lipoate Copolymers by Photoiniferter Polymerization of Isopropyl Lipoamide	Li An WANG, National Yang Ming Chiao Tung University, Taiwan	POS9-0014
PS2_157	Biobased Antifatigue Organohydrogel with Superior Ionic Conductivity Enabled by Lignin Nanostructures	Vu Thanh TRUONG, Stockholm University, Sweden	POS9-0034
PS2_158	Self-Catalytic Recycled Polyols for the Preparation of Sustainable Polyurethane Foams	Ya-Wen HSUEH, National Chung Hsiung University, Taiwan	POS9-0043
PS2_159	Photo-Healable and Stretchable Fibers from Upcycled TPEE and Azopolymers by Harnessing Solid-to-Liquid Photoisomerization	Yen Shen HSU, National Yang Ming Chiao Tung University, Taiwan	POS9-0072
PS2_160	Polarity-Rigidity Modulation of Bio-Based Copolyesters for Water-Responsive Transient Adhesives with Recyclability and Biodegradability	Yujin CHOI, Kyung Hee University, Korea	POS9-0085
PS2_161	Thermal Protection of Proteinase K via Silica Nanoparticle Immobilization and Trehalose Encapsulation for Accelerated PLA Biodegradation	Youngjin KIM, Kyung Hee University, Korea	POS9-0086
PS2_162	Hydrophobic and Flexible Vegan Leather Derived from Bacterial Cellulose Through Vegetable Oil Impregnation	Jungwon LIM, Kyung Hee University, Korea	POS9-0089
PS2_163	Biochar as an Enzyme Carrier for Biodegradation Enhanced PLA: Catalytic Activity Preservation Through Melt Extrusion	Soyeong LEE, Kyung Hee University, Korea	POS9-0096
PS2_164	Evaluation of Physical Property and Biodegradability of Immiscible PBAT/PHBH Blend Films	Atsunori ISHIDA, The University of Tokyo, Japan	POS9-0102
PS2_165	Development of Bio-Polyurethane Elastomers Through Chemical Modification of Poly(3-hydroxybutyric acid) via Alcoholysis	Ji-Heon KIM, Sogang University, Korea	POS9-0107
PS2_166	Segment-Selective Design of Zwitterionic Biodegradable Polyurethane with Tunable Service Lifetime	Younghwan AHN, Sogang University, Korea	POS9-0108
PS2_167	Enzyme Embedding for Improving the Biodegradability of poly(ethylene succinate) Melt-Spun Fibers	Kei ISHIDA, The University of Tokyo, Japan	POS9-0114
PS2_168	Direct Liquid Feeding of Nanocellulose via Melt Extrusion Enhancing the Mechanical Properties of Polybutylene Adipate Terephthalate (PBAT)	Hongrun CHEN, Stockholm Univeristy, Sweden	POS9-0117
PS2_169	Amine-Functionalized Nanocellulose Aerogels from Biomass Waste as Adsorbents for CO2 Capture	Yernazar KUAN, Korea Advanced Institute of Science and Technology, Korea	POS9-0149
PS2_170	Marine-Inspired Hydrogel Platform for Energy-Efficient Carbon Removal and Aragonite Recovery	Jeongsik PARK, Sogang University, Korea	POS9-0156
PS2_171	Comparison of Coating and Blending of Aramid Nanofiber for Flame Retardant Acrylate Plastics	Taehyeon KIM, Sogang University, Korea	POS9-0157
PS2_172	Entropy-Assisted Self-Catalyzed Polyurethane Vitrimer for Multifunctional Elastomers and Reactive Compatibilization of Mixed Urethane Waste	Haemin KIM, Sogang University, Korea	POS9-0158
PS2_173	Recovering Mechanical Properties of Recycled Polystyrene Through Long-Chain Branching	Yujin HA, Inha University, Korea	POS9-0170
PS2_174	Cardanol-Based Reprocessable Poly(urethane-urea) Foams via Hindered Urea Bonds	Ga Yeon AHN, Inha University, Korea	POS9-0171

PS2_175	Designing Closed-Loop Chemically Recyclable Polyurethane Foam	SeoJeong HUR, Inha University, Korea	POS9-0172
PS2_177	Comparative Analysis of Bulk and Surface Erosion Behaviors in Semi-Crystalline PLLA and Amorphous PDLLA Under Accelerated Enzymatic Conditions	Jaeyoung PARK, Korea University, Korea	POS9-0182
PS2_178	Diacid-Encoded Molecular Interactions in Bio-Based Poly(urethane-amide) Elastomers: Linking Hard-Segment Structure to Mechanical Performance	Giyeon JEONG, Sogang University, Korea	POS9-0199
PS2_179	Sustainable Synthesis of CO ₂ -derived Non-Isocyanate Polyurethane and Its Chemical Recycling via Amine-Mediated Depolymerization	Jihwan PARK, KAIST, Korea	POS9-0239
PS2_180	Sustainable Upcycling of Rice Bran into Nitrogen-Doped Carbon Nanodots with Enhanced Antioxidant Activity	Minah PARK, Korea University, Korea	POS9-0267
PS2_181	Graft Architecture-Controlled Toughening and Marine Biodegradation of PHB/CNC Nanocomposites	Sunoo HWANG, Korea Advanced Institute of Science and Technology, Korea	POS9-0273
PS2_182	Saponin-Assisted Fabrication of High-Performance and Antifouling Reverse Osmosis Membranes	Wonhee JO, Korea University, Korea	POS9-0299
PS2_183	Sustainable Fabrication of Gold Adsorbents from poly(vinyl chloride) Waste Plastics	Jiyun CHUNG, Korea University, Korea	POS9-0301
PS2_184	High-Performance Loose RO Membranes via DMSO Activation for PFAS Removal	Mingyu KIM, Korea University, Korea	POS9-0302
PS2_185	Fabrication of Chitin Membranes for dye/salt Separation	Chan Hee JUNG, Korea Univeristy, Korea	POS9-0305
PS2_186	Closed-Loop Recyclable Thermosets Enabled by Dynamic Fe(III)-Vicinal Diol Coordination and Metathesis Depolymerization	Seulchan LEE, Korea Advanced Institute of Science and Technology, Korea	POS9-0340
PS2_187	Environmental Fate and Risk Assessment of Biodegradable Polymer Intermediates	Sang Eun KANG, Chungnam National University, Korea	POS9-0354
PS2_188	Hydrogen-Bond-Assisted Chemical Recycling of Trithiocarbonate-Terminated PMMA via Diverse Alcohol Additives	Jaewon LEE, Soongsil University, Korea	POS9-0358
PS2_189	Circular Bio-Based Polymer Networks with Magnetic Responsiveness for Environmental Remediation	Francisca L. ARANDA, Universidad de Concepción, Chile	POS9-0363
PS2_190	Base-Catalyzed Degradation of Vinyl Polymers via Generation of Anionic Species from C-H Bonds on the Backbones	Tomoki WATANABE, Nagoya University, Japan	POS9-0388
PS2_191	Mechanochemical Atom Transfer Radical Polymerization Unlocks Functional Methacrylate Lifecycles of Controlled Synthesis, Force-Activated Degradation, and Depolymerization	Gyeongjin KWON, Yonsei University, Korea	POS9-0441
PS2_192	Magnetic Membranes as Innovative Materials for Bioethanol Dehydration	Lukasz JAKUBSKI, Silesian University of Technology, Poland	POS9-0446
PS2_193	Solvent-Minimized Esterification of Cellulose Nanofibrils for Superhydrophobic Biofillers in Biodegradable Polymer Composites	Pham Thanh Trung NINH, KAIST, Korea	POS9-0448
PS2_194	Design and Manufacturing of Sustainable High-Performance Biopolymer Films Reinforced by Architected Particle Networks	Byeunggon KIM, North Carolina State University, United States	POS9-0471
PS2_195	Sustainable Mass-Culture System for Natural Polymer-Based Rice Cultured Meat	Bumgyu CHOI, Yonsei University, Korea	POS9-0489

PS2_196	Upcycling Low-Molecular-Weight Polyethylene into High-Performance Hot-Melt Adhesives via Dynamic Boronic Ester Networks	Naon CHOI, Pusan National University, Korea	POS9-0507
PS2_197	Surface Defect Repair of Recycled Carbon Fibers Using Carbon Dots for Composite Applications	Donguk KIM, Ulsan National Institute of Science and Technology, Korea	POS9-0508
PS2_198	Structure-Property Relationships in Reprocessable Microphase-Separated polar/nonpolar Copolymers Based on Diels-Alder Chemistry	Kyung Rok HAN, Kyungpook National University, Korea	POS9-0533
PS2_199	Scalable Fabrication of Uniform Poly(ethylene imine)- Based Hydrogel Particles	Eunsuk HWANG, Pohang University of Science and Technology (POSTECH), Korea	POS9-0548
PS2_200	PET/PBT/PTMEG Random Compatibilizers for PET/PBT/TPU Ternary Blends	Junyong SEO, Korea University, Korea	POS9-0550
PS2_201	Upcycling Silicon Sludge into LiDAR-Detectable Organic-Inorganic Hybrid Black Coatings	Suk JEKAL, Inha University, Korea	POS9-0552
PS2_202	Fiber-Grade Polyurethane Elastomers with Green-Solvent Reprocessability and Mechanical Robustness Through Dynamic Covalent Topology Engineering	Minjeong KIM, Korea University, Korea	POS9-0590
PS2_203	Catalyst-Free Covalent Adaptable Polyurethane Foams for Closed-Loop Recycling and Programmable Mechanical Performance	Jiyun HAN, Pusan National University, Korea	POS9-0747
PS2_204	Lignocellulosic Biomass-Based PVA Composite Films for Flexible Ammonia Sensors	Aun ABBASS, Korea University of Technology and Education, Korea	POS9-0755
PS2_205	Adaptive Soft Electronics Platform Based on Reprocessable Dynamic Polymer Frameworks	Jihun NOH, Sungkyunkwan University, Korea	POS9-0765
PS2_206	Development of PLA/Pulp Biocomposites for Daytime Radiative Cooling in Building Exteriors	Seunghyuk HONG, Chung-Ang University, Korea	POS9-0769
PS2_207	Fabrication of Dual-DNA-Functionalized Janus Particles via Organic Emulsion-Based Masking	Keonwoo LEE, POSTECH, Korea	POS9-0817
PS2_208	Degradation Behavior of PLA, PHA, PBS, and PBAT by Marine Tidal Flat-Derived Marinobacterium Maritimum	Jihyeon JUNG, Incheon National University, Korea	POS9-0880
PS2_209	Barcode-Encoded Superabsorbent Polymer Microparticles for Simultaneous Biomarker Enrichment and Multiplex Analysis	Hyojeong KIM, Incheon National University, Korea	POS9-0881
PS2_210	Sustainable Passive Radiative Cooling Films Based on Variable-Stiffness Poly(thiourethane) Vitrimers	Seonuk KIM, Chungang University, Korea	POS9-0893
PS2_211	Empirical Study on the Anaerobic Degradation and Biogasification of Biodegradable Plastics via Co-Digestion with Organic Waste	Hwicheol SHIN, Incheon National University, Korea	POS9-0968
PS2_212	Chitin from a Zero-Waste Shrimp Shell Biorefinery as a Building Block for Advanced Sustainable Materials	Yasmini PORTES, Stockholm University / WISE Materials, Sweden	POS9-0982
PS2_213	Star-Polyelectrolyte Interfaces Regulate Ion-Water Transport for Continuous Solar-to-Vapor Conversion in Hypersaline Brines	Jie ZHU, Fudan University, China	POS9-1082
PS2_214	Quaternary Ammonium Tetrachloroferrate Ionic Liquids as Thermo-Responsive, Magnetically Recoverable, and Antibacterial Multifunctional Draw Solutes for Forward Osmosis	Aida ZHAMANSHALOVA, Dong-A University, Korea	POS9-1143
PS2_215	Synthesis of High-Performance Fluoroalkyl-Centered Biodegradable Polymers with Outstanding Transparency and Mechanical Properties	Hyebhin YOON, Korea Research Institute of Chemical Technology, Korea	POS9-1232

PS2_216	Balancing Strength and Ductility in Poly(butylene succinate) Copolyesters Through BHET Incorporation	Sang Uk PARK, Ajou University, Korea	POS9-1240
PS2_217	Depolymerization of Boron-Pendant Polymers Triggered by Chain-End Activation	Satoshi AKABANE, Kyoto University, Japan	POS9-1306
PS2_218	Introduction of Ether Bonds as a Design Strategy for Sustainable Nylon	Chen GU, Institute of Industrial Science, The University of Tokyo, Japan	POS9-1319
PS2_219	Effect of LCB Density on Strain Hardening Behavior and Processing Stability of PHA/LCB-PBAT Blends	Jiseok JUNG, Pusan National University, Korea	POS9-1408
PS2_220	MXene-Integrated Alginate-Salt Hydrogel Beads for Solar-Driven Atmospheric Water Harvesting	Tae Hui PARK, Chungnam National University, Korea	POS9-1427
PS2_221	Disulfide-Induced Interparticle Adhesion of Crystalline Cellulose	Subin LEE, Chung-Ang University, Korea	POS9-1459
PS2_222	Dihydroxy-Functionalized Aliphatic Polycarbonates: Geometry of Hydroxy Groups Dictates Their Hydrolysis Behavior	Hyungju LEE, Kyoto Institute of Technology, Japan	POS9-1469
PS2_223	Sustainable Fabrication of Cellulose Organic Solvent Nanofiltration Membranes Modified with Metal-Phenolic Networks	Jaehee LEE, Kyung Hee University, Korea	POS9-1506
PS2_224	Mechanical Processing of Cellulose from Spent Coffee Grounds for Extrusion-Based 3D Printing	Sewon PARK, Chung-Ang University, Korea	POS9-1520
PS2_225	Design of Hydrocarbon Polymer Exchange Membranes with Tunable Interstitial Alkyl Chain Lengths for Enhanced Electrochemical Performance	Yerim LEE, Korea Institute of Science and Technology, Korea	POS9-1619
S10. AI-assisted Design and Simulation of Polymers (PSK50 Student Presentation Award Applicants)			
PS2_226	Ion Transport Mechanisms in Polymer Electrolytes Confined Within Covalent Organic Framework Nanochannels	Jaebeom JANG, Dongeui University, Korea	POS10-0218
PS2_227	Multi-Fidelity Bayesian Optimization for Cost-Efficient Simulation-Based Design: A Study on Optimization of Block Copolymer Directed Self-Assembly	Seungha JEONG, Kyung Hee University, Korea	POS10-0230
PS2_228	Molecular Dynamics-Based Property Prediction and Screening of PFAS-Free Binders for Lithium-Ion Battery Cathodes	Yu Sung OH, Kyung Hee University, Korea	POS10-0231
PS2_229	Theoretical Study of Thermoplastic/Thermoset Blends Incorporating Covalent Adaptable Networks	Gyudong NOH, Dong-Eui University, Korea	POS10-0238
PS2_230	An Informatics Approach for Polymer Dielectric Constants	Hyuntae PARK, Seoul National University, Korea	POS10-0542
PS2_231	Matched-Pair MD Analysis of Functional-Group Effects in PVA-Inspired Cryoprotective Polymers	Jinhoon KIM, Seoul National University of Science and Technology, Korea	POS10-0612
PS2_232	Dissipative Particle Dynamics Simulation of Sequence-Dependent Micellar Network Formation in PLGA-PEG-PLGA Hydrogels	Sangheon LEE, Seoul National University of Science and Technology, Korea	POS10-0613
PS2_233	Molecular Dynamics Simulation Study on Binding Affinity of Integrin and RGD-QWNT for Enhanced Biosensor Performance	Junhee KIM, Korea University, Korea	POS10-0768
PS2_234	Multiscale Simulation of Concentration-Dependent Molecular Clustering and Its Role in the Air-Stability of Viologen Analytes for AORFB	Yoonku LEE, Korea University, Korea	POS10-0864

PS2_235	Comparative Molecular Dynamics Study of Hydrogen-Bond Networks in Hydrogel and Glycolgel Systems	Yedam KANG, Korea University, Korea	POS10-0921
PS2_236	Iterative Electrochemical Doping Simulations in OECTs	Seonghyeon KANG, Seoul National University, Korea	POS10-0986
PS2_237	Toward Agentic Polymer Data Factories for Scalable Foundation Model Training	Ayodele FALETI, Seoul National University, Korea	POS10-1000
PS2_238	Molecular Simulation of Positive-Tone Tin-Oxo Resists Based on Lewis Acid-Base Interactions for EUV Lithography	Sehyun YUN, Daegu Gyeongbuk Institute of Science and Technology, Korea	POS10-1080
PS2_239	Competing Local and Network-Scale Associations in Salt-Screened Coacervate Adhesives	Raehyun JUNG, DGIST (Daegu Gyeongbuk Institute of Science and Technology), Korea	POS10-1279
PS2_240	Data-Driven Screening of Anion Exchange Membranes Using Quantum Chemical and Structural Descriptors of Extended Ion Exchange Groups	Nana TERASOBA, Tokyo Metropolitan University, Japan	POS10-1309
PS2_241	Computational Study on Vanadium Nitride-Enabled Bidirectional Polysulfide Redox in Sodium-Sulfur Batteries	Seongwon LEE, Hanyang University, Korea	POS10-1402
PS2_242	Molecular-Level Design of Li-Free Dopants for Space-Grade Perovskite Solar Cells	Dong Gyu LEE, Hanyang University, Korea	POS10-1457
PS2_243	Molecular-Level Understanding of Dual-Electrode Interphase Stabilization by a Molecularly Functionalized Separator	Jiyeon LEE, Hanyang University, Korea	POS10-1463
PS2_244	Computational Insights into p-Type Doping and Sn ²⁺ Protection by Bifunctional Polymer in Tin-Lead Perovskites	Sang Wook PARK, Hanyang University, Korea	POS10-1465
PS2_245	Mechanistic Insights into Boron-Regulated Solvation Structure and Li ⁺ Transport in Lithium Metal Batteries via Molecular Simulations	Yubhin CHO, Hanyang University, Korea	POS10-1470
PS2_246	Molecular Insights into Temperature-Dependent Interfacial Decomposition of Elastomeric Electrolytes in Lithium Metal Batteries	Junsu SON, Hanyang University, Korea	POS10-1547
S1. Polymer Synthesis			
PS2_247	"Optimizing Miniemulsion Parameters for Controlled ATRP Polymerization: A Study on Stability and Surfactant/Ligand Effects	Ahmad SHAABAN, Egyptian Petroleum Research Institute (EPRI), Egypt	POS1-0010
PS2_248	Toward Sustainable Polyester Synthesis: A Mild Aqueous Route from α,ω -Diols Using a Recyclable Bromide System	Lou DELUGEAU, LCPO - Bordeaux University - CNRS, France	POS1-0028
PS2_249	Novel Epoxy-Amine Vitrimers with Built-in Catalytic Functionality	Alexander SHAPLOV, Luxembourg Institute of Science and Technology (LIST), Luxembourg	POS1-0129
PS2_250	Synthesis of Urethane Acrylate by Introducing Acrylic Groups to Enhance the Wear Resistance of Bio-Based Polyurethane	Hyun Ju PARK, KIMCO, Korea	POS1-0143
PS2_251	Design of Ferulic Acid-Derived Liquid Crystalline Epoxy Resins	Linh PHAM, Kyungpook National University, Korea	POS1-0245
PS2_252	Synthetic Simplicity in Organic Electronics: From Polymer Design to Post-Polymerisation Functionalisation	Martina RIMMELE, Imperial College London, United Kingdom	POS1-0259
PS2_253	Amide-Containing Methylene Malonamates for Photoredox Decarboxylative Construction of Sequence-Defined Vinyl Architectures	Ju Eun KIM, Seoul National University, Korea	POS1-0360

PS2_254	Synthesis of Sequence Ensemble-Controlled Copolymers by Decoupling Monomer Reactivity with Sequence Determination Steps	Sung Han PARK, KAIST, Korea	POS1-0461
PS2_255	Toward Automation-Guided Discovery of Ion-Active Polymers	Youngho KIM, KAIST, Korea	POS1-0480
PS2_256	Synthesis of Sequence-Controlled Oligomers Using Living Anionic Addition Reaction of 1,1-diphenylethylene Derivatives Possessing Protected Formyl Groups	Tomono ARICHIKA, Institute of Science Tokyo, Japan	POS1-0564
PS2_257	Anionic Polymerization of Functional Styrenes in Hydrocarbon Solvent	Yota SANO, Institute of science tokyo, Japan	POS1-0596
PS2_258	Intrinsic Chain-End Stabilization Enables Chain-Transfer-Agent-Free Living Cationic Ring-Opening Polymerization of Cyclic Enol Ethers	Semin SON, Ajou University, Korea	POS1-0627
PS2_259	Molar Mass- and Sequence-Defined Vinyl Macromolecules via Iterative Photoredox Decarboxylative Addition of α -substituted Acrylates	Dongkyu KIM, Seoul National University, Korea	POS1-0635
PS2_260	High-Refractive-Index Polymers Derived from Chalcogen-Containing Monomers via SuFEx Polymerization	Wooyeol CHUNG, Seoul National University, Korea	POS1-0640
PS2_261	Encoding Nucleobase Sequences to Synthetic Polyesters Using adenine/thymine-containing Isocyanides as Gluing Components for the Passerini IEG	Hyojoo NOH, Seoul National University, Korea	POS1-0659
PS2_262	On the Origin of Polymer-Specific Polarity Inversion in Heavily Doped Conjugated Polymers	Landep AYUNINGTIAS, Gyeongsang National University, Korea	POS1-0669
PS2_263	Simultaneous Improvement of Melt Processability and Ductility in Poly(butylene succinate-co-terephthalate) via In-Situ Esterification with Kraft Lignin	Taekyeong PARK, Chungnam National University, Korea	POS1-0678
PS2_264	Scalable Synthesis of CO ₂ -Derived Bis-Cyclic Carbonates for Sustainable Non-Isocyanate Polyurethane (NIPU)	Daewon BANG, Korea Research Institute of Chemical Technology (KRICT), Korea	POS1-0698
PS2_265	Hyper-NA EUV Patterning Through Morphology-Controlled High-Endurance Metal Containing Block Copolymers	HyoWon LEE, Korea Advanced Institute of Science and Technology, Korea	POS1-0724
PS2_266	Achieving Stable Ambipolar and Unipolar Transport in NDI-DPP-Thiophene Copolymers for Organic Transistors	Karina AYU LARASATI, Gyeongsang National University, Korea	POS1-0726
PS2_267	Polymerization of Alkyl-Substituted [3]dendralenes	Haruto TANAKA, Institute of Science Tokyo, Japan	POS1-0763
PS2_268	Precision Synthesis of Conjugated Multiblock Copolymers via Living Polymerization: Molecular Architecture Control for Enhanced Stretchability and Charge Transport	Hee-Seong YANG, Ajou University, Korea	POS1-0781
PS2_269	Molecular Engineering of indolocarbazole-silane Hosts for Blue Phosphorescent Organic Light-Emitting Diodes with Improved Thermal Stability and Efficiency	Chen GUANCHAO, Gyeongsang National University, Korea	POS1-0784
PS2_270	Chemical Recycling of Multiblock Copolymers to Polyols via Degradable Acetal Junctions	Sunhyeong HONG, Ajou University, Korea	POS1-0785
PS2_271	Topology-Dependent Swelling and Mechanics of PNIPAM Hydrogels	Suraj ASWALE, Pusan National University, Korea	POS1-0799
PS2_272	Rubbery OCR with High Clarity for Foldable Displays via In-Situ Photopolymerization	Sangbin SHIN, Korea Research Institute of Chemical Technology, Korea	POS1-0818
PS2_273	Iterative Photoredox Decarboxylative Addition of Acrylates to High-Molecular-Weight Sequence-Defined Polyolefins	Gyeonghun JEONG, Seoul National University, Korea	POS1-0857

PS2_274	Photoresponsive Fuel Molecules for Energy-Embodied	Jiwoo KIM, Chonnam National University, Korea	POS1-0875
PS2_275	Self-Assembly of Pyridinic Conjugated Mesoporous Networks into Functional Porous Monoliths	Jinuk JEONG, Chonnam National University, Korea	POS1-0876
PS2_276	Depolymerizable Polyurethane Hot-Melt Adhesives for Controlled Debonding	Yewon KANG, Chonnam National University, Korea	POS1-0877
PS2_277	Metal-Free Photocatalytic Aryl Radical Coupling for Donor-Acceptor Molecules	JaeHyuk JANG, DGIST, Korea	POS1-0942
PS2_278	Selenophene-Substituted BDT-Based Copolymers: Implications for Molecular Packing and Charge Transport	Violeta Aprilia PRAMITANIA, Gyeongsang National University, Korea	POS1-0969
PS2_279	Ester-Substituted Thiophene Incorporated PM6 Based Copolymers for Ternary Organic Solar Cells	Donny Maruli PASARIBU, Gyeongsang National University, Korea	POS1-0971
PS2_280	Rational Design and Synthesis of a DTBDT-Based Polymeric Guest Donor for Efficient Ternary Organic Solar Cells	Jinwoong HONG, Gyeongsang National University, Korea	POS1-0981
PS2_281	Molecularly Defined Helical Rod-Coil Amphiphiles Reveal the Role of PEG Topology in Self-Assembly Block Copolymers	Seung Hyun BAEK, Seoul National University, Korea	POS1-1022
PS2_282	Nitrile- and Ether-Rich Covalent Organic Framework Nanofillers for High-Performance SPEEK-Based Proton Exchange Membranes	Jiyong CHOI, Seoul National University, Korea	POS1-1112
PS2_283	Vapor-Deposited Polyimide Thin Films via Copolymerization of Functionalized Parylene Dimers	Chaeyeong SON, Chungnam National University, Korea	POS1-1141
PS2_284	Biodegradable Parylene-Coated Paper Straw for Sustainable Paper Applications	Minyeong CHOI, Chungnam National University, Korea	POS1-1145
PS2_285	CO ₂ Carbonylation to Cyclic Carbamate Monomer for Biodegradable Polymer Synthesis over Metal Supported Ceria	Nagendra KULAL, King Fahd University of Petroleum and Minerals, Saudi Arabia, Saudi Arabia	POS1-1162
PS2_286	Exposure-Dependent Solubility Switching in Photoacid Generator-Free, Positive-Tone Photoresists Based on Imidazolium-Mn(II) Carboxylate Coordination	Seung Hoon LEE, Kyungpook National University, Korea	POS1-1167
PS2_287	Synthesis of Carboxyl-Functionalized PI-TPU/Ionic Liquid Composites for Flexible Iontronic Pressure Sensors	Changhee CHAE, Chungnam National University, Korea	POS1-1215
PS2_288	Cerium Ions-Coordinated 2D Crown Ether Framework as a Radical Scavenging Filler for PEMFC Membranes	Min Hong LEE, Seoul National University, Korea	POS1-1287
PS2_289	Molecularly Tunable PHSt-Based Polymers Through Controlled Protection and Functionalization for Photoresist Applications	Dongjin SHIN, Chungbuk National University, Korea	POS1-1294
PS2_290	Chemically Multifunctional Covalent 2D Crown Ether Framework	Dong Joo SEO, Seoul National University, Korea	POS1-1305
PS2_291	Synthesis and Mechanochemistry of Poly(bicyclo[n.2.0]alkenes)	Moritz HANSEN, Stanford University, United States	POS1-1345
PS2_292	Substituted Group Effects in Liquid Jones Oxidation of Primary Alcohols: Foundation for Eco-Friendly Conversion of Seaweed-Derived 3-HP to Malonic Acid	Yunseul PARK, Kongju National University, Korea	POS1-1356
PS2_293	TBAB-Assisted Organic-Flux Crystallization of Pyromellitic Acid-Based Polyimide Covalent Organic Frameworks(PI-COFs)	Myeongjae LEE, Korea Institute of Science and Technology (KIST), Korea	POS1-1372

PS2_294	Hydroxyl-Functionalized Dianhydride-Based Polyimide Membranes for Hydrogen Purification	Sangin YUN, Korea Institute of Industrial Technology (KITECH), Korea	POS1-1384
PS2_295	Aminotriazole Modified PE-g-GMA Filter for Selective Metal Ion Adsorption	Jung-Nam LEE, Korea National University of Transportation, Korea	POS1-1394
PS2_296	Preparation and Characterization of Hydrogel Materials Containing Methacryloxypropyl Polyhedral Oligomeric Silsesquioxane (MA-POSS) via Electron-Beam Irradiation	Minjae KIM, Korea National University of Transportation, Korea	POS1-1397
PS2_297	Formation of PPV Thin Films via CVD for Potential Application as Hole Transport Layers in Perovskite Solar Cells	Won Kyung PARK, Chungnam National University, Korea	POS1-1399
PS2_298	Synthesis of Metal-Free Low-Molecular-Weight Poly(2,6-dimethyl-1,4-phenylene oxide) and Methacrylic Anhydride End-Capping for Enhanced Processability and Resin Miscibility	So Hyun PARK, Korea Institute of Industrial Technology; Sung Kyun Kwan University, Korea	POS1-1413
PS2_299	Castor Oil-Derived C7 and C18 Platform Monomers for Biomass Resource Carbon (BRC)-Based Polymer Synthesis	Sunil KUMAR, Kangwon National University, Korea	POS1-1420
PS2_300	Molecular Structure Engineering of PVDF-Based Polymers for High-Performance Electroactive Materials	Hajin KWON, KRICT, Yonsei Univ., Korea	POS1-1425
PS2_301	Porosity Controlled Polyimide Separators via NIPS for Improved Electrolyte Affinity and Ionic Conductivity	Min Jae KIM, Inha University, Korea	POS1-1428
PS2_302	Effect of Composition Ratios on the Long-Lasting Antifogging Properties of Amphiphilic Ladder-like Polysilsesquioxanes	Hyeonho NOH, Korea National University of Transportation, Korea	POS1-1430
PS2_303	Synthesis of Photoresist Polymers Through a Multistep Flow Process Involving Copolymerization, End-Group Removal, and Deacetylation	Gyeonghui KIM, Inha University, Korea	POS1-1431
PS2_304	Preparation of Hydrophobic Silicone and Hydrophilic Acrylic Hybrid Materials via Electron Beam Irradiation for Medical Adhesives	Hyeonho NOH, Korea National University of Transportation, Korea	POS1-1435
PS2_305	Well-Dispersed AgNP Immobilization on APTES-Modified Zinc Silicate Glass for Durable Antibacterial PLA Composites	Jongjin PARK, Seoul National University, Korea	POS1-1449
PS2_306	The Power of High Pressure: Visible-Light-Mediated and Oxygen-Tolerant ATRP for the Synthesis of Ultra-High Molecular Weight Polymers	Roksana WYGODA, University of Silesia in Katowice, Poland	POS1-1507
PS2_307	Thermally Conductive Acrylic Thermosets Based on a Star-Shaped Aromatic Curing Agent	Soo Jeong JEONG, Sookmyung Women's University, Korea	POS1-1575
PS2_308	Effects of Silane-Based Interfacial Modifiers on the Interfacial and Dielectric Properties of Glass Fiber-Polymer Composites	Juyoung SHIN, Sookmyung Women's University, Korea	POS1-1579
PS2_309	Synthetic Progress Toward PBLG-PLA Alternating Rod-Coil Polymers with Discrete and Disperse α -Helical Rod Segments	Yeon Hee CHO, Synthetic Polymer Chemistry LAB, Seoul National University, Korea	POS1-1600
PS2_310	Novel Low-Band Gap Non-Fullerene Acceptors Based on IDIC Core as Potential Photovoltaic Materials	Radhiha RAVINDRAN, Gyeongsang National University, Korea	POS1-1630
PS2_311	Controlled C-N Bond Construction in Polymers via Post-Polymerization Amidation Strategies	Seung Beom BAEK, Institute for Basic Science, Korea	POS1-1665
S2. High-End Characterization/Polymer Physics/Properties			
PS2_312	Composition-Dependent Properties of Colorless and Transparent Copoly(amide imide)s for Advanced Flexible and Optoelectronic Devices	Yun Sang SHIN, JeonJu University, Korea	POS2-0188

PS2_313	Molecular Design of Sulfone-Containing Colorless and Transparent Polyimides	Yun Sang SHIN, JeonJu University, Korea	POS2-0189
PS2_314	Re-Entrant Porous Cellulose Structures of Reversible Networks Prepared by LCST-Induced Shrinkage	Taein LEE, Chung-ang University, Korea	POS2-0197
PS2_315	Layer-by-Layer Morphology of Spray-Deposited Polymer Blends	Camille MAIRHOFER, Fibre and Polymer Technology, KTH Royal Institute of Technology, Sweden	POS2-0424
PS2_316	Design of Modified Silicone Oils for Advanced Heat-Resistant and Wear-Resistant Grease Lubricants	Jeong Yeon YOO, Inha University, Korea	POS2-0620
PS2_317	Toward Backbone-Degradable Hydrogels with Enhanced Mechanical Performances via Step-Growth Polymerization	Jaehyeok IM, Seoul National University, Korea	POS2-0775
PS2_318	Accurate Tg Characterization and Enthalpy Relaxation Mapping of Aerospace Epoxies via Modulated DSC (MDSC)	Taeyee KIM, Waters Materials Sciences, Korea	POS2-0787
PS2_319	Development of Polyurethane Coating Formulations with Enhanced Impact Resistance for Display Panel Applications	Jiye KWON, Korea Research Institute of Chemical Technology, Korea	POS2-0823
PS2_320	Rigid DOBNA-Based Host Engineering for Efficient Energy Transfer and Long-Lifetime Red Solution-Processed OLEDs	Yeonjoo LEE, Gyeongsang National University, Korea	POS2-0987
PS2_321	Molecular Design of Low-Dielectric Polyimides Based on Spiro Diamines	Seong Jae HONG, Yeungnam University, Korea	POS2-1099
PS2_322	Extreme Energy Absorption in Thin-Film and Mat Materials Under Ultra-High-Rate Micro-Projectile Impact	Jinho HYON, Agency for Defense Development, Korea	POS2-1119
PS2_323	Radiation-Induced Free Volume Evolution in Cyanate Ester Resin Characterized by Positron Annihilation Lifetime Spectroscopy	Youngsu JEONG, Korea Atomic Energy Research Institute / HANARO Utilization Division, Korea	POS2-1195
PS2_324	Unexpected Two-Step Crystallization Driven by Isolated In-Chain Groups in Polyethylene	Nontarin ROOPSUNG, Institute of Industrial Science, The University of Tokyo, Japan	POS2-1209
PS2_325	When Phase Separation Is Not the End: Hidden Interfacial Dynamics of Aqueous Poly(N-isopropylacrylamide)	Katarzyna Fedoruk PISKORSKA, University of Silesia in Katowice, Poland	POS2-1229
PS2_326	From Helices to Superhelices Through Hierarchical Assembly Across Competing Pathways	Kyeongim HONG, ICMM-CSIC, Korea	POS2-1617
S3. Processing / Fabrications (Emerging Horizons in Polymer Processing and Fabrication)			
PS2_327	New Process of Polymerization Engineering	Oludotun DIPEOLU, National Open University of Nigeria, Nigeria	POS3-0028
PS2_328	Development and Characterization of Recycled Polyolefin Blends for Profile Filler in Wet-Type Submarine Cable	Young Seob JONG, Intechnology Co., Ltd., Korea	POS3-0115
PS2_329	Effects of Polyamide 12 Content on the Morphological, Thermal, and Mechanical Properties of Polyolefin Compound	Young Seob JONG, Intechnology Co., Ltd., Korea	POS3-0116
PS2_330	Fabrication of Shape Erasable Liquid-Core Hydrogel Fiber	Yunjin LEE, Daegu Gyeongbuk Institute of Science & Technology, Korea	POS3-0203
PS2_331	Support-Polymer Tg as a Design Parameter for Pore Stabilization in Electrospun Preceramic Fibers	Seung-O CHOI, Soongsil University, Korea	POS3-0309

PS2_332	Automated In-Situ Characterization of Lipid Nanoparticles via Computer Vision-Based Flow RiboGreen Assay	Gi-Su NA, POSTECH, Korea	POS3-0314
PS2_333	Reprocessable xPDMS/h-BN Cooling Composites Enabled by Dynamic Siloxane Bond Exchange	Daeyul KWON, Chung-Ang University, Korea	POS3-0541
PS2_334	Graphene Oxide Passivation of Conductive Pathways in Sulfur-Rich Polymer/MXene Composites for High-Output Triboelectric Nanogenerators	Sungsu KIM, Hanyang University, Korea	POS3-0655
PS2_335	Tunable Interfacial Compatibilization of Poly(3-Hydroxybutyrate-co-3-Hydroxyhexanoate)/Thermoplastic Starch Blends via the Calcium-Induced Egg-Box Formation of Methylated Pectin	Seungjou NA, KITECH, Korea	POS3-0667
PS2_336	Influence of Biaxial Stretching Ratio on the Mechanical and Oxygen Barrier Properties of TPS/PBAT/Talc Biodegradable Films	Jaebin LEE, KITECH, Korea	POS3-0679
PS2_337	A Solvent-Free Core-Shell Encapsulation Strategy for High-Performance RF Substrates	Hyungkyu KIM, KIST, Korea	POS3-0739
PS2_338	Enhanced Impregnation of CNT Film-Based Composite via Electrochemical Processing	Tae Kyeom KIM, Pusan National University, Korea	POS3-0805
PS2_339	Effect of Recycled Nylon and Phosphorus-Based Flame Retardant Content on the Thermal, Physical, and Flame Retardant Properties of Recycled Nylon Flame-Retardant Composites	Jeong Min LIM, Korea Research Institute of Chemical Technology, Korea	POS3-0811
PS2_340	Effects of Varying-Sized Layered Double Hydroxide Particles on Thermal, Mechanical Properties, and Cytotoxicity of Polyvinyl Chloride Compound	Jieun LEE, KIMCO, Korea	POS3-0871
PS2_341	3D-Printed Mold Optimization for Precise Microcellular Structure Control and Enhanced Elastomeric Performance in Physical Foaming Process	Jieun LEE, KIMCO, Korea	POS3-0954
PS2_342	Programming Multimodal Actuation Under Constant Light in Spirally Aligned Liquid Crystal Polymers Through Geometric Design	Jaegyeong LEE, Hanyang University, Korea	POS3-0973
PS2_343	Effect of Activation on the Electrochemical Performance of Shellac/PAN-Derived Carbon Nanofibers	Min Seo KIM, Kumoh National Institute of Technology, Korea	POS3-1033
PS2_344	Characterization of Mechanical and Thermal Properties of Polyamide-Based Composites for Automotive Electronic Components	Hyun Ju PARK, KIMCO, Korea	POS3-1062
PS2_345	Single-Step Macroscopic Alignment of Block Copolymer Nanopatterns via Convective Self-Assembly	HongGu KWON, Kongju National University, Korea	POS3-1161
PS2_346	Flexible Polymer Micropattern with Thermochromic Property	Jaeyu LEE, Chungnam National University, Korea	POS3-1227
PS2_347	High-Pressure CO ₂ Texturing of PMMA/CQD/Ti ₃ C ₂ T _x Composite Films: Surface Hydrophobicity and Retained Sensing Function	Shih Han WANG, National Yunlin University of Science and Technology, Taiwan	POS3-1540
PS2_348	In-Situ ZnO-mediated ZIF-8 Growth on Porous Cellulose Microspheres Fabricated via Membrane Emulsification	Mingyeong KIM, Korea Institute of Industrial Technology, Korea	POS3-1591
PS2_349	Interface Engineering of MXene for Multi-Responsive Intelligent Electronic Platforms Using Structurally Robust Korean Traditional Paper	Dineshkumar MANI, Hanyang University, Korea	POS3-1637
S4. Colloids, Interfaces, and Molecular Assemblies for Functional Soft Materials			
PS2_350	Interfacial Structural Design of Charged Polymer Nanoparticles and Their Functional Applications	Xiayun HUANG, Fudan University, China	POS4-0007

PS2_351	Tunable Self-Wrinkling Polymer Films for Anti-Glare LED Surfaces Enabled by Fuming-Induced Interfacial Polymerization	Han Yu HSUEH, National Chung Hsing University, Taiwan	POS4-0023
PS2_352	Tuning the Mechanics of Capillary Suspension Gels via Particle Bidispersity	Ahmed JARRAY, Multi Scale Mechanics (MSM), MESA+ Institute for Nanotechnology, University of Twente, Netherlands	POS4-0034
PS2_353	Metamaterials from Plasma-Treated Block Copolymer Monolith	Chien CHEN, National Tsing Hua University, Taiwan	POS4-0037
PS2_354	Plumber's Nightmare Phase from Self-Assembly of Polylactide-Containing Asymmetric Triblock Terpolymers	Pin Chia CHEN, National Tsing Hua University, Taiwan	POS4-0038
PS2_355	DIRECTED ASSEMBLY OF LARGE-SIZED GOLD NANOPARTICLES BY BLOCK COPOLYMERS: FABRICATION AND PHOTOTHERMAL PROPERTIES	Haiying HUANG, State Key Laboratory of Polymer Science and Technology, Changchun Institute of Applied Chemistry, Chinese Academy of Sciences, China	POS4-0125
PS2_356	Post-Polymerization Programming of Polyurethane Thermogels with Tunable Gelation and Oxidation-Responsive Disassembly Behaviors	Haoxiang ZENG, The University of Sydney, Australia	POS4-0159
PS2_357	Electrospun Adsorptive Membranes for Efficient CMP Wastewater Treatment	Jaewook LEE, POSTECH, Korea	POS4-0167
PS2_358	3D Printing Using Water-Dispersible Photoinitiators and Their Removal via Post-Processing	Jae Hyeon HEO, Kumoh National Institute of Technology, Korea	POS4-0278
PS2_359	Anisotropic Liquid Crystal Networks for Optical Applications	Kwang-un JEONG, Jeonbuk National University, Korea	POS4-0298
PS2_360	Dual-Layer UV-Curable Anti-Slip Coatings for Enhanced Adhesion and Surface Safety of Steel Structures	Minseok ROH, Kumoh National Institute of Technology, Korea	POS4-0321
PS2_361	Silver Microcubes as SEIRAS Substrate	Seoyoung OH, Kumoh National Institute of Technology, Korea	POS4-0327
PS2_362	Folding Kinetics Controlled Soft Actuators for Flexible Display Applications	Giyeon HAN, Chung-Ang University, Korea	POS4-0371
PS2_363	Bio-Templated Chiral Metal-Organic Framework for Enantioselective Chemoresistive Sensing	Minkyu KIM, Dankook University, Korea	POS4-0374
PS2_364	Salt-Induced Silica Nanoparticle Aggregation Decouples Mechanical Reinforcement and Ionic Transport in Polymer Nanocomposite Electrolytes	Seunghan YUN, Seoul National University, Korea	POS4-0403
PS2_365	Revealing Hidden Confinement-Induced Packing Transitions in Spherical Block Copolymer Thin Films by Suppressing Terracing	Sung Kwan TAE, Institute of Chemical Processes, Seoul National University, Korea	POS4-0458
PS2_366	Sequence-Defined Oligoureia-Peptide Hybrids as Self-Assembling Platforms for Early Diagnosis of Neurodegenerative Diseases	Junghyun CHO, KAIST, Korea	POS4-0463
PS2_367	Structural Dynamics of Adaptive Condensates During Bidirectional Phase Transitions	Gaeun PARK, GIST InnoCORE AI-Nano Convergence Institute for Early Detection of Neurodegenerative Diseases, GIST, Korea	POS4-0485
PS2_368	Reconfigurable EGaIn Nanoinks Stabilized by Conjugated Polymers in Water	Eunjin CHOI, Gwangju Institute of Science and Technology; GIST InnoCORE AI-Nano Convergence Institute for Early Detection of Neurodegenerative Diseases, GIST, Korea	POS4-0496
PS2_369	Sequence Programming of Electronic Properties in Self-Assembling Peptides	Gyurin KIM, Gwangju Institute of Science and Technology, Korea	POS4-0520
PS2_370	Fullerene-Based Nanosurfactant for Automatic Homeotropic Alignment of Liquid Crystals	Chan LEE, Jeonbuk National University, Korea	POS4-0536

PS2_371	DNA Origami Lattice-Assisted Fabrication of Patchy Gold Nanoparticles	Jisung KIM, POSTECH, Korea	POS4-0553
PS2_372	Multicompartment Colloidal Particles via Self-Assembly of Mixed Graft Bottlebrush Block Copolymers	Minjoon BAEK, Soongsil University, Korea	POS4-0589
PS2_373	Phase-Separation Kinetics in Binary Polymer Brushes via Tunable Mobile Anchors	Maurya MANOJ KUMAR, DGIST, Korea	POS4-0595
PS2_374	Fabrication of Anisotropic Low-Melting-Point Alloy Particles for Directional Electrical and Thermal Transport	Min-Young CHOI, Seoul National University of Science and Technology, Korea	POS4-0610
PS2_375	Fabrication of Bicontinuous Porous Carbon Particles via Polymerization-Induced Microphase Separation in Emulsion Droplets	Younghyeon AHN, Korea Advanced Institute of Science and Technology (KAIST), Korea	POS4-0634
PS2_376	Encapsulation of Photonic Crystal-Based Materials for Electrophoretic Displays	Nahyeon PARK, Sejong University, Korea	POS4-0648
PS2_377	Plasma-Assisted Ball Milling for Surface Activation and Dispersion Enhancement of ATO Nanoparticles for Near-Infrared-Reflective Automotive Window Films	So-Jung HWANG, Seoul National University of Science and Technology, Korea	POS4-0674
PS2_378	Synthesis, Self-Assembly, and Hydrolytic Degradation of PEG-Based Biodegradable Block Copolymers	Junho LEE, Dankook University, Korea	POS4-0686
PS2_379	Gas Sensing Properties for Composites of Metal-Organic Frameworks and Nanofiber Polymers	Byeonggwon KIM, Chungnam National University, Korea	POS4-0696
PS2_380	PEG-DHLA Ligand Engineering for Water-Dispersible and Protein-Stable Quantum Dots	Jiwoo CHOI, Dankook University, Korea	POS4-0706
PS2_381	Synthesis and Micellar Development of Poly[(2-isopropyl-ran-2-ethyl)-2-oxazoline]-block-poly(tert-butyl methacrylate) as Functional Bioengineered Surfactants for ARDS Treatment	Seoyeon CHOI, Dankook University, Korea	POS4-0711
PS2_382	Complex Coacervate Core Glycogels Formed by ABA Triblock Copolymers	Alphonse Kodjo ABOTSI-DJOGBESSI, Hongik University, Korea	POS4-0729
PS2_383	Humidity-Resistant Triboelectric Nanogenerators Enabled by Hofmeister Salting-Out Effect in Quaternized Block Copolymer Films	Jinwoo CHOI, Hanyang University, Korea	POS4-0738
PS2_384	Template-Assisted MXene Patterning for Large-Area Sensor Arrays	Haejin JEONG, KIST, Korea	POS4-0758
PS2_385	Molecular Doping-Induced Spontaneous Defect Healing and Enhanced Structural Order in Uniaxially Aligned P3HT Films	Seyoun HWANG, Hanyang University, Korea	POS4-0776
PS2_386	Self-Assembled Thiophene-Conjugated Trialkoxy Benzamide Films for Negative Dispersion Broadband Antireflective Applications	Changhyeon SUNG, Jeonbuk National University, Korea	POS4-0791
PS2_387	Interfacial Self-Assembly of Metal Halide Complexes Mediated by a Crown Ether-Based Bola Amphiphile	Eunseo KO, Hanyang University, Korea	POS4-0800
PS2_388	Self-Healing and Wringing Diacetylene-Functionalized Organogels for Smart Soft Materials	Hyerim LEE, Jeonbuk National University, Korea	POS4-0814
PS2_389	Organic Solar Cells, Polymer Acceptors	Song XUYAO, Gyeongsang National University, Korea	POS4-0819
PS2_390	Oil-Free Sunscreen Templates Enabled by Aqueous Two-Phase System Emulsion	EunSuk LEE, Seoul National University of Science and Technology, Korea	POS4-0821

PS3. Poster Session 3

October 1 (Thu), Room 301 (Grand Ballroom), BEXCO

Board No. 보드번호	Title	Presenter	Abstract No. 초록번호
S4. Colloids, Interfaces, and Molecular Assemblies for Functional Soft Materials			
PS3_001	Dimensionality-Dependent electronic-ionic Transport in a Cs ₄ PbBr ₆ /CsPbBr ₃ Heterojunction	Chaeyeon PARK, Hanyang University, Korea	POS4-0843
PS3_002	Microplastic Separation via Interfacial Trapping in Aqueous Multiphase Systems	YeRin KIM, Seoul National University of Science and Technology, Korea	POS4-0903
PS3_003	Oxide-Reinforced Gallium-Based Liquid Metal Composites for Structural-Stable Electrodes	Da-In PARK, Seoul National University of Science and Technology, Korea	POS4-0904
PS3_004	Directional ANF/BN Porous Scaffolds via Sequential Freeze Casting	Hyeonjoong KIM, Seoul National University of Science and Technology, Korea	POS4-0907
PS3_005	Polydopamine-Functionalized Boron Nitride Embedded in Freeze-Dried 3D Agarose Networks for Thermally Conductive and Electrically Insulating PDMS Composites	Hanbit KYUNG, Seoul National University of Science and Technology, Korea	POS4-0917
PS3_006	Softness-Driven Packing and Rheology of PEG Nanogel Assemblies	Seoyoon CHUN, Seoul National University, Korea	POS4-0926
PS3_007	Design of Self-Assembly in Bottlebrush Random Copolymers by Controlling Grafting Density	Hey SHIN, Soongsil University, Korea	POS4-0985
PS3_008	Flexible Optical Paints Based on Self-Crosslinkable Cyanostilbene Side-Chain Polysiloxanes	Yebeen SEONG, Jeonbuk National University, Korea	POS4-1020
PS3_009	Effect of Polymer Adsorption State on the Properties of Poly(ethylene glycol)/Cellulose Nanocrystal Composites	Taeyoon LEE, Seoul National University, Korea	POS4-1031
PS3_010	Effect of Anionic Environment on the Hydration and Interfacial Properties of Poly(2-methoxyethyl Acrylate)	Aki KASHIWAZAKI, Tohoku University, Japan	POS4-1058
PS3_011	Rational Design of Short Peptide Mimics for Dual-Functional Cryoprotection	Yingjie BU, Gwangju Institute of Science and Technology, Korea	POS4-1063
PS3_012	Flow-Through Catalytic Remediation of Water Pollutants Using PVA/Alginate Hydrogels Immobilized with Pd and Pd-Pt Nanoplates	Fenni Woro HASTUTI, Pukyong National University, Korea	POS4-1083
PS3_013	Carboxymethyl cellulose/alginate/chitosan-based Polyelectrolyte Complex Hydrogel with Irregularly Shaped Multi-Metallic Nanowires for Efficient Continuous-Flow Cr(VI) Remediation	Songhee CHOI, Pukyong National University, Korea	POS4-1084
PS3_014	Hierarchical Multi-Scale Physical Unclonable Functions Based on Colloid Lithography and Block Copolymer Directed Self-Assembly	Hyun Su YUN, Kongju University, Korea	POS4-1090
PS3_015	Hierarchical Superhydrophobic Surfaces Fabricated via Colloidal Crystal Templating and Block Copolymer Self-Assembly	Hyun Soo KIM, Kongju National University, Korea	POS4-1091
PS3_016	Polydopamine-Integrated cellulose/graphene Oxide Monoliths for Continuous-Flow Iodine Capture and Photothermal-Enhanced Cr(VI) Reduction	Hyeon Jeong KIM, Pukyong National University, Korea	POS4-1092

PS3_017	Defect-Engineered Chiral Plasmonic Nanoparticle Assemblies for Multichannel Optical Physical Unclonable Functions	Siwon LEE, Sungkyunkwan University, Korea	POS4-1098
PS3_018	Enhanced Stability and Linearly Polarized Emission from CsPbI ₃ Perovskite Nanoplatelets Through A-Site Cation Engineering	Woo Hyeon JEONG, Sungkyunkwan University, Korea	POS4-1147
PS3_019	Interfacially Engineered Amphiphilic MXene Inks via Alkoxide Functionalization for Versatile Solvent Processing	Seongeun LEE, Korea Institute of Science and Technology, Korea	POS4-1160
PS3_020	Fabrication of Negative-Order Liquid Crystal Elastomer Shells for Thermal Actuation	Dayeon JEONG, Kyungpook National University, Korea	POS4-1218
PS3_021	Polyphosphoester-Based Bottlebrush Block Copolymers as Fully Degradable Stable Micellar Carriers	Hyori BAE, Soongsil University, Korea	POS4-1251
PS3_022	Micellization of SIS Triblock Copolymer Toward Improving Ionic Conductivity of All-Solid-State Battery Binders	Ahjeong KIM, Department of Organic Materials Engineering, Korea	POS4-1434
PS3_023	Stable and Highly Loaded Indocyanine Green J-Aggregate Nanosuspensions for Photothermal Therapy	Yeeri KIM, Seoul National University, Korea	POS4-1453
PS3_024	Interfacial Engineering of Zwitterionic Copolymer Brushes Toward Biofouling-Resistant PDMS Surfaces	Jeong Won KIM, University of Ulsan, Korea	POS4-1479
PS3_025	Developing a Universal Approach to Nanostructured Ionic Materials: From Discovery to Scaleable Production	James JENNINGS, University of Graz, Austria	POS4-1483
PS3_026	Rapid and General DNA Functionalization of Hydrophobic Nanoparticles	Murali GOLLA, Ewha Womans University, Korea	POS4-1495
PS3_027	In-Drop Metal Leaching from Cathodes for Continuous Recycling of Spent Batteries	Yewon KIM, Chungnam National University, Korea	POS4-1510
PS3_028	Fabrication of Highly Ordered Cu Nanopatterns via Interfacial Control of Vertically Oriented PS-b-PMMA	Kyongmoon KO, Chungnam National University, Korea	POS4-1521
PS3_029	Tunable Rheological Properties and Controlled Curcumin Release in Pickering Emulsions Stabilized by Self-Assembled Chitosan Particles	Rizwan Ahmed BHUTTO, King Fahd University of Petroleum and Minerals, Saudi Arabia, Saudi Arabia	POS4-1551
PS3_030	Self-Assembly of Discrete PEG-b-PLA Block Copolymers into Cubosome Nanostructures	Se Hwa PARK, Seoul National University, Korea	POS4-1576
PS3_031	Mechanochemically Synthesized Defect-Rich Cu-BTC for Selective CO ₂ Conversion to Formic Acid	Muhammad ZAHID, Interdisciplinary Research Center for Refining and advanced chemicals, King Fahd University of Petroleum and Minerals, Saudi Arabia	POS4-1580
PS3_032	Magnetically Encoded Plasmonics for Physical Information Encryption	Jinhyeok PARK, Ajou University, Korea	POS4-1599
PS3_033	Multi-Scale Interfacial Structural Dynamics of Smart Polymers via Neutron Probes	Qing CHEN, China Spallation Neutron Source, China	POS4-1626
S5. Polymers for Electronics and Photonics			
PS3_034	Spin-Coated Integration of a Visible-Transparent Organic Photodiode on Si-CMOS for Dual VIS-NIR Imaging	Sangjun LEE, POSTECH, Korea	POS5-0061
PS3_035	Visible-Light Photoprogrammable Transistors Enabled by an All-Organic Visible-Light Photoprogrammable Transistors Enabled by an All-Organic Triplet-Sensitized Diarylethene Channel Layer Triplet-Sensitized Diarylethene Channel Layer	Jieun KWON, POSTECH, Korea	POS5-0062

PS3_036	Enhancing Organic Photodiodes Performance in the SWIR Region by Tuning Open/Closed-shell Characteristics	Seung-Gi LEE, POSTECH, Korea	POS5-0063
PS3_037	High-Performance High-k Dielectric Metal Oxide TFTs with Azide-Functionalized Coordination Ligands	Suryeon JANG, POSTECH, Korea	POS5-0064
PS3_038	Improving Neuromorphic Memory in Organic Electrochemical Transistors Through Electrostatic Ion Trapping	Junseo KIM, POSTECH, Korea	POS5-0065
PS3_039	Ligand-Engineered Stretchable Metal Oxide Transistors Using Styrene-Functionalized β -Diketone Ligands	Hyo Min AHN, POSTECH, Korea	POS5-0066
PS3_040	Rational Molecular Design of π -Extended Thiazolothiazole for High-Performance UV-OPDs Seamlessly Integrated with CMOS	Won Jun PYO, Pohang University of Science and Technology, Korea	POS5-0067
PS3_041	Two-Terminal SWIR Photosynaptic Memory for Neuromorphic Vision	Chan SO, Postech, Korea	POS5-0068
PS3_042	Photocrosslinked Organic-Inorganic Hybrid PbS Quantum Dot Avalanche Photodiodes for CMOS-Compatible SWIR LiDAR Sensors	Kyobin PARK, POSTECH, Korea	POS5-0073
PS3_043	Structural Simplification of High-Bandwidth Organic Photodiodes for Next-Generation Optical Wireless Communication	Young Gyoung LEE, POSTECH, Korea	POS5-0080
PS3_044	Isotropic Nanomanufacturing of 3D Curvilinear Optoelectronics via Deterministic Chaos-Driven Gyroscopic Thermal Evaporation	Taek Min KIM, POSTECH, Korea	POS5-0082
PS3_045	Aggregation-Controlled Vacuum-Deposited Non-Fullerene Acceptors for High-Performance Organic Photodetectors	Geoneop CHOI, Pohang University of Science and Technology, Korea	POS5-0148
PS3_046	Kinetically Induced Phase Transition of Plastic Crystals	Chankyu KIM, Sogang University, Korea	POS5-0184
PS3_047	Production of Mesoporous Carbon Nitrides and Their Photocatalytic Properties for Degradation of Organic Pollutants	Sungmin BAE, Inha University, Korea	POS5-0205
PS3_048	Two-Step Process to Produce Porous Carbon Nitrides as Efficient Photocatalysts to Degrade Organic Pollutants	Heesun KIM, Inha University, Korea	POS5-0207
PS3_049	Two-Dimensional C ₃ N ₄ Nanoplatelets Produced by Unzipping of C ₃ N ₄ Nanotubes	Bogyeong KIM, Inha University, Korea	POS5-0214
PS3_050	Side-Chain Engineering of Polythiophene for Enhanced NO ₂ Affinity in Organic Gas Sensors	Seunghyeon JEON, Incheon Nat'l Univ., Korea	POS5-0279
PS3_051	Cluster Engineering of UiO-66-NH ₂ MOFs for High-Performance P3HT-Based Organic Gas Sensors	Seulgi CHOI, Incheon National University, Korea	POS5-0349
PS3_052	Kinetically Arrested Interfacial Polyelectrolyte Complexes Enable Ionic Rectification in Ionoelastomer Heterojunctions	Younghoon KIM, Ulsan National Institute of Science & Technology, Korea	POS5-0359
PS3_053	Highly Sensitive and Selective Organic Gas Sensors with Surface-Modified Activated Carbon	Shin Hyuk KANG, Incheon National University, Korea	POS5-0394
PS3_054	Enhancement of Gas Adsorption Performance in P3HT Films via Chemical and Structural Effects of PVA	Se Jin PARK, Incheon National University, Korea	POS5-0395
PS3_055	Water-Assisted Formation of Amine-Bridged Polymeric Carbon Nitride: A Structural Insight into the Photocatalytic Performance for H ₂ Evolution Under Visible Light	Seyeop JEONG, Inha University, Korea	POS5-0426

PS3_056	Pure-FAPbI ₃ Perovskite Solar Cells via Butanol-Assisted Printing-Friendly Sequential Deposition	Minwoo LEE, Gwangju Institute of Science and Technology(GIST), Korea	POS5-0474
PS3_057	Chiral Conjugated Polymer Nanowire Templates for CPL-Responsive Ferroelectric Encoding in Soft Electronics	Jeongmin JI, Gwangju Institute of Science and Technology (GIST), GIST InnoCORE AI-ACE Nano Convergence Institute for Fuel Production of Next Generation Biomass, Korea	POS5-0538
PS3_058	Elucidating Self-Doping in Donor-Acceptor Conjugated Polyelectrolytes: From Molecular Origin to Applications	Sangmin CHAE, RIKEN, Japan	POS5-0544
PS3_059	Development of Y6 Derivatives for Near-Infrared Organic Photodetectors	Chaehyun BYUN, Gyeongsang National University, Korea	POS5-0560
PS3_060	Elucidation of Energetic Disorder Evolution and Degradation Mechanisms in Organic Photovoltaic Through Transient Spectroscopic Analysis	Changwoo PARK, Kyungpook National University, Korea	POS5-0573
PS3_061	Impact of Backbone Regioregularity on the Conformation and Aggregation Behavior of PTQ11 Copolymers	Hyeonwoo JUNG, RIKEN, Japan	POS5-0574
PS3_062	Benzobisoxazole-Based Semicrystalline Polymer Donors for High-Gain, Low-Noise Photomultiplication-Type Organic Photodetectors	Sumin LEE, Hanyang University, Korea	POS5-0602
PS3_063	Modulation of Crystallinity Through Solvent-Induced Pre-Aggregation for Thick-Film Organic Photovoltaics	Damin LEE, Kyungpook National University, Korea	POS5-0619
PS3_064	Directly Written Laser-Induced Graphene Ionogels for Wide-Temperature Stretchable Micro-Supercapacitors	Ha Young LEE, KRICT (Korea research institute of chemical technology), Korea	POS5-0665
PS3_065	Delamination Suppression in Stretchable Ceramic/Polymer Composites	Junho HAN, UNIST, Korea	POS5-0684
PS3_066	Continuous Fullerene Derivatives Network Formed via Adhesion-Controlled Transfer Process for High-Performance Organic Photodiodes in Wireless Communication	Woongsik JANG, University of California Santa Barbara, United States	POS5-0700
PS3_067	Metal Synapses for Fingerprinting Amyloid- β Assembly-State-Dependent Plasticity Distortion	Hwandong JANG, Gwangju Institute of Science and Technology, Korea	POS5-0703
PS3_068	Harnessing sp ² -Hybridized Branched Side Chains to Control Pre-Aggregation in Organic Solar Cells	Seokhwan JEONG, UNIST, Korea	POS5-0708
PS3_069	Biodegradable Nanoporous Polymer-Based Composites for a Flexible Electronic Devices	Byeonggwon KIM, Chungnam National University, Korea	POS5-0732
PS3_070	Modulation of Thermoelectric Performance in PEDOT:PSS OECT Using Electrolyte-Gated Doping	Sanghee LEE, Kookmin University, Korea	POS5-0737
PS3_071	Collectively Reinforced Brick-and-Mortar Nanocomposite Films with High EMI Shielding Performance and Superior Folding Reliability	Subin PARK, Korea Institute of Industrial Technology, Korea	POS5-0759
PS3_072	Unveiling Diffusion Kinetics in Selective Contact Evaporation Printing for Multi-Scale Organic Crystal Architectures	Seeun KWON, Hannam University, Korea	POS5-0794
PS3_073	Thin-Film Fabrication and Electrochemical Characterization of PEDOT:carrageenan Composites	Hyejun PARK, Chung-Ang University, Korea	POS5-0795
PS3_074	Controlling Unusual NIR Red-Shifted Absorption in Quinoid-Based Chiral π -Conjugated Molecules Through Side-Chain Engineering	Jeongjin HONG, Gwangju Institute of Science and Technology, Korea	POS5-0807
PS3_075	Achieving Conduction Charge Transport Pathways to Improve Impurity Scattering in Doped Conjugated Polymers	Byeongjin KIM, POSTECH RIST 3- dong, Korea	POS5-0808

PS3_076	Systematic Development of Open-Shell Quinoid-Proquinoid Copolymers by Balancing Crystallinity and Spin Properties for Efficient Spin Transport	Younghyo KIM, Advanced Photonics Research Institute (APRI), Gwangju Institute of Science and Technology (GIST), Korea	POS5-0827
PS3_077	In-Situ Clear Detection Under Reverse Bias via Noise Floor Suppression in Nano-Modulated Ternary Blend Organic Photodetectors	Jihyun LIM, Chung-Ang University, Korea	POS5-0831
PS3_078	N-Benzylideneaniline-Derived Transparent Photoaddressable Polymers for AR Holographic Optics	Mungeun KIM, Korea University, Korea	POS5-0834
PS3_079	Side-Chain Positional Isomerism Enhances Intrinsic Photostability in Conjugated Polymers	Eonji KIM, Korea Research Institute of Chemical Technology, Korea	POS5-0877
PS3_080	Oxidative Chemical Vapor Deposition of Poly(3,4-ethylenedioxythiophene) Films for All-Plastic Organic Electrochemical Synaptic Transistors	Hanheum CHO, Sungkyunkwan University, Korea	POS5-0882
PS3_081	Chirality Imprinting in a Solution-Processable Conjugated Polymer by Circularly Polarized Light	Junho BAE, Chung-Ang University, Korea	POS5-0894
PS3_082	Channel Conductivity-Dependent Performance and Charge Transport Behaviors of Self-Doped Conjugated Polymer OECTs	Seunguk LEE, Kookmin University, Korea	POS5-0938
PS3_083	Impact of π -Extended Thiophene-Based Ligands with Varying Halogen Substituents on the Chiroptical Properties of Chiral Hybrid Perovskites	Min Jae KWON, Hanyang University, Korea	POS5-0951
PS3_084	Molecular Aggregation Engineering of Indacenodithiophene-Based Non-Fullerene Acceptors for High-Performance Vacuum-Deposited Organic Photodiodes	Gyuri KIM, Hanyang University, Korea	POS5-0953
PS3_085	Vertically Integrated Conjugated Polymer Transistors for Monolithic 3D CMOS Inverters	Sanghwa SONG, Korea Maritime and Ocean University, Korea	POS5-0957
PS3_086	Conjugated Polymers with Perovskite Quantum Dots via Polarity Adjustment for Ultralow Energy Consumption Photosynaptic Transistors	Wei Cheng CHEN, National Taipei University of Technology, Taiwan	POS5-0967
PS3_087	Basilar Membrane-Inspired Piezo-Ionotropic Polymer Sensor for Broadband Multi-Resonance Acoustic Sensing	Limin DENG, KAIST, Korea	POS5-0970
PS3_088	Continuous Jumping and Efficient Liquid Manipulation by Mechanical Instability-Driven Buckling of Magnetic Soft Robots	Kijun YANG, Hanyang University, Korea	POS5-0976
PS3_089	Artificial Layers with Sulfur-Rich Polymers-Cellulose Nanocrystal Composite Enables Uniform Metal Plating and Dendrite Inhibition	Gyeong Yeon BAE, Hanyang University, Korea	POS5-0995
PS3_090	Electro-Steric Ion Confinement in Polyelectrolyte Networks for Robust Nonvolatile Artificial Synapse	Jinbo KIM, Yonsei University, Korea	POS5-1002
PS3_091	Anisotropic Thermal Interface Material with Magnetically Programmable Heat Pathways Based on Polydopamine-Mediated Fe3O4/carbon Nanotube Polymer Nanocomposites	Dongyeong GIM, Gwangju Institute of Science and Technology, Korea	POS5-1008
PS3_092	Size-Controlled Microbeads via Oil-in-Oil Pickering Emulsion for High-Percolation Network in Polymer Composites	Gyeongmo KIM, Gwangju Institute of Science and Technology, Korea	POS5-1009
PS3_093	Long-Term Stable Organic Electrochemical Transistors via Bio-Derived Choline Ionic Liquids	Jiyeon OH, Sungkyunkwan University, Korea	POS5-1079
PS3_094	Asymmetrically Crosslinked Janus π -Ion Films as Reusable, Stackable Building Blocks for Reconfigurable Organic Electrochemical Transistors	YongHee KIM, Pukyong National University, Korea	POS5-1096
PS3_095	Highly Stretchable and Transparent Piezo-Ionic Sensors Based on Double-Network Hydrogels for Wearable Electronics	Seunga JUNG, University of Seoul, Korea	POS5-1116

PS3_096	Highly Stretchable Microsupercapacitor Achieved with Porous Nanocomposite Electrodes and Zwitterionic Ionogels	Hyowon LEE, University of Seoul, Korea	POS5-1118
PS3_097	Advanced Polymer Materials for Ionotronic Sensory Platforms	Jin Han KWON, KAIST, Korea	POS5-1177
PS3_098	Ion-Trapping Binder for Ceramic-Coated Separators to Suppress Transition Metal Crossover in NCM811 Batteries	Juyoung LEE, Seoul National University, Korea	POS5-1181
PS3_099	Efficient Dark Current Suppression in Organic Photodiodes via Solid Additive-Assisted Morphology Control	ChangWoo YU, Kongju National University, Korea	POS5-1182
PS3_100	Hole-Transport Layer Engineering for Improved Charge Transport in FAPbI ₃ -Based Perovskite Solar Cells	Gyeongho JO, Kongju National University, Korea	POS5-1187
PS3_101	Cross-Linked Polyimide Gate Dielectric Through Thermal Click Chemistry for Low Temperature Processable Organic Thin Film Transistor	Chae Yeong LIM, Kyung Sung Univ., Korea	POS5-1189
PS3_102	Alternative Solvent Engineering for Enhanced Yield of Semiconducting Single-Walled Carbon Nanotubes and High-Performance Field-Effect Transistors	Minseo BAEK, Kongju National University, Korea	POS5-1191
PS3_103	Highly Efficient Perovskite Color Conversion Devices via Strong Absorption and Photon Recycling	Boseung JE, POSTECH, Korea	POS5-1193
PS3_104	Green Solvent Engineering for Solution-Sheared TIPS-Pentacene Thin-Film Transistors	Dongil HO, Kongju National University, Korea	POS5-1198
PS3_105	Volatile Coordination-Driven Surface Reconstruction for Stable Tin Perovskite Transistors	Geonwoong PARK, POSTECH, Korea	POS5-1200
PS3_106	Stretchable MXene/Silver Nanowire (MXAg) Electrodes for Stretchable Light-Emitting Diodes in Biophotonic Applications	Jonghyeong CHO, Pohang University of Science and Technology (POSTECH), Korea	POS5-1212
PS3_107	Development of a Two-Layer pH-Responsive CLC-IPN Hydrogel Platform for Enzyme-Based Sweat Biosensing	Jeongmin CHOI, Kyungpook National University, Korea	POS5-1219
PS3_108	Optimizing Hole-Transport Interfaces in Infrared Photodetectors Using Phosphonic Acid-Modified Conjugated Polymers	Ji Hyeon LEE, Gwangju Institute of Science and Technology; Korea Institute of Science and Technology, Korea	POS5-1226
PS3_109	Phenolic Benzothiazolium-Based Nonlinear Organic Crystals for Terahertz Generation	WooShin KIM, Ajou University, Korea	POS5-1227
PS3_110	Hybrid Electrolyte-Doped OFETs for Neuromorphic Applications	Yu Jung PARK, University of Seoul, Korea	POS5-1232
PS3_111	Optical Study of MoS ₂ /Cu ₂ O Heterostructures to Elucidate Auger-Assisted Non-Linear Electron Uphill Transfer	Jaewon HEO, Korea Institute of Science and Technology, Korea	POS5-1234
PS3_112	High-Yield and High-Purity Sorting of Semiconducting Single-Walled Carbon Nanotubes Using Conjugated Polymers and Molecular Cofactors	Chaeyeon HAN, Kongju National University, Korea	POS5-1249
PS3_113	Centrifugation-Controlled Semiconducting SWNT Separation for Purity-Dependent Field-Effect Transistor Performance	DongHyun SONG, Kongju National University, Korea	POS5-1255
PS3_114	Controlling Synaptic Plasticity via Metal-Organic Frameworks to Develop an Artificial Retina for Information Filtering	Yoojin OH, Yonsei University, Korea	POS5-1259
PS3_115	Thermally Engineered Sodium-Embedded Alumina with Programmable Synaptic Plasticity for Neuromorphic Transistors	Saeyoun PARK, Yonsei University, Korea	POS5-1261

PS3_116	All-in-one Organic Ligand for Emitting Perovskite Nanocrystals: Efficient Dispersion, Photocurable and Charge Transporting Capability	Jaejong RYU, Sungkyunkwan University, Korea	POS5-1267
PS3_117	Enhanced Circularly Polarized Electroluminescence from Polymer Spin-LEDs via Halogenated Chiral Perovskite Interlayers	Wonbin CHOI, Seoul National University, Korea	POS5-1269
PS3_118	Flexible Multistate Vertical Organic Electrochemical Transistor Arrays for Neuromorphic Computing	Inho LEE, Ajou University, Korea	POS5-1274
PS3_119	Revisiting the Role of Oxidation in Stable and High-Performance Lead-Free perovskite-IGZO Junction Field-Effect Transistors	Seonkwon KIM, Yonsei University, Korea	POS5-1281
PS3_120	Photomultiplication Type Perovskite Photodetectors Enabled by Dopant Controlled Bulk Heterojunction	Taegeon LEE, University of Seoul, Korea	POS5-1283
PS3_121	Room-Temperature Aqueous Synthesis of Pd-Core Pt-Shell Nanowires for Electrochemical Non-Enzymatic Sweat Glucose Sensing	Hanseon BAE, POSTECH, Korea	POS5-1291
PS3_122	Improving Conductivity and Stability in Molecularly Doped Conjugated Polymers	Juhyung PARK, KU Leuven, Belgium	POS5-1295
PS3_123	Physical Implementation of Reinforcement Learning Through Signal Summation in a Dual-Input Synaptic Transistor: Photoinduced Dipole Inversion of an Au(I) Complex with Charge-Trapping cPVP	Chiho YOUN, Yonsei University, Korea	POS5-1297
PS3_124	Self-Aligned Sol-Gel Metal Oxide Heterojunction Transistors for Scalable Multivalued Logic Devices	Seokhyeon BAEK, Ajou University, Korea	POS5-1304
PS3_125	Understanding Electrothermal Mechanisms Toward Soft Thermal Management Systems and Wearable Electronics	Janghee HAN, Korea Advanced Institute of Science and Technology (KAIST), Korea	POS5-1317
PS3_126	Optical Direct Writing of Solution-Processed SnO _x Thin-Film Transistors	Jeongjae PARK, Ajou University, Korea	POS5-1328
PS3_127	Hydrophilicity-Modulated Iodine-Containing Alternating Copolymer Resists for Extreme Ultraviolet (EUV) Lithography	Seoyeon LEE, Inha University, Korea	POS5-1363
PS3_128	Slot-Die-Printed MoS ₂ Transistor Arrays with a Sodium-Modified Alumina Dielectric	DongYeon WON, Yonsei University, Korea	POS5-1404
PS3_129	Engineering Exciton Lifetime of Pt(II) Complex and Templating Layer for Solution-Processed Deep-Blue Phosphor-Sensitized Fluorescent OLEDs with 36% External Quantum Efficiency	Raja KUMARESAN, Institute for Plastic Information and Energy Materials Sustainable Utilization of Photovoltaic Energy Research Center (EPIC), Pusan National University, Korea	POS5-1406
PS3_130	Toward Solution-Processed Deep-Blue OLEDs: Ligand-Core Engineering of Pt(II) Sensitizers	Hyerin KIM, Pusan National University, Korea	POS5-1411
PS3_131	Organic Non-Volatile Floating-Gate Memory Using Self-Assembled Gold Nanodot Arrays	Jong In YOON, Inha University, Korea	POS5-1415
PS3_132	Modulating Ionic Hydration States via Hydrogel Network Architecture for Controlled Solvation	Juyoung KANG, Hong Chul Moon, Korea	POS5-1417
PS3_133	A Monolithically Integrated Fibriform Organic Electrochemical Schmitt Trigger for Noise-Immune Smart Textiles	Jongsu LEE, Korea Institute of Science and Technology, Korea	POS5-1439
PS3_134	Process Optimization of PbS Quantum Dot Near-Infrared Photodetectors	Bumseok KO, Nanophotonic System Research Center, Korea Institute of Science and Technology, Korea	POS5-1486
PS3_135	Development of High-Performance N-Type OECTs with Small Molecule OMIECs	Yeonjoo KIM, Seoul National University, Korea	POS5-1492

PS3_136	High-Photovoltage Perovskite-Organic Tandem Photovoltaics for Bias-Free Solar Ammonia Production	Dong Hyun SHIN, Pusan National University, Korea	POS5-1497
PS3_137	Tailored Polyphosphazene Matrices with Cross-Linking Functionality for High-Performance Solid Polymer Electrolytes with Enhanced Ionic Conductivity	Seungyoon LEE, Seoul National University, Korea	POS5-1535
PS3_138	Translating Skin-Conformal Electrode Materials into Low-Noise Wireless Biopotential Systems	Joohwan SHIN, Gnothi Inc., Korea	POS5-1537
PS3_139	Ultrasonication-Induced Ambient Synthesis of Perovskite Nanowires with Block Copolymer for Photodetector Applications	Doyun KIM, University of Seoul, Korea	POS5-1544
PS3_140	Interfacial Molecular Engineering for Patternable and Stretchable Metal Electrodes	Sangwook NAM, Ajou University, Korea	POS5-1549
PS3_141	High-Performance Short-Wave Infrared Organic Photodetectors Based on Indanone-Derived Narrow-Bandgap Non-Fullerene Acceptors with Strong Electron-Withdrawing Groups	Hajeong KIM, Chungnam National University, Korea	POS5-1577
PS3_142	Advancing SWIR Organic Photodetectors Through Molecular Design	Junhwan LEE, Chungnam National University, Korea	POS5-1590
PS3_143	Aggregation-Suppressed Linear Donor Polymer with Deep HOMO Level for Efficient Organic Photovoltaics	Jinlong PANG, Chungnam National University, Korea	POS5-1591
PS3_144	Side-Chain Engineering of Non-Fullerene Acceptors for Enhanced Photoresponse in SWIR Organic Photodetectors	Kyubeen KWAK, Chungnam National University, Korea	POS5-1592
PS3_145	Design of a Multifunctional Self-Healing Polymer Binder Based on Dynamic Disulfide Crosslinking for Silicon Anodes	Dahye RYU, Chungnam National University, Korea	POS5-1594
PS3_146	Suppressing Li ⁺ Migration from Li-TFSI-Doped HTLs via a PEI-Crown Ether Complex for Enhanced Long-Term Stability of Perovskite Solar Cells	Dongwook LEE, University of Seoul, Korea	POS5-1597
PS3_147	Interface and Surface Modifications of Organic Photocathodes for High-Performance and Stable solar-to-H ₂ Conversion	Eui Hyun SUH, Kyungpook National University, Korea	POS5-1648
PS3_148	Simulation-Assisted Design of Transparent Photoresponsive Fluorinated Coating Films with Topographic Recovery for LiDAR Sensor Covers	Jinho KIM, Yonsei University, Korea	POS5-1666
PS3_149	High-Performance Polymers Bearing Polycyclic Aromatic Hydrocarbon Chromophores for Optoelectronic and Energy Storage Applications	Yu Ruei KUNG, National Taipei University of Technology, Taiwan	POS5-1674
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PS3_150	Self-Healable Solid-State Bio-Polymer Electrolytes for Next-Generation Lithium-Ion Batteries	Kaushiki AHUJA, UPES, India	POS6-0113
PS3_151	High-Configurational Entropy Materials for Zn-Based Battery System	Dohoon PARK, Korea Science Academy of KAIST, Korea	POS6-0128
PS3_152	Facile One-Pot Fabrication of Te@Carbon Core-Shell Nanocables with Superior Photothermal Conversion Performance	Tae Kyu AN, Korea National University of Transportation, Korea	POS6-0146
PS3_153	Suppressing Parasitic Reactions in Lithium Metal Batteries via an In-Situ Single-Ion Polymer Coating	Susung YUN, Korea Advanced Institute of Science and Technology, Korea	POS6-0228
PS3_154	POSS-PEG-based Pore-Filling Solid Electrolyte with Enhanced Li ⁺ Transport and Interfacial Stability for Solid-State Lithium-ion Batteries	Bokyung PARK, Sungkyunkwan University, Korea	POS6-0233

PS3_155	Solvation Structure Modulation in Aqueous Electrolytes: Stabilizing Pyrazine Radicals and Facilitating Electrochemical Nitrate Reduction	Min Young SEO, Korea University, Korea	POS6-0235
PS3_156	Tailoring Product Selectivity in Electrochemical CO ₂ Reduction with Polyampholyte Ionomers	Sang Jun PARK, Sogang University, Korea	POS6-0240
PS3_157	Enhanced Cycling Stability of Lithium-Oxygen Batteries Through a Bilayer Solid Polymer Electrolyte with Independent Anode Protection and Cathode Activation	Hyocheol LEE, Sungkyunkwan University, Korea	POS6-0293
PS3_158	Effect of Cyclic Fluorinated Monomer Structure on the Optical and Gas Transport Properties of Perfluorosulfonic Acid Ionomers	Hyeongyeol SEO, Soongsil University, Korea	POS6-0306
PS3_159	Synthesis of Dispersible Imine-Based COF-nanoparticles for Enhanced Photocatalytic Reactivity	Laurens CAYE, Max-Planck-Institute for Polymer Research, Germany	POS6-0352
PS3_160	Molecular Engineering of Elastomeric Electrolytes with Enhanced Mechanical Robustness for Lithium Metal Batteries	Dongguk KANG, Kumoh National Institute of Technology, Korea	POS6-0418
PS3_161	Hybrid Organic/Inorganic Layer Interphase via Capillary Rise Infiltration for Stabilizing Li Plating from Li-Rich to Anode-Free Operation	Jejun BAEK, Sungkyunkwan University, Korea	POS6-0435
PS3_162	Tailoring Ion Transport and Dimensional Stability in Poly(aryl piperidinium) Anion Exchange Membranes via Molecular Branching	Kyuha LEE, Jeonbuk National University, Korea	POS6-0436
PS3_163	Enhanced Low-Humidity Performance of Block Copolymer-Based Composite Proton Exchange Membranes	Seonwoo YANG, Korea Advanced Institute of Science and Technology, Korea	POS6-0453
PS3_164	Effect of GO-QPEI Incorporation on the Properties of Quaternized Naphthalene-Based Poly(arylene ether ketone) Anion Exchange Membranes	Ji Young CHU, Jeonbuk National University, Korea	POS6-0518
PS3_165	Inorganic Buffer and Sputtered ITO Diffusion Barrier for Suppressing Lamination-Induced Degradation in PSCs	Sehun OH, Pusan National University, Korea	POS6-0607
PS3_166	Surfactant Effects on Structure-Property Relationships in Nanoparticles for Solar Hydrogen Production	Hyunjoo LEE, Kyunpook National University, Korea	POS6-0629
PS3_167	Enhancing Selectivity of Electrochemical CO ₂ Reduction in Highly Acidic Media via Local Environment Modulation Using Block Copolymer-Templated Carbons with Abundant Inner Cavities	Jaeyoung CHOI, KAIST, Korea	POS6-0637
PS3_168	Solvent-Free Fabrication of Thick LiFePO ₄ Electrodes Using a Thermoplastic Binder for High-Energy-Density Li-ion Batteries	Dae Kyom KIM, Chung-Ang University, Korea	POS6-0660
PS3_169	iCVD-Deposited Imidazole Copolymer Interfaces for Surface-Directed MOF Growth	Chae Hwan CHO, Korea Institute of Science and Technology, Korea	POS6-0661
PS3_170	Alkyl Chain Engineering of NDI-Based Conjugated Polymer Cathodes for Improved Capacity and Rate Performance	Soyoung KIM, Kumoh National Institute of Technology, Korea	POS6-0680
PS3_171	Microphase-Engineered Ether-Free Proton Exchange Membranes for High-Performance Water Electrolysis	Camthach TO, Sungkyunkwan University, Korea	POS6-0691
PS3_172	Tunable Electrochemical Prelithiation of Li-In Alloy Anodes for Stable Sulfide-Based All-Solid-State Batteries	Yiwen LU, Sungkyunkwan University, Korea	POS6-0727
PS3_173	Natural Polymer-Based Aerogels for Moisture-Electric Energy Generation	Eunae SHIN, Korea Institute of Industrial Technology (KITECH), Korea	POS6-0735
PS3_174	Photocatalytic Polymer Nanoparticles with Localized Pyridine Microenvironments for Selective CO ₂ -to-Formate Conversion	Juwon HA, DGIST, Korea	POS6-0762

PS3_175	Cellulose Acetate- and Sulfosuccinic Acid-Modified SPEEK Composite Electrolyte Membranes with Controlled Dimensional Stability and Ion Transport Properties	Dongjun LEE, Hannam University, Korea	POS6-0809
PS3_176	Polyampholyte Methylcellulose Incorporating Lithium Chloride as a High Performance Quasi-Solid Polymer Electrolyte for Supercapacitors with High Rate Capability	Haewon JIN, Hannam University, Korea	POS6-0860
PS3_177	Perovskite Quantum Dot Surface Engineering for High-Efficiency and Stable Tin-Lead Perovskite Solar Cells	Dohun BAEK, University of Seoul, Korea	POS6-0866
PS3_178	Enabling Electrode Manufacturing Through a PFAS-Free Binder for High-Performance Lithium-ion Batteries	So Young NAM, Korea Electronics Technology Institute, Korea	POS6-0874
PS3_179	Performance Enhancement of Flexible Thermoelectric Generators via PEDOT:PSS-Based Materials Coated on 3D Spacer Fibers	Saehaneul JUNG, Kookmin University, Korea	POS6-0877
PS3_180	In-Situ Crosslinked Quasi-Solid-State Electrolytes with Molecular Weight-Controlled Networks for High-Performance Lithium Metal Batteries	Hyunjun JOO, Sungkyunkwan University, Korea	POS6-0884
PS3_181	Coupling Thermodiffusion and Thermogalvanic Effects in Anhydrous Ionogels for Enhanced Thermoelectric Performance	Seokyeon JANG, Kookmin University, Korea	POS6-0936
PS3_182	Phase-Separated Ionogel Binders for Durable High-Capacity Silicon Anodes	Xuejian ZENG, Sungkyunkwan University, Korea	POS6-0943
PS3_183	Bifunctional Polyamide-Imidazole Binder for Silicon Anodes via Dual Interactions with Silicon Particles and Conductive Carbon	Seungeon OH, Hanyang University, Korea	POS6-0944
PS3_184	Illumination-Dependent Charge Carrier Dynamics and Photoelectrochemical Performance of LaFeO ₃	DongHwan HWANG, Pusan National University, Korea	POS6-0946
PS3_185	Lamination-Compatible SnO ₂ /Reactive-Sputtered ITO/Ag Rear Contacts for Stable Perovskite Solar Cells	Sehun OH, Pusan National University, Korea	POS6-1108
PS3_186	Surface-Oxygen-Tuned LLZTO in EMITFSI-Based Polymer Ionogel Electrolytes for Stable Solid-State Lithium Batteries	Harim SEO, Kyungpook National University, Korea	POS6-1111
PS3_187	Enhancing Ionic Transport and Interfacial Stability in Lithium Metal Batteries via Dual-Anion Ionic Liquid Electrolytes	Jemin LEE, Kyungpook National University, Korea	POS6-1114
PS3_188	Temperature-Responsive Ionic Conductors for Energy Applications	Hyeon Woo YANG, Korea Advanced Institute of Science and Technology, Korea	POS6-1163
PS3_189	Impact of Interfacial Roughness on Light Absorption of Perovskite Solar Cells	Kyungmin LEE, Pohang University of Science and Technology (POSTECH), Korea	POS6-1168
PS3_190	Four-Dimensional (x, y, z, t) Photoluminescence Dynamics Revealing Efficient Vertical Charge Transport in Perovskite Solar Cells	Boseung JE, POSTECH, Korea	POS6-1194
PS3_191	Controlled Ion Intercalation Kinetics for High-Performance Electrochromic Displays	Neul Kyum HA, KAIST, Korea	POS6-1224
PS3_192	Impact of Nanostructuring on BBL Organic Anodes for Lithium-Ion Batteries	Seohee YE, Hongik University, Korea	POS6-1227
PS3_193	PFAS-free Phosphorylated Cellulose Nanofibril Membranes as Sustainable Polymer Electrolytes for Microbial Fuel Cells	Kunhee LEE, Seoul National University, Korea	POS6-1232
PS3_194	Impact of Electrochemical Stability on Performance Deterioration in Alkaline Redox Flow Batteries Utilizing Different Benzoquinone Derivatives	Jinu JUN, Sungkyunkwan University, Korea	POS6-1242

PS3_195	Transition Metal Ion Regulated Tannic Acid Based Metal-Phenolic Networks for Engineering a Protective Interphase on Zn Anodes	Hae Lim OH, Sungkyunkwan University, Korea	POS6-1290
PS3_196	Supramolecularly Reinforced Silicon Anodes via UPy Functionalization	Suhyeon RHO, Seoul National University, Korea	POS6-1318
PS3_197	Property of Polyurethane Binder with Different Crosslinking Densities for Silicon Anode	Subin PARK, Pusan National University, Korea	POS6-1443
PS3_198	Interfacial Engineering of Ba1-xSrxTiO3/PDMS Nanohybrids for Efficient, Thermally Stable, and Skin-Adherent Triboelectric Nanogenerators	Mirza Mahmood BAIG, University of Ulsan, Korea	POS6-1455
PS3_199	Magnetic Self-Assembly of 3D Micro Pillar Electrodes for Highly Effective Supercapacitors	HaeBin JEONG, Chung-Ang University, Korea	POS6-1456
PS3_200	Dielectrophoretic Assembly of Silver Nanowires for Surface-Area-Enhanced Micro-Supercapacitors	Yeonwoo KIM, Chung-Ang University, Korea	POS6-1473
PS3_201	Separator-Programmed Electric-Double-Layer Ordering for Stabilizing Thin Lithium-Metal Batteries	Eunbin JANG, Kyungpook National University, Korea	POS6-1489
PS3_202	Cobalt Hydroxide-Decorated Polyaniline on Nickel Foam for Efficient Oxygen Evolution Reaction	Animesh ROY, Advanced Technology Research Centre, Korea University of Technology and Education, Korea	POS6-1552
PS3_203	Hindered Phenols Enable Dual-Action Passivation and Reactive Oxygen Species Mitigation for Perovskite Solar Cells	Qurrotun Ayuni Khoirun NISA, Pukyong National University, Korea	POS6-1601
PS3_204	Effect of Interstitial Alkyl Chain Engineering of Anion Exchange Membrane for Improved CO2 Electrolyzer	Haeryang LIM, Korea Institute of Science and Technology, Korea	POS6-1621
PS3_205	Development of Zwitterion-Functionalized Polymer Electrolyte Membranes with Enhanced Ionic Conductivity	Bumseo PARK, Hannam University, Korea	POS6-1651
PS3_206	Structural Engineering of PTFE-Carbon Conductive Binder Domains for High-Loading Dry Electrodes for Lithium-Ion Batteries	Byeongcheol KANG, Korea Institute of Science and Technology (KIST), Korea	POS6-1670
S7. Innovations in Polymeric Composites: From Design and Processing to Industrial Applications			
PS3_207	Novel PLA/wood Flour WPC Composite with Biomimetic Compounds for Laser Assisted Electroless Metallization	Krzysztof MORACZEWSKI, Kazimierz Wielki University, Poland	POS7-0011
PS3_208	Polymerization of α -Lipoic Acid on Silica Particles for Adhesive Applications	Sein KIM, Pusan National University, Korea	POS7-0343
PS3_209	Effect of Vinyl Silane on the Thermal Durability of Silica-Reinforced PFAS-Free Polyurethane Composites for	Seonghye KIM, Korea Institute of Materials Convergence Techno	POS7-0406
PS3_210	Synthesis of Fluorine-Free Binders for Dry Electrode of Battery Using Sustainable Bio-Derived Molecule	Min Jeong KIM, Kumoh National Institute of Technology, Korea	POS7-0449
PS3_211	Enhanced Gas Barrier Performance of Polypropylene Composites Through Low-Loading Boron Nitride Nano	Hyung-Ho CHOI, Chungnam National University, Korea	POS7-0486
PS3_212	Enhanced Filler Dispersion and Reinforcement of Silica/Carbon Black-Filled SBR Composites Using an Inverse	In-Hee KIM, Chungnam National University, Korea	POS7-0506
PS3_213	Low-Dielectric Polyimide Films Prepared via Physical Entanglement and Thermal Decomposition of Poly(2-hy	Geun Seong LEE, Chungnam National University, Korea	POS7-0522

PS3_214	Tailoring the Dielectric Properties of Thermotropic Liquid Crystalline Polymers Through ZIF-8 Incorporation	Seung-Hyeon LEE, Chungnam National University, Korea	POS7-0534
PS3_215	Effects of BNNS Surface Chemistry on the Thermal, Dielectric, and Mechanical Properties of Epoxy Composites	Yujin NOH, Chungnam National University, Korea	POS7-0537
PS3_216	Hybrid Reinforcement of Polyimide Nanocomposites Using Aramid Nanofibers and POSS for Flexible Circuit	Seung-Hun CHAE, Chungnam National University, Korea	POS7-0587
PS3_217	Suppression of Initial Solvent-Driven Nonequilibrium Effect on Polymer Nanocomposites with Physically Adsorbed Nanoparticles	Seong Min JO, Seoul National University, Korea	POS7-0611
PS3_218	Bio-Based Transparent Self-Healable Polyimides Enabled by Isosorbide-Derived Dianhydride Design and Aliphatic Diamine	Woojin KANG, Inha University, Korea	POS7-0622
PS3_219	Enhanced Absorption-Dominant EMI Shielding of Polyimide Composite Films via Thermal Oxidation of Carbon Nanotubes	Min Hye WOO, Chungnam National University, Korea	POS7-0643
PS3_220	Rheological Percolation Threshold of 2D Anisotropic Graphene Oxide/PEG Nanocomposites	Juseong KIM, Seoul National University, Korea	POS7-0650
PS3_221	Synthesis of Polypeptide-Coated Copper(II) Chloride Hydroxide and Its Application to Heterojunction Films	Junhyeong IM, Dong-Eui University, Korea	POS7-0695
PS3_222	Synthesis and Functional Modification of Metal Oxide Nanoparticles for Applications in Electrochemical Storage	HanHo KIM, Kookmin University, Korea	POS7-0709
PS3_223	PEG-based Polyurethane - 2D Materials GO Composite Coatings for Dual Gas Barrier Applied for Reducing CO ₂ Emissions	Thi Thuy Van TRAN, Jeju National University, Korea	POS7-0721
PS3_224	pH-Dependent Conformation and Rheological Behaviors of Sodium Carboxymethyl Cellulose: An Integrated Study	Seoahn KIM, Seoul National University, Korea	POS7-0752
PS3_225	Real-Time Gelation Point Analysis of a Standard Aerospace Epoxy System Using Oscillation Windowed Chirp	Taeyee KIM, Waters Materials Sciences, Korea	POS7-0783
PS3_226	DMD-based Maskless Lithography of Three Dimensional Boron Nitride Microarchitectures for Enhanced Thermal Conductivity	Hyorin KIM, Ulsan National Institute of Science and Technology, Korea	POS7-0797
PS3_227	Structure-Controlled Percolation Networks in Monolithic Stretchable Functional Fibers for Wearable Electronics	Yeonseo LEE, KU-KIST, Korea	POS7-0839
PS3_228	Highly Stretchable and Tough Pressure-Sensitive Adhesives Enabled by Dynamic Polyrotaxane Cross-Linkers	Ganga Ratnamma RUDRU, Pusan National University, Korea	POS7-0865
PS3_229	Influence of Metal β -Diketonate Addition on Thermal Stability and Cytotoxicity in Calcium-Zinc Stabilized Poly(vinylidene fluoride)	Jieun LEE, KIMCO, Korea	POS7-0873
PS3_230	Thermally Conductive Epoxy Molding Compounds with BN/MgO Hybrid Fillers for Electronic Packaging	Minbeom SIM, Kyungpook National University, Korea	POS7-0876
PS3_231	Vertically Aligned Boron Nitride/MgO Hybrid Filler Networks for Thermally Conductive Epoxy Composites	NanHee LEE, Kyungpook National University, Korea	POS7-0885
PS3_232	Rheological Flow Engineering of h-BN/epoxy Fluids for Semiconductor Packaging	Hansam PARK, Hanyang University, Korea	POS7-0897
PS3_233	Preparation of Fiber-Reinforced Vitrimers and Their Laminated Composites for Semiconductor Substrates	Seongho KIM, Hanyang University, Korea	POS7-0979

PS3_234	Highly Durable and Self-Healing Superhydrophobic Polyurethane Nanocomposites for Anti-Icing Application	Myongkeon OH, Korea Institute of Industrial Technology, Korea	POS7-1007
PS3_235	Effects of Optimized Thermal Annealing on the Microstructure and Mechanical Property of FDM-Printed PA	Hogeun NAM, KITECH, Korea	POS7-1296
PS3_236	A New Paradigm for Polyurethane CMP Pads: Tailoring Phase Mixing and Surface Polarity via CO ₂ -based Po	Hajeong KANG, R&D Center, EnPulse Co., Ltd., Korea	POS7-1380
PS3_237	Development of Structurally Stabilized Graphene-Based Composite Aerogels for Piezoresistive Sensing	Hyeonho CHOI, Inha University, Korea	POS7-1405
PS3_238	Multifunctional Interfacial Design of Ladder-like Polysilsesquioxane (LPSQ)-Silica Hybrids for Advanced EV T	Jeong Ju BAEK, Korea Institute of Industrial Technology, Korea	POS7-1437
PS3_239	Effects of Hard Magnetic Particle Alignment on the Magnetic Actuation of Micropillars and Micro Soft Robo	Hyeseong KIM, Hanyang University, Korea	POS7-1452
PS3_240	Enhancing Thermal Conductivity of Non-Conductive Films Through Freeze-Cast Alumina Filler Network Form	Minwook JEON, Seoul National University, Korea	POS7-1467
PS3_241	Ionic Liquid-Functionalized Reduced Graphene oxide-aramid Nanofiber Composite Films for Enhanced Ther	Rimsha Qurrat Ul AIN, Korea National University of Transportation	POS7-1484
PS3_242	Colorimetric Gas Sensor Based on Biocompatible Porous Microbeads	Saehaneul JUNG, Kookmin University, Korea	POS7-1541
PS3_243	Stretchable Liquid Metal-Based Gas Barrier Films Enabled by Vacuum-Assisted Filling and Internal Post-Supp	Jihye KIM, KIST, Korea	POS7-1546
PS3_244	Development of Mold-in-Colored LFI Composite Hood with Non-Destructive Defect Prediction Using A-Scar	Jinwoo LEE, Hyundai Motor Company, Korea	POS7-1594
PS3_245	Synthesis and Their Applications in Flexible Sensors of Self-Healing Polyurethane-Based Ion-Absorbing Mate	Chao CHEN, Chungnam National University, Korea	POS7-1624
PS3_246	The Impact of Topology and Chain Length of Polyvinylpyrrolidone Macromolecules on the Crystallization of	Ewa OZIMINA KAMINSKA, Medical University of Silesia in Katow	POS8-0025
S8. Frontiers of Functional Polymers in Biology and Medicine			
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PS3_248	Synthesis of Rice Straw Dialdehyde Cellulose and Its Application in Hydrogels	Kritsana RUENJAI, Chiang Mai University, Thailand	POS8-0100
PS3_249	Single-Administration Polymeric Microneedles for Prime-and-Boost Vaccination Against Hepatitis B Through Biphasic Antigen Delivery	Seojin BYUN, Gachon University, Korea	POS8-0226
PS3_250	Novel Insect Repellents and Polymer Films for Their Control Release	Sergey ZVEREV, Disinfectology Institute of F.F. Erismann Federal Scientific Center of Hygiene of the Federal Service for	POS8-0249
PS3_251	Fabrication and Evaluation of Polyphosphate-Modified Starch Powder as a Rapid Hemostatic Agent	Eun yeong KIM, Kumoh National Institute of Technology, Korea	POS8-0336
PS3_252	Fabrication and Characterization of Biodegradable Ultrafine Fibrous Sheets with Improved Tissue Adhesion Properties for Preventing Lung Air Leakage	Jihye HAN, Kumoh National Institute of Technology, Korea	POS8-0344

PS3_253	Fabrication of ChS-APBA/EGCG Functionalized Biodegradable PHB/PEO Ultrafine Fiber Sheets for Wound Healing	Nayoung PARK, Kumoh National Institute of Technology, Korea	POS8-0346
PS3_254	Composite Formation of Alpha-Lipoic Acid and Quercetin by Reverse Drowning-out Crystallization	Inhyeok KIM, Chung-Ang University, Korea	POS8-0413
PS3_255	Chemically Defined Polymer Surfaces Direct Collective Motility and Epithelial Repair in Human Intestinal Stem Cells	Seonghyeon PARK, KAIST, Korea	POS8-0427
PS3_256	DNA Rigidity-Dependent Force Transmission for Mechanophore Activation	Minjeong KU, Pukyong National University, Korea	POS8-0524
PS3_257	Polymeric Nanoparticles Containing Lumogen® Dyes for Cellular Detection and Imaging	Sujin JEONG, Pukyong National University, Korea	POS8-0525
PS3_258	Conformation-Dependent Regenerative Effects of PDRN in 3D-Bioprinted Muscle Scaffolds	Soyun LEE, Pukyong National University, Korea	POS8-0526
PS3_259	DNAOD: A Predictive Tool for DNA Absorption Spectra Using a Three-Base Nearest-Neighbor Model	Banyu Firdaus SOERIAWIDJAJA, Pukyong National University, Korea	POS8-0527
PS3_260	Multi-Scale MD Simulations Uncover lipid-DNA Self-Assembly and Membrane Insertion Driving Cellular Uptake	Eunryul JEON, Pukyong National University, Korea	POS8-0528
PS3_261	DNA-Mediated Force Transmission for Precise and Efficient Mechanophore Activation	Soobin LEE, Pukyong National University, Korea	POS8-0529
PS3_262	Development of a Real-Time, Non-Contact Photothermal Characterization Platform Featuring Integrated Accurate Temperature Measurement and Thermal Imaging	Dohyun KIM, Pukyong National University, Korea	POS8-0531
PS3_263	Poly(tannic acid) Coatings for Enhanced Osseointegration of Dental Implants	Hwain MYEONG, Pusan National University, Korea	POS8-0551
PS3_264	Composite of Bentonite and Tannic Acid as Hemostatic Biomaterials	Yejin KIM, Pusan National University, Korea	POS8-0555
PS3_265	Methacrylated Polysaccharide Hydrogels Crosslinked with Slide-Ring Polyrotaxane for Wound Healing Enhancement	Chaeyoung LEE, Pukyong National University, Korea	POS8-0569
PS3_266	Neutral Buoyancy-Based Simulated Microgravity Accelerates Vascular Network Formation and Maturation in Engineered 3D Tissues	Ho Yong KIM, Dankook University, Korea	POS8-0651
PS3_267	Choline Transporter-Targeted Nanoparticles for Dual Inhibition of Lipid Metabolism and DNA Methylation in Cancer Therapy	Naehak PARK, The Catholic University of Korea, Korea	POS8-0657
PS3_268	iCVD Polymer Thin Films on Meso-Macroporous Hydrogels for Enhanced Extracellular Vesicle Isolation	Sangsoo CHOI, Korea Institute of Science and Technology (KIST), Korea	POS8-0717
PS3_269	Injectable Nanogel Jams for Affinity-Free Nm Scale Molecular Diffusion Barriers	Gwangho SONG, Seoul National Univ., Korea	POS8-0826
PS3_270	Amphiphilic Ionogel Enabling Synergistic Universal Interface Adhesion and Reversible Debonding	Won Hyuk CHOI, Hanyang University, Korea	POS8-0829
PS3_271	Development of Photocurable Dental Resins Using Methacrylate-Functionalized Polyrotaxane	Seoyoung HAN, Pusan National University, Korea	POS8-0847
PS3_272	Photocurable MAPPSU-based di(meth)acrylate Resins with Enhanced Mechanical Properties for Dental Restorative Applications	Soojung LEE, Pusan National University, Korea	POS8-0849

PS3_273	Development of Slide-Ring Hydrogels Using Polymerizable Polyrotaxane Crosslinkers	Seoyoung HAN, Pusan National University, Korea	POS8-0851
PS3_274	Development of Polyrotaxane-Crosslinked Collagen Hydrogels with Enhanced Mechanical Properties via Movable Crosslinking Networks	Wonbin YI, Pusan National University, Korea	POS8-0854
PS3_275	3D-printable PEKK-based Dimethacrylate Resins with Multifunctional Groups for high-performance Dental Applications	Soojung LEE, Pusan National University, Korea	POS8-0870
PS3_276	Designing Affine Poly(A) Tails for Vector Stabilization and Protein Expression	Jin KIM, Korea University, Korea	POS8-0913
PS3_277	Curcumin Polymer-Decorated Cationic Hydroxyethylcellulose Nanocomplexes for Lung Cancer Therapy	Soye KIM, Seoul National University, Korea	POS8-0914
PS3_278	Triphenylphosphonium-Modified Cationic Hydroxyethylcellulose Polyplexes for Mitochondria-Targeted Gene Delivery and Cancer Therapy	Seoyeon PARK, Seoul National University, Korea	POS8-0919
PS3_279	Thermal Diffusion of Alkyl Derivatives of IR3535 in Anisotropic Hydroxypropyl Cellulose Films	Ekaterina ANOSOVA, Disinfectology Institute of F.F. Erisman FSCH of Rospotrebnadzor, Russia	POS8-0950
PS3_280	Lactate Oxidase-Coated Nanorobots for Autonomous Therapy in Inflammatory Bowel Disease	Hye Hyun KIM, Hanyang University, Korea	POS8-0966
PS3_281	Polymeric Materials with Repellent Activity Against Ixodid Ticks Based on Alkyl Butyl Acetyl Aminopropionate Derivatives	Evita KOSTENKO, Disinfectology Institute of F.F. Erisman FSCH of Rospotrebnadzor, Russia	POS8-0975
PS3_282	Thrombin-Coated Cross-Linked Chitosan-Gelatin Based Hemostatic Biomaterials	Dae Ryeong LEE, Any Medi-Healthcare Co.,Ltd., Korea	POS8-1072
PS3_283	A Molecularly Imprinted Hierarchical SnO ₂ Nanowire/Fe ₂ O ₃ Nanoflake Photoelectrochemical Sensor for Selective Fibrinogen Detection Under Visible-Light Illumination	Seok Jin LIM, Kyungpook National University, Korea	POS8-1169
PS3_284	Methacryl Glycol Chitosan Thermogel Matrix for Chondrocyte Spheroid Encapsulation and Cartilage Regeneration	Young Ju LEE, Chungnam National University, Korea	POS8-1216
PS3_285	Engineering a Multiresponsive Injectable Hydrogel via a Double-Injection Strategy for Diabetic Wound Healing	Chenyu ZHANG, West China Hospital, Sichuan University, China	POS8-1249
PS3_286	A Multilayer-Functionalized Acellular Dermal Scaffold for Improved Abdominal Wall Reconstruction	Yue LIU, West China Hospital, Sichuan University, China	POS8-1268
PS3_287	Effects of Single-Segmented Nanofibers in Composite Spheroids and Their Effect on Bio-Activities	Bada SHIN, Dong-A University, Korea	POS8-1276
PS3_288	Programmable Microfluidic Flow Through Collective Magnetic Oscillation of Microstructures	Hojun MOON, Hanyang University, Korea	POS8-1280
PS3_289	Network Control of Gelatin-Carboxymethyl Cellulose-Chitosan Bioinks for Extrusion Bioprinting	Sunggeun LEE, Seoul National University of Science and Technology, Korea	POS8-1289
PS3_290	Kaolin-Containing Kombucha Nanocellulose Bioink for Repair of Complex Tissue Defects	Sunggeun LEE, Seoul National University of Science and Technology, Korea	POS8-1292
PS3_291	A Multilayered 3D Bioprinted Scaffold with Engineered Extracellular Vesicles for Microenvironment Remodeling and Functional Recovery Following Severe Spinal Cord Injury	Duck Hyun SONG, Korea Institute of Science and Technology, Korea	POS8-1301
PS3_292	Multifunctional Platform Based on Cancer-Targeted Polymeric Nanoparticles for Enhanced Radiotherapy	Eunsun LEE, KIRAMS, Korea	POS8-1374

PS3_293	Organelle Localization-Induced Bio-Orthogonal Polymerization (OLIBOP) for Photostable Super-Resolution Live-Cell Imaging	Injun HWANG, UNIST, Korea	POS8-1418
PS3_294	Charge-Reversing Poly(L-alanine-co-L-choline glutamate) Thermogel for Sustained Drug Delivery	Rama Moorthy APPA, Ewha Womans University, Korea	POS8-1421
PS3_295	Design and Characterization of GOD/PtNP-Loaded Glucose-Responsive Microneedles for Controlled Drug Release	Panalee POMSEETHONG, Kangwon National University, Korea	POS8-1423
PS3_296	Visible-Light-Cured Carboxymethyl Chitosan Hydrogels with Tunable Crosslinking Density for Localized Photothermal and Photodynamic Therapy of Triple-Negative Breast Cancer	Wheemoon CHO, Institute of Cell and Tissue Engineering, The Catholic University of Korea, Korea	POS8-1426
PS3_297	Preparation and Characterization of Dual pH/Reduction-Responsive Self-Assembled Nanoparticles for Drug Delivery	Yuyuan GUO, Kangwon National University, Korea	POS8-1432
PS3_298	NIR-triggered Upconversion Nanocarriers for Synergistic Photodynamic Therapy and Carbon Monoxide Therapy	Jongseon CHOI, Chungnam National University, Korea	POS8-1442
PS3_299	NIR-triggered Upconversion Nanoreactor for Combined Photodynamic Therapy and Cancer Immunotherapy	Jongseon CHOI, Chungnam National University, Korea	POS8-1445
PS3_300	An Integrated Colorimetric and Strain-Sensing Hydrogel Platform for Multimodal Health Monitoring	Won Woo KIM, Chungnam National University, Korea	POS8-1446
PS3_301	Multifunctional NIR-Activated Upconversion Nanoplatform for MRI-Guided Synergistic Photodynamic and Photothermal Therapy	Ajith Kumar LAKSHMANAN, Chungnum National University, Korea	POS8-1461
PS3_302	Biodegradable Iodinated Polymeric Nanoparticle as Computed Tomography Contrast Agent for Direct Imaging of Cerebral Thrombus	Hyunjae PARK, Seoul National University, Korea	POS8-1466
PS3_303	Novel Liquid Embolic Compositions with Inherent Radiopacity for Endovascular Embolization with Iodinated Cyanoacrylate Structure	Subin PARK, Seoul National University, Korea	POS8-1468
PS3_304	Versatile Lysine-Based Bioreducible Nanogel for Chemical Drug and Nucleic Acid Therapeutics Delivery	Hana CHO, Soonchunhyang University, Korea	POS8-1512
PS3_305	Hyaluronic acid/polynucleotide Double-Network Hydrogels for Wound Healing	Kuen Yong LEE, Hanyang University, Korea	POS8-1513
PS3_306	High-Performance Dual-Crosslinked Nanocomposite Hydrogel for Multimodal Sensing Applications	Saw Htet Thura LIN, Chungnam National University, Korea	POS8-1538
PS3_307	Cryo-Enhanced Upconversion Photodynamic Therapy	Anara MOLKENOVA, Pusan National University, Korea	POS8-1644
PS3_308	Smart Triboelectric Bilayer Vascular Graft with Integrated Drug Delivery and Real-Time Hemodynamic Sensing	Kavitha GOPAL, GIST, Korea	POS8-1646
PS3_309	In Situ Nanomicelle-Forming Microneedles for Enhanced Antitumor Vaccination	Pin LIU, Sungkyunkwan University, Korea	POS8-1653
PS3_310	Microneedle-Mediated Lymph Node Targeting for Synergistic Chemoimmunotherapy of Metastatic Cancer	Pin LIU, Sungkyunkwan University, Korea	POS8-1654
PS3_311	Enhanced Cellular Delivery of siRNA via Micelle-Templated Dendritic Gold Nanoparticles	Byung Deok KIM, Sungkyunkwan Univ., Korea	POS8-1655
PS3_312	Anti-VEGF Encapsulated Sustained-Release Microspheres for Treating Subfoveal Choroidal Neovascularization	Kangmin KIM, Sungkyunkwan University, Korea	POS8-1656

PS3_313	Enhanced Tumor Therapy via Optimized 4-FTP-DA-PEI/Thiol-siRNA Delivery System	Kangmin KIM, Sungkyunkwan University, Korea	POS8-1657
PS3_314	Upcycling Banana Pseudostem Residues into Sustainable PBAT Biocomposites: A Circular Material Approach	Ming Te CHEN, National Chung Hsing University, Taiwan	POS9-0052
S9. Polymer Technology for Sustainability			
PS3_315	Grafting poly(amino acid)s onto Cellulose Nanofibers for Developing Biodegradable Superabsorbent Materials	Yongjun CHO, Korea Advanced Institute of Science and Technology, Korea	POS9-0162
PS3_316	Marine-Biodegradable Oligo(L-lactide) as a Plasticizer for Enhancing the Properties of Cellulose Acetate	Jin Ho SEOK, University of Tokyo, Japan	POS9-0165
PS3_317	Recoverable Polyanion Bead Catalysts for Selective Depolymerization and Fiber Recovery from Polyester Composites	Yongsu KIM, Sogang University, Korea	POS9-0183
PS3_318	Gluten-Derived Multifunctional Monomers Enabling Sustainable High-Performance Polymer Networks	Sebin JIN, Sogang University, Korea	POS9-0193
PS3_319	Influence of Plastic Surface Hardness on Micro- and Nanoplastic Generation Under Hydrodynamic Forces	Hyungrai KIM, Sogang University, Korea	POS9-0232
PS3_320	Research on Degradation Technology of Biodegradable Plastics (PBAT) Using Photoacid Generators (PAGs)	MinSoo BAEK, Pusan National University, Korea Research Institute of Chemical Technology, Korea	POS9-0290
PS3_321	Rice Paper-Derived Adsorbent for Gold Recovery	Seung Su SHIN, Korea University, Korea	POS9-0297
PS3_322	Li-Selective Fully Aromatic Polyamide Membranes Fabricated by Dual Strategies-Cosolvent Addition and Solvent Activation	Myung-Seok LEE, Korea University, Korea	POS9-0300
PS3_323	Preparation of Nitrogen-Doped Porous Carbon from Chemically Modified Lignin for Efficient CO ₂ Capture	Nattawut RODTUK, Inha University, Korea	POS9-0313
PS3_324	Bio-Based Multifunctional Soybean Oil as a Reactive Compatibilizer for PBAT/TPS Blends	Dinesh, Inha University, Korea	POS9-0319
PS3_325	Chemical Recycling of a Vanillin-Derived Bio-Based Polymer	Hyo Su SEO, Inha University, Korea	POS9-0330
PS3_326	Cross-Linked Poly(carbosiloxane) (cPC ₂ SO): A Depolymerizable Silicone Platform for Closed-Loop Recycling and 3D Printing	Yubin KANG, Kumoh National Institute of Technology, Korea	POS9-0447
PS3_327	Electroactive Bacterial Cellulose Nanofiber/PEDOT:PSS Composites for Environmental Remediation: Electroadsorptive Removal of Dyes and Heavy Metals	Lenny SEGURA, Universidad de Concepción, Chile	POS9-0513
PS3_328	Property Modification of Recycled PA66: Effects of Tannic Acid and Nucleation Agent	Sosan HWANG, Sustainable Materials R&D Center, Mobility Materials R&D Division, Net Zero Technology R&D Laboratory,	POS9-0519
PS3_329	One-Step Carbon Homologation of Cyclopentene to Cyclohexene via Tandem Isomerization-Metathesis	Jeongwoo LEE, KAIST, Korea	POS9-0600
PS3_330	Structure-Property Relationships of Recycled Polypropylene Modified with Functional Additives	Kaeun LEE, KITECH, Korea	POS9-0654
PS3_331	Thermally Processable Recycled Paper Foams Using Starch as a Natural Binder and Water-Blown Matrix	Dasom SON, Korea institute of industrial technology, Korea	POS9-0677

PS3_332	TEMPO-modified Nanocellulose-Based Dynamic Polymer Networks for Sustainability	In Woo CHEONG, Kyungpook National University, Korea	POS9-0722
PS3_333	High-Strength Acoustic Foams Enabled by Multiscale Plant-Fiber Interactions	James Sangmin CHOO, Seoul National University, Korea	POS9-0723
PS3_334	Nitrogen Site Incorporation in Lignin-Based Adsorbents for CO ₂ Capture	Jakkree INTRANUWONG, Inha University, Korea	POS9-0742
PS3_335	Crosslinker-Free Thiol-Epoxy Vitrimers Enabling Closed-Loop Chemical Recycling	Sohee KIM, Kumoh National Institute of Technology, Korea	POS9-0771
PS3_336	Bio-Based Dual-Imine Epoxy Vitrimers: High-Performance Reprocessable Thermosets for Sustainable Composite Applications	Kundan Singh RAWAT, Convergence Research Center for Recyclable Air mobility Materials and Platform, Korea Institute of Science and Technology (KIST), Korea	POS9-0793
PS3_337	Vanillin-Derived Polyimine Covalent Adaptive Networks with Self-Healing, Reprocessability, and Closed-Loop Chemical Recycling	Chetna TEWARI, Convergence Research Center for Recyclable Air mobility Materials and Platform, Korea Institute of Science and Technology (KIST), Korea	POS9-0796
PS3_338	Solid-State CO ₂ Capture Based on the Self-Assembly of poly(acrylic acid)-N,N'-di-tert-butylethylenediamine (DBEDA) Ionic Complexes	Jinsoon CHOI, Gwangju Institute of Science and Technology (GIST), Korea	POS9-0822
PS3_339	Pore-Confined Carbonic Anhydrase in a Nanoporous Covalent Framework Membrane for Cooperative CO ₂ Capture Enhancement	Marbert PINQUE, Gwangju Institute of Science and Technology, Korea	POS9-0845
PS3_340	Synthesis of Bio-Based Waterborne Polyurethane Resins and Evaluation of Lightfastness and Color Stability According to UV Stabilizer Blends	Hyun Ju PARK, KIMCO, Korea	POS9-1017
PS3_341	Development of Cellulose-Reinforced Bio-Composites for Sustainable and Highly Durable Footwear	Hyun Ju PARK, KIMCO, Korea	POS9-1018
PS3_342	Mechanochemical Depolymerization of Hard to Recycle Plastics as Opportunity for a Circular Economy	Christian SCHMITT, Karlsruhe Institute of Technology, Germany	POS9-1066
PS3_343	Bicontinuous Polyelectrolyte Complex Membranes for Redox-Mediated Electrochemical Recovery of Bio-Derived Volatile Fatty Acids	Wangsuk OH, Yonsei University Mirae campus, Korea	POS9-1077
PS3_344	Development of Fully Bio-Based Rosin-THFDM Tackifiers for Acrylic PSAs: Effect of Ester Composition on Adhesion Properties	Quang Linh TRAN, UST, KITECH, Korea	POS9-1126
PS3_345	Fully Renewable Lignin-Graft-Castor Oil Copolymers: Tailoring Properties from Nature's Building Blocks	Sundol KIM, Dankook University, Korea	POS9-1137
PS3_346	Chitosan Nanowhisker-Coated Cellulose Microbeads for pH-Triggered Release of Vitamin C	Hoik LEE, Korea Institute of Industrial Technology, Korea	POS9-1172
PS3_347	Mechanically Reinforced Biodegradable Nanocomposite Films Enabled by a Dual-Functional Additive	Chaeun MOON, Korea Institute of Industrial Technology (KITECH), Korea	POS9-1273
PS3_348	Fabrication of Sustainable Superhydrophobic Antibacterial Natural Textiles	Taehoo CHANG, Incheon National University, Korea	POS9-1275
PS3_349	Recycling of Waste Polyurethane Foam via Esterification	Yerin PARK, KITECH, Korea	POS9-1409
PS3_350	Biodegradable Membranes Using PBAT(polybutylene adipate-co-terephthalate) and PBS(polybutylene succinate) Biopolymer Blends	Jeongsun HWANG, Kyunghee University, Korea	POS9-1441
PS3_351	Composition-Dependent Thermo-Mechanical Degradation and Quantitative Degradation Indices of PP/HDPE Blends Under Repeated Extrusion	Sungwoong CHOI, KITECH, Korea	POS9-1487

PS3_352	Life Cycle Assessment of Recycled Polypropylene Derived from End-of-Life Vehicles for Automotive Recycling	Seongho RYU, Korea Automotive Technology Institute, Korea	POS9-1584
PS3_353	Structure-Property Relationships in Cellulose Microbeads Through Different Crosslinking Architectures	Su Kyoung LEE, Korea Institute of Industrial Technology, Korea	POS9-1590
S10. AI-assisted Design and Simulation of Polymers			
PS3_354	Computational Study of Complex Coacervate Core Micelle Structures and Properties	Je-Yeon JUNG, Seoul National University, Korea	POS10-0556
PS3_355	Accelerating Surface Passivation Material Design for High-Efficiency Perovskite LEDs via Reinforcement Learning	Dongbeen LEE, Gwangju Institute of Science and Technology, Korea	POS10-1452
PS3_356	AI-Assisted Multiscale Polymer Simulation Workflow: From the Atomistic Scale to Phase-Separated Morphology Representation	Haein LEE, Resonac Corporation, Japan	POS10-0074